Computing the Exact Distribution of Predictive Regression*

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Abstract

We provide an exact finite sample analysis of a predictive regression whose regressor follows a stationary AR(1) process, giving integral expressions for the densities and distribution functions of the estimated slope coefficient, the OLS t-ratio, and the R^2. The obstacle to using such results in practice has been computational: existing methods require the eigenvalues of $2T \times 2T$ matrices, where T is the sample size, an $O(T^3)$ operation, repeated at many quadrature points for the t-ratio and the R^2. We develop algorithms that exploit the near-Toeplitz structure of the problem to evaluate every integrand in $O(1)$ operations, independent of T, so that the cost of an exact distribution no longer grows with the sample size: the distribution of the slope costs milliseconds and those of the t-ratio and the R^2 cost seconds to minutes, at any T. The same algorithms deliver exact moments, including those of the t-ratio, for which no low-dimensional integral representation was previously available. As a by-product of independent interest, we prove a structural property of the matrices involved that identifies the correct branch of the complex square root at every point, removing the error-prone tracking of branch-cut crossings that numerical inversion of characteristic functions typically requires.

Keywords: predictive regression, exact distribution, autoregressive process, quadratic forms in normal variables

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1. Introduction

Predictive regressions, in which the return of an asset is regressed on a lagged predictive variable, are a workhorse of empirical finance and macroeconomics. It has long been understood that when the predictor is persistent and its innovations are correlated with returns, the OLS estimator of the slope coefficient is biased in finite samples and the usual t-test can be badly oversized (Kendall, 1954; Marriott and Pope, 1954; Stambaugh, 1999). A large literature responds to this problem with alternative asymptotic frameworks and bias corrections, including local-to-unity asymptotics (Campbell and Shiller, 1988; Poterba and Summers, 1988; Cutler et al., 1991; Valkanov, 2003; Torous et al., 2004; Kiviet and Phillips, 2012; Phillips, 2015), bias-adjusted estimators (Amihud and Hurvich, 2004; Amihud et al., 2009; Xu, 2020), conditional or Bonferroni-type procedures (Cavanagh et al., 1995; Campbell and Yogo, 2006; Lanne, 2002; Jiang et al., 2019), and instrumental-variable procedures robust to persistence (Kostakis et al., 2015).

An alternative and, in principle, more direct route is exact finite sample analysis. Under the standard assumption that the innovations are i.i.d. bivariate normal and the predictor follows a stationary AR(1) process, the estimated slope coefficient, its t-ratio, and the regression R^2 are all functions of ratios of quadratic forms in normal random variables, and their distributions can be written as finite-dimensional integrals using classical inversion results (Gil-Pelaez, 1951; Geary, 1944; Imhof, 1961; Shephard, 1991). Stambaugh (1999) used exactly this approach to characterize the distribution of the estimated slope coefficient. Why, then, is exact finite sample inference almost never carried out in applied work? The answer is computational. The integrands of these inversion formulas involve determinants and traces of $2T \times 2T$ complex matrices, where T is the sample size. Direct evaluation of each integrand costs $O(T^3)$ operations, and while a one-time eigenvalue decomposition reduces the per-point cost for the distribution of the slope coefficient, no such shortcut is available for the t-ratio and R^2, whose distributions require triple integrals with matrices that change with the variables of integration. For monthly samples of the size typical in return predictability studies (T in the hundreds or above a thousand), the computational burden is prohibitive.

This paper removes the computational obstacle. Our contribution is a set of algorithms

that evaluate every integrand arising in the exact distribution theory of predictive regressions in $O(1)$ operations, a count that does not grow with the sample size. The key observation is that, after an analytical reduction, every determinant and trace that appears is built from a symmetric tridiagonal Toeplitz matrix that is modified by a rank-one update and a low-dimensional border. The determinant and the required bilinear forms of the inverse of a tridiagonal Toeplitz matrix admit closed-form expressions, and the rank-one and border corrections can be absorbed analytically. Consequently, the characteristic functions and their derivatives can be computed in closed form. What this changes is not the dimension of the inversion integrals for $\hat{\beta}$, $t(\hat{\beta})$, and R^2—that is fixed by the inversion theorem and is low to begin with—but the cost of their integrands, which becomes independent of T and, in absolute terms, negligible.

Turning the closed-form expressions into a reliable algorithm requires some care, and a second contribution of the paper is a treatment of the numerical issues that arise. The closed-form determinant of a tridiagonal Toeplitz matrix becomes numerically degenerate when the two roots of its characteristic equation coalesce, which happens precisely in the empirically relevant region where the predictor is highly persistent; we derive series expansions that restore full accuracy there. In addition, the square root of a complex determinant must be evaluated as a continuous function of the variables of integration to avoid branch-cut discontinuities; we show how to track the phase of each factor so that the integrand remains continuous. Finally, the integrands have removable singularities at the origin, and we provide the closed-form limits needed to handle them.

The branch-cut problem deserves a further word, because our treatment of it differs from the prevailing practice. When the eigenvalue decomposition is not computed, the usual remedy is to detect the discontinuities numerically and repair the phase ex post, which is time consuming and not fool-proof. No repair is needed here. The matrices arising from these characteristic functions are automatically *accretive*—their Hermitian parts are positive definite along the integration path—and we *prove* that accretivity keeps every factor of the assembled phase off the branch cut. The argument is given in Section 3.3.2 and the Appendix.

With these tools, the exact distribution of $\hat{\beta}$ requires a univariate integral with an $O(1)$

integrand, and those of $t(\hat{\beta})$ and R^2 at most a triple integral with an $O(1)$ integrand. When the innovations of the return and of the predictor are uncorrelated, the distributions of $t(\hat{\beta})$ and R^2 collapse to mixtures of noncentral t and noncentral beta distributions with mixing weights driven by a single quadratic form, and a double integral suffices.

The same machinery computes exact moments. Each moment of $\hat{\beta}$, $t(\hat{\beta})$, and R^2 can be written as a low-dimensional integral of derivatives of the joint moment generating function of the underlying quadratic forms, and that moment generating function is the same determinant as the characteristic function, evaluated at real instead of imaginary arguments—so the $O(1)$ evaluation carries over verbatim, and in fact becomes simpler, since at real arguments the matrix is positive definite and the branch difficulties cannot arise at all. For $\hat{\beta}$ and R^2, integral representations of the moments were available in principle from the literature on ratios of quadratic forms (Sawa, 1978; Magnus, 1986; Meng, 2005; Bao and Ullah, 2007); we make them practical. For $t(\hat{\beta})$ no such representation was available—the denominator involves the determinant of a 2×2 Gram matrix rather than a product of quadratic forms, and existing approaches lead to an infinite series of double integrals whose convergence deteriorates exactly where predictive regressions are interesting. We derive a representation that linearizes this determinant by a matrix-variate gamma integral over the cone of 2×2 positive definite matrices: every moment of $t(\hat{\beta})$ becomes a three-dimensional integral, and odd moments, including the mean, collapse to two-dimensional ones.

We put the resulting algorithms to work on a question that exact computation is uniquely suited to answer: how well do the asymptotic distributions in common use approximate the finite sample distributions they stand in for? We compare the exact densities of $\hat{\beta}$, $t(\hat{\beta})$, and R^2 with the standard fixed-ϕ theory and with the local-to-unit-root theory that treats the predictor as nearly integrated, under the null of no predictability and under an alternative calibrated to the weak predictability found in return regressions. Under the null, the fixed-ϕ approximation is excellent for a moderately persistent predictor but badly inaccurate for a highly persistent one—the empirically relevant case—even at T close to a thousand, while the local-to-unit-root approximation is accurate across the range of persistence we consider. Under the alternative, the usual fixed-alternative local-to-unit-root statement is a

rate of divergence rather than a usable finite sample approximation: it scales away the $O(1)$ sampling noise that carries the dispersion of the statistic. Retaining that term restores a distribution, and so restored the approximation is accurate at three of our four configurations. It is the fixed-ϕ limits that are strained under the alternative: they describe the distribution of R^2 well only once the signal $T\rho^2$ is large, which at the two percent population R^2 of return regressions takes a sample of roughly a thousand observations. Both frameworks also change form discontinuously at $\beta = 0$, whereas the exact distributions do not, so neither branch can be trusted when predictability is weak. Since the applied researcher does not know which asymptotic regime her data inhabit, and since the exact distribution requires no such choice, the case for computing it rests on being able to do so quickly—which is what the algorithms of this paper provide. A MATLAB suite implementing every algorithm of the paper is publicly available.

The remainder of the paper is organized as follows. Section 2 presents the model and the integral expressions for the exact distributions. Section 3 develops the $O(1)$ algorithm for the distribution of the estimated slope coefficient. Section 4 develops the corresponding algorithms for the t-ratio and R^2. Section 5 derives the exact moments of the three statistics and shows that the same machinery, applied to the moment generating function, computes them. Section 6 compares the exact finite sample distributions with their asymptotic approximations, and Section 7 concludes. Proofs are in the Appendix.

Throughout the paper, we use I_T to stand for a $T \times T$ identity matrix, 0_T and 1_T to stand for the $T \times 1$ vectors of zeros and of ones, $0_{m\times n}$ to stand for the $m \times n$ matrix of zeros, $e_{i,n}$ to stand for the i-th column of I_n, $\Gamma(a)$ is the gamma function, $\mathrm{B}(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$ is the beta function, $(a)_n = a(a + 1) \dots (a + n - 1)$ is the Pochhammer symbol, and $\lfloor x \rfloor$ stands for the greatest integer less than or equal to x. For a complex number z, $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ denote its real and imaginary parts, $|z|$ its modulus, $\arg(z) \in (-\pi, \pi]$ its principal argument, and $z^{\frac{1}{2}}$ its principal square root. For a complex matrix C, $\mathrm{Re}\, C$ denotes its entrywise real part and C^* its conjugate transpose, and for a Hermitian matrix P we write $P \succ 0$ to mean that P is positive definite. Every matrix to which we apply this notation is complex symmetric, so that $\mathrm{Re}\, C$ is real symmetric and coincides with the Hermitian part $(C + C^*)/2$;

a matrix with $\operatorname{Re} C \succ 0$ is called *accretive*, equivalently x^*Cx has positive real part for every $x \neq 0$.

2. Finite Sample Analysis of Predictive Regressions

2.1. The Model

Our setup of predictive regression is identical to the one adopted by Stambaugh (1999). Let r_t be the return (which can be raw return or excess return) of an asset at time t and z_{t-1} be a predictive variable at time $t-1$. We assume

$$r_t = \alpha + \beta z_{t-1} + u_t, \quad t = 1, \ldots, T, \tag{1}$$

where T is the sample size, and z_t follows a stationary AR(1) process

$$z_t - \mu_z = \phi(z_{t-1} - \mu_z) + v_t, \quad t = 1, \ldots, T-1, \tag{2}$$

with mean μ_z, AR(1) coefficient ϕ satisfying $|\phi| < 1$, and initial observation drawn from the stationary distribution, $z_0 \sim \mathcal{N}(\mu_z, \sigma_v^2/(1-\phi^2))$. In addition, we assume (u_t, v_t) is i.i.d. bivariate normally distributed with mean zero and covariance matrix

$$\operatorname{Var}\begin{bmatrix} u_t \\ v_t \end{bmatrix} = \begin{bmatrix} \sigma_u^2 & \sigma_{uv} \\ \sigma_{uv} & \sigma_v^2 \end{bmatrix}, \tag{3}$$

and we write $\rho_{uv} = \sigma_{uv}/(\sigma_u \sigma_v)$ for the correlation of the innovations. Based on the property of an AR(1) process, the variance of z_t is given by $\sigma_z^2 = \sigma_v^2/(1-\phi^2)$.

The OLS estimates of β and α in (1) are given by

$$\hat{\beta} = \frac{\sum_{t=1}^{T}(r_t - \bar{r})(z_{t-1} - \bar{z})}{\sum_{t=1}^{T}(z_{t-1} - \bar{z})^2}, \tag{4}$$

$$\hat{\alpha} = \bar{r} - \hat{\beta}\bar{z}, \tag{5}$$

where $\bar{r} = \sum_{t=1}^{T} r_t/T$ and $\bar{z} = \sum_{t=1}^{T} z_{t-1}/T$.

For the test of $H_0 : \beta = 0$, one often relies on the OLS t-ratio of $\hat{\beta}$, which is defined as

$$t(\hat{\beta}) = \frac{\hat{\beta}}{s(\hat{\beta})}, \tag{6}$$

where

$$s^2(\hat{\beta}) = \frac{\sum_{t=1}^{T} \hat{u}_t^2}{(T-2)\sum_{t=1}^{T}(z_{t-1}-\bar{z})^2} \tag{7}$$

and $\hat{u}_t = r_t - \bar{r} - \hat{\beta}(z_{t-1} - \bar{z})$.

In addition, one can also measure the goodness-of-fit of a predictive regression by computing its sample R^2, which is defined as

$$R^2 = \frac{\left[\sum_{t=1}^{T}(r_t - \bar{r})(z_{t-1} - \bar{z})\right]^2}{\sum_{t=1}^{T}(r_t - \bar{r})^2 \sum_{t=1}^{T}(z_{t-1} - \bar{z})^2}. \tag{8}$$

To prepare for our finite sample analysis, we rewrite the expressions of $\hat{\beta}$, $t(\hat{\beta})$ and R^2 in matrix notation. Letting $r = [r_1,\ r_2, \ldots, r_T]'$ and $z = [z_0,\ z_1, \ldots, z_{T-1}]'$, we can express $\hat{\beta}$, $t(\hat{\beta})$ and R^2 as

$$\hat{\beta} = \frac{z'Mr}{z'Mz}, \tag{9}$$

$$t(\hat{\beta}) = \frac{(T-2)^{\frac{1}{2}} z'Mr}{[(z'Mz)(r'Mr) - (z'Mr)^2]^{\frac{1}{2}}}, \tag{10}$$

$$R^2 = \frac{(z'Mr)^2}{(z'Mz)(r'Mr)}, \tag{11}$$

where $M = I_T - (1_T 1_T')/T$.

Note that $\hat{\beta}$, $t(\hat{\beta})$ and R^2 are invariant to the intercept α and the predictor mean μ_z, since both enter the statistics only through constant shifts of r and z that the demeaning matrix M annihilates; we therefore set $\alpha = \mu_z = 0$ without loss of generality.

The two scale parameters σ_u and σ_v can also be removed, but doing so changes what plays the role of the slope, so it is worth being explicit. Divide the return by σ_u and the predictor by σ_v,

$$r_t^* \ = \ \frac{r_t}{\sigma_u}, \qquad z_t^* \ = \ \frac{z_t}{\sigma_v}. \tag{12}$$

The starred variables satisfy the same system (1)–(2) with the same autoregressive coefficient ϕ and innovations $u_t^* = u_t/\sigma_u$ and $v_t^* = v_t/\sigma_v$ that have unit variances and the same correlation ρ_{uv}; only the slope changes, from β to

$$\tilde{\beta} \ = \ \frac{\sigma_v}{\sigma_u}\beta. \tag{13}$$

Because $t(\hat{\beta})$ and R^2 are invariant to separate rescalings of the return and of the predictor, they take the same value whether they are computed from the original or from the starred data. The estimated slope is not invariant: the slope estimated from the starred data is $(\sigma_v/\sigma_u)\hat{\beta}$, which is why the normalized statistic b introduced in Section 2.2 below is defined as $(\sigma_v/\sigma_u)(\hat{\beta}-\beta)$, the centered slope estimate of the standardized regression. From here on we work with the standardized variables and drop the stars, so that $\sigma_u = \sigma_v = 1$ throughout and the population slope is $\tilde{\beta}$ rather than β. No generality is lost: the four parameters T, ϕ, ρ_{uv} and $\tilde{\beta}$ determine everything that follows.

2.2. Finite Sample Analysis of Estimated Slope Coefficient

A finite sample analysis of $\hat{\beta}$ in a predictive regression with an AR(1) regressor had already been conducted by Stambaugh (1999). Specifically, he presented the cumulative distribution function of $\hat{\beta}-\beta$ as well as its moments. In this subsection, we present not just the cumulative distribution function of $\hat{\beta}-\beta$ but also its density function. Instead of presenting the exact distribution of $\hat{\beta}-\beta$, we choose to present the exact distribution of

$$b = \frac{\sigma_v}{\sigma_u}(\hat{\beta}-\beta). \tag{14}$$

This is the centered slope estimate of the standardized regression (12), so its exact distribution depends only on ϕ, ρ_{uv} and T, and not on σ_u or σ_v. Once we have the cumulative distribution and density functions of b, we can obtain those of $\hat{\beta}$ based on the following identities:

$$\mathbb{P}[\hat{\beta} < c] = \mathbb{P}\left[b < \frac{\sigma_v}{\sigma_u}(c-\beta)\right], \tag{15}$$

$$f_{\hat{\beta}}(c) = \frac{\sigma_v}{\sigma_u} f_b\left(\frac{\sigma_v}{\sigma_u}(c-\beta)\right). \tag{16}$$

The key to obtaining the exact distribution of b is to write it as a ratio of quadratic forms in normal random variables. Letting $x = [u',\ z']'$ where $u = [u_1,\ u_2, \ldots, u_T]'$, we can express b as

$$b = \frac{z'Mu}{z'Mz} = \frac{x'Ax}{x'Bx}, \tag{17}$$

where

$$A = \begin{bmatrix} 0_{T\times T} & M/2 \\ M/2 & 0_{T\times T} \end{bmatrix}, \qquad B = \begin{bmatrix} 0_{T\times T} & 0_{T\times T} \\ 0_{T\times T} & M \end{bmatrix}. \tag{18}$$

Under the assumptions of $\alpha = \mu_z = 0$ and $\sigma_u = \sigma_v = 1$, we have $x \sim \mathcal{N}(0_{2T}, V_x)$, with

$$V_x = \begin{bmatrix} I_T & \rho_{uv}G \\ \rho_{uv}G' & H \end{bmatrix}, \tag{19}$$

where H is a symmetric Toeplitz matrix with its (i,j)-th element equal to $\phi^{|i-j|}/(1-\phi^2)$ and G is an upper triangular Toeplitz matrix with its first row being $[0,\ 1,\ \phi, \ldots, \phi^{T-2}]$.

Using the results from Gil-Pelaez (1951) and Imhof (1961), we can obtain the cumulative distribution function of b as

$$\mathbb{P}[b < c] = \mathbb{P}[x'(A - cB)x < 0] = \frac{1}{2} - \frac{1}{\pi}\int_0^\infty \frac{\mathrm{Im}(\phi_X(t))}{t}\mathrm{d}t, \tag{20}$$

where $\phi_X(t)$ is the characteristic function of $X = x'(A - cB)x$, which is given by

$$\phi_X(t) = |I_{2T} - 2\mathrm{i}t(\tilde{A} - c\tilde{B})|^{-\frac{1}{2}}, \tag{21}$$

where $\tilde{A} = V_x^{\frac{1}{2}} A V_x^{\frac{1}{2}}$, $\tilde{B} = V_x^{\frac{1}{2}} B V_x^{\frac{1}{2}}$, and $\mathrm{i} = \sqrt{-1}$. Here and in (22) below, the power $-\frac{1}{2}$ denotes the square root that is continuous in t with $\phi_X(0) = 1$—equivalently, the product of principal roots in (23)—which in general differs from the principal root of the assembled determinant; Section 3.3.2 treats this distinction at length. Numerical methods for evaluating this integral are given by Imhof (1961) and Lu and King (2002).

For the density function of b, we can use the result from Geary (1944) to show that

$$f_b(c) = \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left(\phi_X(t)\mathrm{tr}((I_{2T} - 2\mathrm{i}t(\tilde{A} - c\tilde{B}))^{-1}\tilde{B})\right)\mathrm{d}t, \tag{22}$$

where Broda and Paolella (2009) provide an algorithm for computing this integral using real arithmetic.

There are two different ways of computing the integrands in (20) and (22). One is to directly compute $\phi_X(t)$ by evaluating the determinant. However, since computing the

determinant of a dense $n \times n$ matrix requires $O(n^3)$ operations with standard algorithms,[1] it is quite time consuming to compute $\phi_X(t)$ directly. Instead, researchers typically first compute the eigenvalues of $\tilde{A} - c\tilde{B}$ and then compute $\phi_X(t)$ using

$$\phi_X(t) = \prod_{j=1}^{2T} (1 - 2it\lambda_j)^{-\frac{1}{2}}, \tag{23}$$

where λ_1 to λ_{2T} are the eigenvalues of $\tilde{A} - c\tilde{B}$. Once these eigenvalues are computed, the integrands in (20) and (22) can be computed with $O(n)$ operations, where $n = 2T$ is the order of the matrices involved.[2] This is the approach adopted by the algorithms of Imhof (1961) and Broda and Paolella (2009). Nevertheless, computing the eigenvalues of a dense symmetric $n \times n$ matrix also requires $O(n^3)$ operations with standard algorithms, so this method is still quite inefficient for large T because we need to compute the eigenvalues of a $2T \times 2T$ matrix. Due to the inefficiency of the computation of these numerical integrals, it is quite rare to find applied researchers computing the finite sample distribution of $\hat{\beta}$ in their empirical work. In Section 3, we show that the integrands in (20) and (22) can in fact be computed with $O(1)$ operations, without ever forming a matrix.

2.3. Finite Sample Analysis of t-ratio and R^2

For the finite sample analysis of $t(\hat{\beta})$ and R^2, we first define two ratios of quadratic forms in $y = [r',\ z']'$:

$$\mathcal{R}_1 = \frac{z'Mr}{z'Mz} = \frac{y'Ay}{y'By}, \tag{24}$$

$$\mathcal{R}_2 = \frac{r'Mr}{z'Mz} = \frac{y'Cy}{y'By}, \tag{25}$$

where A and B are defined in (18) and

$$C = \begin{bmatrix} M & 0_{T\times T} \\ 0_{T\times T} & 0_{T\times T} \end{bmatrix}. \tag{26}$$

[1] Determinant computation can in principle be carried out in matrix-multiplication time $O(n^\omega)$ with $\omega < 2.38$, but such algorithms are not used in numerical practice. Throughout the paper, operation counts refer to the standard algorithms implemented in numerical linear algebra libraries.

[2] For the computation of the integrand in (22), we also need to obtain the eigenvectors of $\tilde{A} - c\tilde{B}$.

Using $\mathcal{R}_1$ and $\mathcal{R}_2$, we can express $t(\hat{\beta})$ and R^2 as

$$t(\hat{\beta}) = \frac{(T-2)^{\frac{1}{2}}\mathcal{R}_1}{(\mathcal{R}_2 - \mathcal{R}_1^2)^{\frac{1}{2}}}, \tag{27}$$

$$R^2 = \frac{\mathcal{R}_1^2}{\mathcal{R}_2}. \tag{28}$$

With the standardization (12) in force, so that $\alpha = \mu_z = 0$, $\sigma_u = \sigma_v = 1$ and the population slope is $\tilde{\beta}$ of (13), we have $y \sim \mathcal{N}(0_{2T}, V_y)$ with

$$V_y = \begin{bmatrix} I_T + \tilde{\beta}\rho_{uv}(G + G') + \tilde{\beta}^2 H & \rho_{uv}G + \tilde{\beta}H \\ \rho_{uv}G' + \tilde{\beta}H & H \end{bmatrix}. \tag{29}$$

Since the distribution of y only depends on T, ϕ, ρ_{uv} and $\tilde{\beta}$, the finite sample distributions of $t(\hat{\beta})$ and R^2 are also determined by just these four parameters.

The joint cumulative distribution function of $\mathcal{R}_1$ and $\mathcal{R}_2$ is given by

$$F_{\mathcal{R}_1,\mathcal{R}_2}(r_1, r_2) = \mathbb{P}[\mathcal{R}_1 < r_1,\ \mathcal{R}_2 < r_2] = \mathbb{P}[X_1 < 0,\ X_2 < 0], \tag{30}$$

where $X_1 = y'(A - r_1 B)y$ and $X_2 = y'(C - r_2 B)y$. Using Theorem 5 of Shephard (1991), we can show that

$$\begin{aligned} F_{\mathcal{R}_1,\mathcal{R}_2}(r_1, r_2) = &\frac{F_{\mathcal{R}_1}(r_1) + F_{\mathcal{R}_2}(r_2)}{2} - \frac{1}{4} \\ &- \frac{1}{2\pi^2}\int_0^\infty \int_0^\infty \frac{\mathrm{Re}(\phi_{X_1,X_2}(s,t) - \phi_{X_1,X_2}(s,-t))}{st}\mathrm{d}s\mathrm{d}t, \end{aligned} \tag{31}$$

where $\phi_{X_1,X_2}(s,t)$ is the joint characteristic function of (X_1, X_2), and its explicit expression is given by (see, for example, Magnus (1986, Lemma 5))

$$\phi_{X_1,X_2}(s,t) = |\mathcal{D}|^{\frac{1}{2}}, \tag{32}$$

where

$$\mathcal{D} = \mathcal{D}(s,t) = (I_{2T} - 2\mathrm{i}s(\tilde{A} - r_1\tilde{B}) - 2\mathrm{i}t(\tilde{C} - r_2\tilde{B}))^{-1}, \tag{33}$$

$\tilde{A} = V_y^{\frac{1}{2}} A V_y^{\frac{1}{2}}$, $\tilde{B} = V_y^{\frac{1}{2}} B V_y^{\frac{1}{2}}$, and $\tilde{C} = V_y^{\frac{1}{2}} C V_y^{\frac{1}{2}}$ (the tilded matrices of Section 2.2 redefined with V_y in place of V_x). It is straightforward to show that

$$\frac{\partial |\mathcal{D}|^{\frac{1}{2}}}{\partial r_1} = -\mathrm{i}s|\mathcal{D}|^{\frac{1}{2}}\mathrm{tr}(\mathcal{D}\tilde{B}), \tag{34}$$

$$\frac{\partial|\mathcal{D}|^{\frac{1}{2}}}{\partial r_2} = -\mathrm{i}t|\mathcal{D}|^{\frac{1}{2}}\mathrm{tr}(\mathcal{D}\tilde{B}), \tag{35}$$

$$\frac{\partial \mathrm{tr}(\mathcal{D}\tilde{B})}{\partial r_1} = -2\mathrm{i}s\mathrm{tr}((\mathcal{D}\tilde{B})^2), \tag{36}$$

$$\frac{\partial \mathrm{tr}(\mathcal{D}\tilde{B})}{\partial r_2} = -2\mathrm{i}t\mathrm{tr}((\mathcal{D}\tilde{B})^2). \tag{37}$$

Using these identities, we obtain

$$\frac{\partial^2 \phi_{X_1,X_2}(s,t)}{\partial r_1 \partial r_2} = -st|\mathcal{D}|^{\frac{1}{2}}\left[(\mathrm{tr}(\mathcal{D}\tilde{B}))^2 + 2\mathrm{tr}((\mathcal{D}\tilde{B})^2)\right]. \tag{38}$$

Differentiating $F_{\mathcal{R}_1,\mathcal{R}_2}(r_1, r_2)$ and using (38), we obtain the joint density function of $(\mathcal{R}_1, \mathcal{R}_2)$ as

$$f_{\mathcal{R}_1,\mathcal{R}_2}(r_1, r_2) = \frac{1}{2\pi^2}\int_0^\infty \int_0^\infty \mathrm{Re}(h(s,t) + h(s,-t))\mathrm{d}s\mathrm{d}t, \tag{39}$$

for $-\sqrt{r_2} \leq r_1 \leq \sqrt{r_2}$ and $r_2 > 0$, where

$$h(s,t) = |\mathcal{D}|^{\frac{1}{2}}\left[(\mathrm{tr}(\mathcal{D}\tilde{B}))^2 + 2\mathrm{tr}((\mathcal{D}\tilde{B})^2)\right]. \tag{40}$$

With the joint density of $(\mathcal{R}_1, \mathcal{R}_2)$ available, we can then compute the densities of $t(\hat{\beta})$ and R^2. The following proposition summarizes these results.

Proposition 1. *The density functions of $t(\hat{\beta})$ and R^2 are given by*

$$f_{t(\hat{\beta})}(c) = \frac{T-2}{(T-2+c^2)^{\frac{3}{2}}}\int_0^\infty \sqrt{r_2} f_{\mathcal{R}_1,\mathcal{R}_2}\left(\frac{c\sqrt{r_2}}{\sqrt{T-2+c^2}}, r_2\right)\mathrm{d}r_2, \tag{41}$$

$$f_{R^2}(c) = \frac{1}{2\sqrt{c}}\int_0^\infty \sqrt{r_2}[f_{\mathcal{R}_1,\mathcal{R}_2}(-\sqrt{cr_2}, r_2) + f_{\mathcal{R}_1,\mathcal{R}_2}(\sqrt{cr_2}, r_2)]\mathrm{d}r_2 \quad \textit{for } 0 < c \leq 1. \tag{42}$$

The two densities are tied together by the exact identity $R^2 = t(\hat{\beta})^2/(\nu + t(\hat{\beta})^2)$ with $\nu = T-2$, which follows from (28). Writing w for the argument of the density of R^2, it gives

$$f_{R^2}(w) = \frac{\sqrt{\nu}}{2\sqrt{w}\,(1-w)^{\frac{3}{2}}}\left[f_{t(\hat{\beta})}(w_\nu) + f_{t(\hat{\beta})}(-w_\nu)\right], \qquad w_\nu = \left(\frac{\nu w}{1-w}\right)^{\frac{1}{2}}. \tag{43}$$

One consequence is worth recording, because it is visible in every R^2 density we plot. The density of R^2 is unbounded at the origin for every T, under the null and under the alternative alike, and the divergence is of order $w^{-1/2}$ with a coefficient that is simply the density of the t-ratio at zero:

$$\sqrt{w}\, f_{R^2}(w) \longrightarrow \sqrt{\nu}\, f_{t(\hat{\beta})}(0) \qquad \text{as } w \downarrow 0. \tag{44}$$

The limit is finite and strictly positive, so the singularity is integrable but never disappears. The coefficient is computed at $O(1)$ cost by the algorithm of Section 4. When $\rho_{uv} = 0$ it is available in closed form,

$$\sqrt{w}\, f_{R^2}(w) \;\longrightarrow\; \frac{\mathbb{E}\left[e^{-\frac{1}{2}\tilde{\beta}^2 Q}\right]}{B\left(\frac{1}{2}, \frac{T-2}{2}\right)} \;=\; \frac{\left|I_T + \tilde{\beta}^2 HM\right|^{-\frac{1}{2}}}{B\left(\frac{1}{2}, \frac{T-2}{2}\right)}, \tag{45}$$

where $Q = z'Mz$ and $B(\cdot,\cdot)$ is the beta function. Under the null this is $1/B(\frac{1}{2}, \frac{T-2}{2})$, the value implied by the Beta$(\frac{1}{2}, \frac{T-2}{2})$ distribution of Section 2.4; under the alternative the same singularity survives, attenuated by the Laplace transform of Q evaluated at $\tilde{\beta}^2/2$. Increasing the signal thus damps the spike at the origin but never removes it. The Online Appendix derives (43)–(45) and presents an explicit expression for the determinant in (45).

The singularity is a fact about the statistic and not an artifact of the plots, and it has a consequence for how an R^2 should be read. Because the mode of the exact density sits at zero whatever the signal, a plotted density conveys almost nothing about where the distribution actually is, and the figures below therefore truncate the vertical axis at the body of the distribution. The quantiles do not have this problem, and they are what the reader should take from those figures. At $\phi = 0.99$, $T = 120$ and $\rho_{uv} = -0.856$, for instance, the exact null distribution of R^2 has median 0.0104 and ninety-fifth percentile 0.0533. Half of all samples generated with no predictability whatever therefore produce a reported R^2 above one percent, and one in twenty produces one above five percent—a figure that in this literature would be read as economically substantial. For comparison, at $\rho_{uv} = 0$ the null distribution of R^2 is exactly Beta$(\frac{1}{2}, \frac{T-2}{2})$ (Section 2.4), whose median at $T = 120$ is 0.0038; the correlation between the innovations inflates the median by a factor of nearly three. Whenever a truncated density is displayed in what follows, the corresponding quantiles are quoted alongside it.

Proposition 1 suggests that we need to numerically evaluate a triple integral to compute the density of $t(\hat{\beta})$ or R^2. For this result to be practical, we need to have an efficient method for computing the integrand $h(s,t)$ in $f_{\mathcal{R}_1,\mathcal{R}_2}(r_1, r_2)$. Unlike the case of $\hat{\beta}$ where we can first compute the eigenvalues of a matrix and then evaluate the integrand with $O(n)$ operations, this faster method cannot be applied here because the matrix $\mathcal{D}(s,t)$ depends on both s and

t. Nevertheless, one can somewhat improve the computational efficiency by making a polar transformation of $s = r\cos\theta$ and $t = r\sin\theta$ to obtain

$$f_{\mathcal{R}_1,\mathcal{R}_2}(r_1, r_2) = \frac{1}{2\pi^2}\int_0^{\frac{\pi}{2}}\int_0^{\infty} \mathrm{Re}(h(r\cos\theta, r\sin\theta) + h(r\cos\theta, -r\sin\theta))r\mathrm{d}r\mathrm{d}\theta. \tag{46}$$

Letting $\Lambda = \mathrm{Diag}(\lambda_1, \ldots, \lambda_{2T})$ be a diagonal matrix of the eigenvalues of $\cos\theta(\tilde{A} - r_1\tilde{B}) + \sin\theta(\tilde{C} - r_2\tilde{B})$ and P be the corresponding matrix of eigenvectors, we can then evaluate the terms in $h(r\cos\theta, r\sin\theta)$ using

$$|\mathcal{D}|^{\frac{1}{2}} = \prod_{j=1}^{2T}\frac{1}{(1 - 2\mathrm{i}r\lambda_j)^{\frac{1}{2}}}, \tag{47}$$

$$\mathrm{tr}(\mathcal{D}\tilde{B}) = \mathrm{tr}((I_{2T} - 2\mathrm{i}r\Lambda)^{-1}\hat{B}), \tag{48}$$

$$\mathrm{tr}((\mathcal{D}\tilde{B})^2) = \mathrm{tr}(((I_{2T} - 2\mathrm{i}r\Lambda)^{-1}\hat{B})^2), \tag{49}$$

where $\hat{B} = P'\tilde{B}P$. This suggests that after we obtain P and Λ, we can compute the integrand with $O(n^2)$ operations (because the term $\mathrm{tr}((\mathcal{D}\tilde{B})^2)$ requires $O(n^2)$ operations to compute). However, this eigenvalue decomposition needs to be performed for every value of θ, so while the inner integral in (46) can be computed relatively fast, the evaluation of the outer integral is still very slow.

For the computation of the cumulative distribution functions of $t(\hat{\beta})$ and R^2, we can integrate the densities of $t(\hat{\beta})$ and R^2 to obtain their cumulative distribution functions. However, this would require the evaluation of a quadruple integral and hence it is not practical. In the following proposition, we provide an alternative way of computing the cumulative distribution functions of $t(\hat{\beta})$ and R^2 that requires only the evaluation of triple integrals.

Proposition 2. *The cumulative distribution functions of $t(\hat{\beta})$ and R^2 are given by*

$$\mathbb{P}[t(\hat{\beta}) < c] = \frac{1}{2} - \int_0^{\infty} G\left(\frac{c\sqrt{r_2}}{\sqrt{T - 2 + c^2}}, r_2\right)\mathrm{d}r_2, \tag{50}$$

$$\mathbb{P}[R^2 < c] = \int_0^{\infty}[G(-\sqrt{r_2}c, r_2) - G(\sqrt{r_2}c, r_2)]\mathrm{d}r_2 \quad \textit{for } 0 < c \leq 1, \tag{51}$$

where

$$G(r_1, r_2) = \frac{1}{2\pi^2}\int_0^{\infty}\int_0^{\infty}\frac{\mathrm{Im}(g(s,t) + g(s,-t))}{s}\mathrm{d}s\mathrm{d}t, \tag{52}$$

with $g(s,t) = |\mathcal{D}|^{\frac{1}{2}}\mathrm{tr}(\mathcal{D}\tilde{B})$.

Similar to $f_{\mathcal{R}_1,\mathcal{R}_2}(r_1, r_2)$, we can reduce the computation time of $G(r_1, r_2)$ by using a polar transformation. Nevertheless, even with the polar transformation, the computation of the density and cumulative distribution functions of $t(\hat{\beta})$ and R^2 still requires an $O(n^3)$ eigenvalue decomposition for every value of θ and every (r_1, r_2), and it is prohibitively expensive even for moderately large values of T. In Section 4, we show that the integrands $h(s,t)$ and $g(s,t)$ can also be computed with $O(1)$ operations.

2.4. A Special Case

For the general case, our expressions for the density and cumulative distribution functions of $t(\hat{\beta})$ and R^2 involve triple integrals. In this subsection, we show that for the special case of $\rho_{uv} = 0$, we can simplify these expressions to double integrals. Throughout this subsection, we maintain the standardization (12), under which the population slope is the $\tilde{\beta}$ of (13). When $\rho_{uv} = 0$, we can use (29) to show that $r \sim \mathcal{N}(\tilde{\beta}z, I_T)$ conditional on z. It follows that conditional on z, we have

$$X \equiv \frac{z'Mr}{(z'Mz)^{\frac{1}{2}}} \sim \mathcal{N}\left((z'Mz)^{\frac{1}{2}}\tilde{\beta}, 1\right), \tag{53}$$

$$Y \equiv r'\left(M - \frac{Mzz'M}{z'Mz}\right)r \sim \chi^2_{T-2}, \tag{54}$$

and X and Y are independent of each other. This follows because $M - (Mzz'M)/(z'Mz)$ is an idempotent matrix that has a rank of $T-2$ and it is orthogonal to Mz. As a result, we can easily show that $Y \sim \chi^2_{T-2}$ and it is independent of X.

With X and Y defined, we can write $t(\hat{\beta})$ and R^2 in terms of these two random variables as

$$t(\hat{\beta}) = \frac{X}{\left(\frac{Y}{T-2}\right)^{\frac{1}{2}}}, \tag{55}$$

$$R^2 = \frac{X^2}{X^2 + Y}. \tag{56}$$

With this result, we can easily see that conditional on z, we have

$$t(\hat{\beta}) \sim t_{T-2}(\tilde{\beta}\sqrt{Q}), \tag{57}$$

$$R^2 \sim \text{Beta}_{\frac{1}{2},\frac{T-2}{2}}(\tilde{\beta}^2 Q), \tag{58}$$

where we denote $Q = z'Mz$, i.e., $t(\hat{\beta})$ follows a noncentral t-distribution with $T-2$ degrees of freedom and a noncentrality parameter of $\tilde{\beta}\sqrt{Q}$, and R^2 follows a noncentral beta distribution with parameters $1/2$ and $(T-2)/2$, and a noncentrality parameter of $\tilde{\beta}^2 Q$.

Note that when $\tilde{\beta} = 0$, $t(\hat{\beta})$ follows a central t-distribution with $T-2$ degrees of freedom, and R^2 follows a central beta distribution with parameters $1/2$ and $(T-2)/2$. This suggests that when $\rho_{uv} = 0$, the OLS t-ratio and R^2 have the standard distributions under the null hypothesis of $\tilde{\beta} = 0$, and this is true regardless of the value of ϕ.[3]

For the general case of $\tilde{\beta} \neq 0$, we let $f_\nu^\mu(x)$ and $F_\nu^\mu(x)$ stand for the density and cumulative distribution functions of a noncentral t-distribution with ν degrees of freedom and a noncentrality parameter of μ. In addition, we let $f_{a,b}^\delta(x)$ and $F_{a,b}^\delta(x)$ stand for the density and cumulative distribution functions of a noncentral beta distribution with parameters a and b and a noncentrality parameter of δ. We can then evaluate the unconditional density and distribution functions of $t(\hat{\beta})$ and R^2 as follows:

$$f_{t(\hat{\beta})}(c) = \int_0^\infty f_{T-2}^{\tilde{\beta}\sqrt{q}}(c) f_Q(q) \mathrm{d}q, \tag{59}$$

$$f_{R^2}(c) = \int_0^\infty f_{\frac{1}{2},\frac{T-2}{2}}^{\tilde{\beta}^2 q}(c) f_Q(q) \mathrm{d}q, \tag{60}$$

$$\mathbb{P}[t(\hat{\beta}) < c] = \int_0^\infty F_{T-2}^{\tilde{\beta}\sqrt{q}}(c) f_Q(q) \mathrm{d}q, \tag{61}$$

$$\mathbb{P}[R^2 < c] = \int_0^\infty F_{\frac{1}{2},\frac{T-2}{2}}^{\tilde{\beta}^2 q}(c) f_Q(q) \mathrm{d}q, \tag{62}$$

where $f_Q(q)$ is the density function of Q and it is given by

$$f_Q(q) = \frac{1}{\pi} \int_0^\infty \mathrm{Re}(e^{-\mathrm{i}tq} \phi_Q(t)) \mathrm{d}t, \tag{63}$$

and $\phi_Q(t)$ is the characteristic function of Q, which is defined as

$$\phi_Q(t) = |I_T - 2\mathrm{i}t H^{\frac{1}{2}} M H^{\frac{1}{2}}|^{-\frac{1}{2}}. \tag{64}$$

The above expressions suggest that when $\rho_{uv} = 0$, the density and cumulative distribution functions of $t(\hat{\beta})$ and R^2 can be expressed as double integrals. In Section 4, we show that

[3] In fact, as long as z and r are independent of each other and r follows a spherical distribution, then $t(\hat{\beta}) \sim t_{T-2}$ and $R^2 \sim \mathrm{Beta}(1/2, (T-2)/2)$, regardless of the distribution of z (see, for example, Theorem 5.1.1 of Muirhead (2005)).

the integrand of (63) can also be computed with $O(1)$ operations, so this double integral is extremely cheap to evaluate.

For the case of $\phi = 0$, $Q \sim \chi^2_{T-1}$ and we can even obtain explicit expressions for the density functions of $t(\hat{\beta})$ and R^2. They are given by[4]

$$f_{t(\hat{\beta})}(c) = \frac{f_{T-2}(c)}{(1+\tilde{\beta}^2)^{\frac{T-1}{2}}} \left[{}_2F_1\left(\frac{T-1}{2}, \frac{T-1}{2}; \frac{1}{2}; \vartheta^2\right) + \frac{2\vartheta\Gamma\left(\frac{T}{2}\right)^2}{\Gamma\left(\frac{T-1}{2}\right)^2} {}_2F_1\left(\frac{T}{2}, \frac{T}{2}; \frac{3}{2}; \vartheta^2\right) \right], \quad (65)$$

$$f_{R^2}(c) = \frac{f_{\frac{1}{2},\frac{T-2}{2}}(c)}{(1+\tilde{\beta}^2)^{\frac{T-1}{2}}} {}_2F_1\left(\frac{T-1}{2}, \frac{T-1}{2}; \frac{1}{2}; \frac{\tilde{\beta}^2 c}{1+\tilde{\beta}^2}\right), \quad (66)$$

where $f_{T-2}(c)$ is the density of a t-distribution with $T-2$ degrees of freedom, $f_{\frac{1}{2},\frac{T-2}{2}}(c)$ is the density of a beta distribution with parameters $1/2$ and $(T-2)/2$, ${}_2F_1(\cdot,\cdot;\cdot;\cdot)$ is the Gaussian hypergeometric function, and

$$\vartheta = \frac{\tilde{\beta} c}{(1+\tilde{\beta}^2)^{\frac{1}{2}}(T-2+c^2)^{\frac{1}{2}}}. \quad (67)$$

3. Fast Computation of the Exact Distribution of $\hat{\beta}$

In this section, we show that the integrands in (20) and (22) can be evaluated in closed form, with a number of arithmetic operations that does not depend on T. The development proceeds in three steps. First, we reduce the $2T \times 2T$ determinant that defines the characteristic function to the determinant of a $T \times T$ matrix which, apart from its first and last rows and columns and a rank-one correction, is a tridiagonal Toeplitz matrix (Section 3.1). Second, we collect closed-form expressions for the determinant of a tridiagonal Toeplitz matrix and for the small number of bilinear forms of its inverse that the reduction requires (Section 3.2). Third, we address the numerical issues that arise when these closed-form expressions are implemented in floating point arithmetic (Section 3.3).

[4]These follow by integrating (59) and its R^2 analog against the χ^2_{T-1} density of Q, using the Poisson-mixture representations of the noncentral distributions and summing the resulting series term by term. The same route expresses the cumulative distribution functions of $t(\hat{\beta})$ and R^2 as infinite series of incomplete beta functions.

3.1. Reduction to a Structured Determinant

Let $U = x'Ax$ and $W = x'Bx$, where A, B, and the distribution of x are given in (18) and (19). The joint characteristic function of (U, W) is

$$\varphi_{U,W}(s,t) = |I_{2T} - 2\mathrm{i}sV_x^{\frac{1}{2}}AV_x^{\frac{1}{2}} - 2\mathrm{i}tV_x^{\frac{1}{2}}BV_x^{\frac{1}{2}}|^{-\frac{1}{2}} = |V_x|^{-\frac{1}{2}}\,|V_x^{-1} - 2\mathrm{i}sA - 2\mathrm{i}tB|^{-\frac{1}{2}}, \tag{68}$$

and the integrands in (20) and (22) are obtained from $\varphi_{U,W}$ and its partial derivative with respect to t, both evaluated at $t = -cs$:

$$\mathbb{P}[b < c] = \frac{1}{2} - \frac{1}{\pi}\int_0^\infty \frac{\mathrm{Im}(\varphi_{U,W}(s,-cs))}{s}\mathrm{d}s, \tag{69}$$

$$f_b(c) = \frac{1}{\pi}\int_0^\infty \mathrm{Im}\left(\frac{\partial\varphi_{U,W}(s,t)}{\partial t}\bigg|_{t=-cs}\right)\mathrm{d}s. \tag{70}$$

The reason that (68) is useful is that V_x^{-1} is an (almost) banded matrix even though V_x is dense. Using the well-known inverse of the AR(1) covariance matrix,

$$H^{-1} = \begin{bmatrix} 1 & -\phi & 0 & \cdots & 0 \\ -\phi & 1+\phi^2 & -\phi & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & -\phi & 1+\phi^2 & -\phi \\ 0 & \cdots & 0 & -\phi & 1 \end{bmatrix}, \tag{71}$$

together with the identities

$$GH^{-1} = \begin{bmatrix} -\phi & 1 & 0 & \cdots & 0 \\ 0 & -\phi & 1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -\phi & 1 \\ 0 & \cdots & 0 & 0 & 0 \end{bmatrix}, \qquad GH^{-1}G' = \begin{bmatrix} I_{T-1} & 0_{T-1} \\ 0'_{T-1} & 0 \end{bmatrix}, \tag{72}$$

it is straightforward to verify that

$$V_x^{-1} = \begin{bmatrix} V^{11} & V^{12} \\ V^{21} & V^{22} \end{bmatrix}, \tag{73}$$

where

$$V^{11} = \begin{bmatrix} \frac{1}{1-\rho_{uv}^2} I_{T-1} & 0_{T-1} \\ 0'_{T-1} & 1 \end{bmatrix}, \tag{74}$$

$$V^{12} = -\rho_{uv} V^{11} G H^{-1}, \tag{75}$$

$$V^{22} = H^{-1} + \rho_{uv}^2 H^{-1} G' V^{11} G H^{-1}, \tag{76}$$

so that V^{11} is diagonal, V^{12} is bidiagonal, and V^{22} is tridiagonal. In addition,

$$|V_x| = \frac{|H|}{|V^{11}|} = \frac{(1-\rho_{uv}^2)^{T-1}}{1-\phi^2}. \tag{77}$$

Taking the Schur complement of the $(1,1)$ block of $V_x^{-1} - 2\mathrm{i}sA - 2\mathrm{i}tB$ and simplifying, we obtain

$$\varphi_{U,W}(s,t) = (1-\phi^2)^{\frac{1}{2}} \, |C(s,t)|^{-\frac{1}{2}}, \tag{78}$$

where $C(s,t)$ is the $T \times T$ matrix

$$C(s,t) = H^{-1} - 2\mathrm{i}tM + s^2 M (V^{11})^{-1} M - \mathrm{i}s\rho_{uv}(MGH^{-1} + H^{-1}G'M), \tag{79}$$

where the ρ_{uv}^2 term in V^{22} has been canceled by the corresponding term in the Schur correction. The matrix $C(s,t)$ inherits the structure of its ingredients: it is tridiagonal, except that (i) the presence of $M = I_T - 1_T 1'_T/T$ contributes a rank-one term proportional to $1_T 1'_T$, and (ii) its first and last rows and columns contain boundary corrections. Partitioning $C(s,t)$ into its first row/column, its last row/column, and its interior $(T-2) \times (T-2)$ block, we can write

$$C(s,t) = \begin{bmatrix} c_{1,1} & a' & c_{1,T} \\ a & \Delta & b \\ c_{T,1} & b' & c_{T,T} \end{bmatrix}, \tag{80}$$

where

$$c_{1,1} = 1 + \frac{1 + [T(T-1)-1](1-\rho_{uv}^2)}{T^2} s^2 + 2\mathrm{i}(\rho_{uv}\phi s - t)\left(1 - \frac{1}{T}\right), \tag{81}$$

$$c_{1,T} = c_{T,1} = -\frac{T - \rho_{uv}^2}{T^2} s^2 + \mathrm{i}\,\frac{2t + (1-\phi)\rho_{uv} s}{T}, \tag{82}$$

$$c_{T,T} = 1 + \frac{(T-1)(T-\rho_{uv}^2)}{T^2} s^2 + 2\mathrm{i}\left[\frac{\rho_{uv} s}{T} - \left(1 - \frac{1}{T}\right) t\right], \tag{83}$$

$$a = \alpha_1 1_{T-2} + q\, e_{1,T-2}, \tag{84}$$

$$b = \beta_1 1_{T-2} + q\, e_{T-2,T-2}, \tag{85}$$

$$\Delta = D - \kappa\, 1_{T-2} 1'_{T-2}, \tag{86}$$

with

$$\alpha_1 = \frac{\rho_{uv}^2 - T(1-\rho_{uv}^2)}{T^2} s^2 + \mathrm{i}\, \frac{(1-2\phi)\rho_{uv} s + 2t}{T}, \tag{87}$$

$$\beta_1 = -\frac{T - \rho_{uv}^2}{T^2} s^2 + \mathrm{i}\, \frac{(2-\phi)\rho_{uv} s + 2t}{T}, \tag{88}$$

$$\kappa = \frac{T(1-\rho_{uv}^2) - \rho_{uv}^2}{T^2} s^2 - \frac{2\mathrm{i}}{T}[(1-\phi)\rho_{uv} s + t], \tag{89}$$

and D is the $(T-2) \times (T-2)$ symmetric tridiagonal Toeplitz matrix with

$$D_{i,i} = p \equiv 1 + \phi^2 + (1-\rho_{uv}^2)s^2 + 2\mathrm{i}\phi\rho_{uv} s - 2\mathrm{i}t, \qquad D_{i,i\pm1} = q \equiv -\phi - \mathrm{i}\rho_{uv} s. \tag{90}$$

Recall that $e_{i,n}$ denotes the i-th column of I_n, so that $e_{1,T-2} = [1,\ 0'_{T-3}]'$ and $e_{T-2,T-2} = [0'_{T-3},\ 1]'$ in (84) and (85). Note that the coefficient of $e_{1,T-2}$ in a and of $e_{T-2,T-2}$ in b is exactly the off-diagonal element q of D.

The value of the partition (80) is that the determinant and the required elements of the inverse of a bordered matrix can be expressed in terms of its interior block.

Lemma 1. *Let $\mathcal{A}$ be an $n \times n$ symmetric matrix partitioned as*

$$\mathcal{A} = \begin{bmatrix} a_{11} & b' & a_{1n} \\ b & \mathcal{A}_{22} & c \\ a_{1n} & c' & a_{nn} \end{bmatrix}, \tag{91}$$

where $\mathcal{A}_{22}$ is the interior $(n-2) \times (n-2)$ block. Assume $\mathcal{A}$ and $\mathcal{A}_{22}$ are nonsingular, and that δ_1 defined below is nonzero. Define

$$\delta_1 = a_{11} - b'\mathcal{A}_{22}^{-1} b, \quad \delta_2 = a_{1n} - b'\mathcal{A}_{22}^{-1} c, \quad \delta_3 = a_{nn} - c'\mathcal{A}_{22}^{-1} c, \quad \delta_4 = \delta_1\delta_3 - \delta_2^2. \tag{92}$$

Then

$$|\mathcal{A}| = |\mathcal{A}_{22}|\, \delta_4, \tag{93}$$

and, partitioning $\mathcal{A}^{-1}$ *conformably with* $\mathcal{A}$*,*

$$\mathcal{A}^{11} = \frac{\delta_3}{\delta_4}, \qquad \mathcal{A}^{31} = -\frac{\delta_2}{\delta_4}, \qquad \mathcal{A}^{33} = \frac{\delta_1}{\delta_4}, \tag{94}$$

$$\mathcal{A}^{21} = \frac{1}{\delta_4}\mathcal{A}_{22}^{-1}(\delta_2 c - \delta_3 b), \qquad \mathcal{A}^{23} = \frac{1}{\delta_4}\mathcal{A}_{22}^{-1}(\delta_2 b - \delta_1 c), \tag{95}$$

$$\mathcal{A}^{22} = \mathcal{A}_{22}^{-1} + \frac{1}{\delta_1}\mathcal{A}_{22}^{-1} b b' \mathcal{A}_{22}^{-1} + \frac{\delta_1}{\delta_4}\mathcal{A}_{22}^{-1}\left(c - \frac{\delta_2}{\delta_1} b\right)\left(c - \frac{\delta_2}{\delta_1} b\right)' \mathcal{A}_{22}^{-1}. \tag{96}$$

Applying Lemma 1 to $C(s,t)$, and using the Sherman–Morrison formula on $\Delta = D - \kappa 1_{T-2} 1'_{T-2}$,

$$|\Delta| = |D|(1 - \kappa\, 1'_{T-2} D^{-1} 1_{T-2}), \tag{97}$$

$$\Delta^{-1} = D^{-1} + \frac{\kappa}{1 - \kappa\, 1'_{T-2} D^{-1} 1_{T-2}} D^{-1} 1_{T-2} 1'_{T-2} D^{-1}, \tag{98}$$

we find that $|C(s,t)|$ is completely determined by $|D|$ and by bilinear forms of D^{-1} in the three vectors 1_{T-2}, $e_{1,T-2}$, and $e_{T-2,T-2}$. Specifically, letting $m = T-2$ and defining the four primitives

$$d_1 = 1'_m D^{-1} 1_m, \qquad d_2 = e'_{1,m} D^{-1} 1_m, \qquad d_3 = e'_{1,m} D^{-1} e_{1,m}, \qquad d_4 = e'_{1,m} D^{-1} e_{m,m}, \tag{99}$$

the persymmetry of D implies $e'_{m,m} D^{-1} 1_m = d_2$ and $e'_{m,m} D^{-1} e_{m,m} = d_3$, so that every bilinear form of Δ^{-1} in a and b—and hence $|C(s,t)|$ itself—is an explicit scalar expression in d_1, d_2, d_3, d_4, $|D|$, and the boundary coefficients of (80). For instance, with $w_1 = 1'_m \Delta^{-1} 1_m$, $w_2 = e'_{1,m}\Delta^{-1} 1_m$, $w_3 = e'_{1,m}\Delta^{-1} e_{1,m}$, and $w_4 = e'_{1,m}\Delta^{-1} e_{m,m}$, the Sherman–Morrison formula gives

$$w_1 = \frac{d_1}{1 - \kappa d_1}, \qquad w_2 = \frac{d_2}{1 - \kappa d_1}, \qquad w_3 = d_3 + \frac{\kappa d_2^2}{1 - \kappa d_1}, \qquad w_4 = d_4 + \frac{\kappa d_2^2}{1 - \kappa d_1}, \tag{100}$$

and, using (84)–(85),

$$a'\Delta^{-1} a = \alpha_1^2 w_1 + 2\alpha_1 q w_2 + q^2 w_3, \tag{101}$$

$$a'\Delta^{-1} b = \alpha_1 \beta_1 w_1 + (\alpha_1 + \beta_1) q w_2 + q^2 w_4, \tag{102}$$

$$b'\Delta^{-1} b = \beta_1^2 w_1 + 2\beta_1 q w_2 + q^2 w_3. \tag{103}$$

It is worth putting the pieces together at this point, since the resulting expression is the whole of the $O(1)$ claim for the distribution of b. Lemma 1 applied to (80) gives $|C(s,t)| = |\Delta|\,\delta_4$, and (97) gives $|\Delta| = |D|(1-\kappa d_1)$, so

$$|C(s,t)| \;=\; \underbrace{|D|}_{\text{tridiagonal core}} \cdot \underbrace{(1-\kappa d_1)}_{\text{demeaning}} \cdot \underbrace{\left(\delta_1\delta_3 - \delta_2^2\right)}_{\text{border}}, \tag{104}$$

where, by (101)–(103),

$$\delta_1 = c_{1,1} - a'\Delta^{-1}a = c_{1,1} - \alpha_1^2 w_1 - 2\alpha_1 q w_2 - q^2 w_3, \tag{105}$$

$$\delta_2 = c_{1,T} - a'\Delta^{-1}b = c_{1,T} - \alpha_1\beta_1 w_1 - (\alpha_1+\beta_1) q w_2 - q^2 w_4, \tag{106}$$

$$\delta_3 = c_{T,T} - b'\Delta^{-1}b = c_{T,T} - \beta_1^2 w_1 - 2\beta_1 q w_2 - q^2 w_3, \tag{107}$$

and w_1–w_4 are the four expressions (100) in d_1–d_4 and κ. Reading (104)–(107) from the right, every quantity in it is a scalar: $c_{1,1}$, $c_{1,T}$, $c_{T,T}$, α_1, β_1, κ, p and q are given in closed form in (80)–(90), and d_1–d_4 and $|D|$ are the five quantities that Section 3.2 supplies in closed form from p, q and m. So $|C(s,t)|$—a determinant of a $T \times T$ matrix that changes with the quadrature node—is a fixed sequence of arithmetic operations on a fixed list of scalars, and the number of operations does not depend on T. Everything that remains in this section is about evaluating those five quantities accurately.

3.2. Closed-Form Expressions for the Primitives

The four primitives in (99), together with $|D|$, admit closed-form expressions. The following lemma collects the ones we need.

Lemma 2. *Let D be an $m \times m$ symmetric tridiagonal Toeplitz matrix with diagonal element p and off-diagonal element $q \neq 0$, where p and q may be complex, and suppose $p \neq \pm 2q$ and D is nonsingular. Define*

$$h = (p^2 - 4q^2)^{\frac{1}{2}}, \qquad r_\pm = \frac{p \pm h}{2}, \tag{108}$$

so that $r_+ + r_- = p$ and $r_+ r_- = q^2$. Then

$$|D| = \frac{r_+^{m+1} - r_-^{m+1}}{h}, \tag{109}$$

$$d_3 = \frac{r_+^m - r_-^m}{r_+^{m+1} - r_-^{m+1}}, \tag{110}$$

$$d_4 = \frac{(-q)^{m-1} h}{r_+^{m+1} - r_-^{m+1}}, \tag{111}$$

$$d_2 = \frac{1 + q(d_3 + d_4)}{p + 2q}, \tag{112}$$

$$d_1 = \frac{m + 2q d_2}{p + 2q}. \tag{113}$$

Equations (109)–(111) follow from the classical theory of tridiagonal Toeplitz matrices. Write $|D_k|$ for the determinant of the leading $k \times k$ submatrix of D. Expanding along its last row, whose only nonzero entries are q and p in the final two positions, and expanding the resulting minor along its last column, gives the second-order recursion

$$|D_{k+1}| = p\,|D_k| - q^2 |D_{k-1}|, \qquad |D_0| = 1, \quad |D_1| = p, \tag{114}$$

whose characteristic roots are $r_\pm$; and the elements of D^{-1} are ratios of cofactors that telescope into the stated forms. Equations (112) and (113) follow from the row-sum identity $D 1_m = (p+2q) 1_m - q(e_{1,m} + e_{m,m})$: multiplying through by $e'_{1,m} D^{-1}$ and by $1'_m D^{-1}$, respectively, and solving. A proof is given in the Appendix.

For the density integrand (70) we additionally need $\mathrm{tr}(C^{-1} M)$, which by Lemma 1 requires $\mathrm{tr}(\Delta^{-1})$ and bilinear forms of Δ^{-2}; through (98), these in turn require $\mathrm{tr}(D^{-1})$ and the second-order primitives $1'_m D^{-2} 1_m$, $e'_{1,m} D^{-2} 1_m$, $e'_{1,m} D^{-2} e_{1,m}$, and $e'_{1,m} D^{-2} e_{m,m}$. These are obtained without any new machinery by observing that

$$D^{-2} = -\frac{\partial D^{-1}}{\partial p}, \tag{115}$$

so that each second-order primitive is (minus) the derivative with respect to p of the corresponding closed form in Lemma 2. Carrying out the differentiation yields

$$\mathrm{tr}(D^{-1}) = \frac{m(r_+^{m+1} + r_-^{m+1})}{h(r_+^{m+1} - r_-^{m+1})} - \frac{2q^2 d_3}{h^2}, \tag{116}$$

$$e'_{1,m} D^{-2} e_{1,m} = \frac{d_3(r_+^{m+1} + r_-^{m+1})}{h(r_+^{m+1} - r_-^{m+1})} - \frac{2mq^{2m}}{(r_+^{m+1} - r_-^{m+1})^2}, \tag{117}$$

$$e'_{1,m} D^{-2} e_{m,m} = -(-q)^{m-1} \frac{\frac{2q^2}{h}(r_+^m - r_-^m) - m(r_+^{m+1} + r_-^{m+1})}{(r_+^{m+1} - r_-^{m+1})^2}, \tag{118}$$

$$e'_{1,m}D^{-2}1_m = \frac{d_2 + q(e'_{1,m}D^{-2}e_{1,m} + e'_{1,m}D^{-2}e_{m,m})}{p+2q}, \tag{119}$$

$$1'_m D^{-2}1_m = \frac{d_1 + 2q\, e'_{1,m}D^{-2}1_m}{p+2q}. \tag{120}$$

The same device delivers third-order primitives (D^{-3}) when they are needed in Section 4.

The central point of this subsection is a counting exercise: the evaluation of $|C(s,t)|$, and hence of the characteristic function $\varphi_{U,W}(s,t)$ in (78), requires a fixed number of scalar operations—roughly two dozen complex multiplications and a handful of complex powers—regardless of the sample size T. The same is true of $\mathrm{tr}(C^{-1}M)$. We record this formally.

Proposition 3. *For any (s,t), the characteristic function $\varphi_{U,W}(s,t)$ in (78) and its partial derivative $\partial\varphi_{U,W}(s,t)/\partial t$ can be evaluated with $O(1)$ arithmetic operations, independent of T. Consequently, the cumulative distribution function and the density function of $b = (\sigma_v/\sigma_u)(\hat{\beta} - \beta)$ in (69)–(70) can each be computed as a univariate numerical integral whose total cost is proportional to the number of quadrature points and independent of T.*

By contrast, the standard eigenvalue-based approach costs $O(T^3)$ for the one-time decomposition plus $O(T)$ per quadrature point, and direct determinant evaluation costs $O(T^3)$ per quadrature point. The practical difference is dramatic: for $T = 1{,}000$, the setup cost of the eigenvalue method alone exceeds the entire cost of our algorithm by several orders of magnitude.

3.3. Numerical Issues

The closed-form expressions of Lemma 2 are exact in exact arithmetic, but three distinct numerical issues must be addressed before they can be used in floating point arithmetic. None of the issues is hypothetical: each one produces large errors in empirically relevant regions of the parameter space if left untreated.

3.3.1. Removable Cancellation when $p + 2q \approx 0$

Two of the primitives, d_2 in (112) and d_1 in (113), carry the factor $p + 2q$ in their denominators, so both lose accuracy in floating point arithmetic when this factor is small. From (90),

$$p + 2q = (1-\phi)^2 + (1-\rho_{uv}^2)s^2 - 2\mathrm{i}[(1-\phi)\rho_{uv}s + t], \tag{121}$$

which is small precisely when ϕ is close to one and s and t are close to zero—that is, in the empirically central case of a highly persistent predictor, at the origin of the inversion integral where the integrand matters most. When $p+2q \approx 0$ we also have $r_+ \approx r_-$ (the two characteristic roots coalesce), so (109)–(111) suffer from cancellation as well.

The loss is not marginal. Take $T = 120$ and evaluate d_1 at the left endpoint $s = t = 0$ of the integration path, where $p = 1+\phi^2$ and $q = -\phi$, so that $p+2q = (1-\phi)^2$ and the whole calculation is governed by ϕ alone. Table 1 does this for seven values of ϕ and reports what (113) returns in double precision against the exact value. The quantity being computed is perfectly ordinary throughout, moving smoothly from 16593 to 140420. What (113) returns is not: at $\phi = 0.999$ it parts company with the truth in the ninth digit, at $\phi = 0.99999$ it is wrong by a factor of twenty-two, and at $\phi = 0.999999$ it is wrong by a factor of 2.5 million. What fails is the evaluation of (113) in double precision, not the function it defines. The same happens along the rest of the path whenever s and t are small, and the smaller $1-\phi$ is relative to $1/T$, the wider the stretch of the path on which it happens.

The last column of the table is the remedy this subsection constructs: the same d_1, at the same seven configurations, evaluated in double precision by the expansions (125) instead of by (113). What follows derives those expansions, which are accurate throughout the region where (113) fails, including at $s = t = 0$ with ϕ arbitrarily close to one.

The starting point is to name the function that all five quantities are built from. Substitute $p = 2qx$ in the determinant recursion (114) and divide through by q^{k+1}: the scaled determinants $c_k = |D_k|/q^k$ satisfy $c_{k+1} = 2x\,c_k - c_{k-1}$ with $c_0 = 1$ and $c_1 = 2x$, which is the recursion of the Chebyshev polynomials of the second kind together with their initial conditions. Hence

$$|D| = q^m\,U_m(x), \qquad U_m(\cos\theta) = \frac{\sin((m+1)\theta)}{\sin\theta}, \qquad x = \frac{p}{2q}, \tag{122}$$

and the primitives of Lemma 2 collapse to

$$d_3 = \frac{U_{m-1}(x)}{q\,U_m(x)}, \qquad d_4 = \frac{(-1)^{m+1}}{q\,U_m(x)}, \qquad d_2 = \frac{Q(x)}{2q\,U_m(x)}, \qquad d_1 = \frac{S(x)}{2q\,U_m(x)}, \tag{123}$$

where Q and S are the polynomials defined by

$$U_m(x) + U_{m-1}(x) + (-1)^{m+1} = (x+1)\,Q(x), \qquad m\,U_m(x) + Q(x) = (x+1)\,S(x). \tag{124}$$

Table 1: **Cancellation in the Closed Forms, and its Removal**

The table evaluates $d_1 = 1'_m D^{-1} 1_m$ for $T = 120$ ($m = 118$) at the left endpoint $s = t = 0$ of the integration path, where $p = 1 + \phi^2$ and $q = -\phi$. The exact column is (109)–(113) carried in 100-digit arithmetic; using the same expressions isolates the effect of double precision. The last two columns are the relative errors of the two double precision evaluations: the closed forms in the roots, (109)–(113), and the Chebyshev expansions (125). The algorithm switches between them at $|\tau| = 5/2$, between the fifth and sixth rows. The closed forms are unusable below $|\tau| \approx 10^{-2}$ and accurate when $|\tau| \geq 1$; the expansions are accurate throughout. The two branches therefore overlap over more than two orders of magnitude in $|\tau|$, which is why the placement of the threshold is not delicate.

			relative error in double precision	
ϕ	$\lvert\tau\rvert$	d_1 (exact)	roots (109)–(113)	expansions (125)
0.999999	7.1×10^{-9}	140420.140221	2.5×10^{6}	1.8×10^{-15}
0.99999	7.1×10^{-7}	140421.384327	2.1×10^{1}	1.9×10^{-15}
0.9999	7.1×10^{-5}	140432.054407	1.1×10^{-4}	1.8×10^{-15}
0.999	7.1×10^{-3}	140361.584859	6.2×10^{-9}	1.8×10^{-15}
0.99	7.2×10^{-1}	124113.601736	4.9×10^{-14}	1.9×10^{-15}
0.9812	2.6	94951.037928	1.3×10^{-14}	2.9×10^{-15}
0.925	4.3×10^{1}	16593.445894	1.1×10^{-15}	8.7×10^{-16}

Both identities are instances of the factor theorem: the left sides vanish at $x = -1$, since $U_m(-1) = (-1)^m(m+1)$ and $Q(-1) = (-1)^{m+1}m(m+1)$, so each is divisible by $x+1$ as a polynomial. (The first two forms in (123) restate (110)–(111); the last two are (112)–(113) with the factor $p + 2q = 2q(x+1)$ canceled once against each numerator.)

This is the whole of the cure. The confluent point is $x = -1$, and nothing there is singular: U_m is entire, with $U_m(-1) = (-1)^m(m+1)$. The factor $x+1$ that (112) and (113) carry in their denominators has been divided out in (124), once and on paper, so no cancellation survives into (123). The second- and third-order primitives inherit the same structure, being p-derivatives of the first-order ones ($D^{-2} = -\partial D^{-1}/\partial p$, $2D^{-3} = \partial^2 D^{-1}/\partial p^2$), so every quantity the algorithm needs is a rational function of U_m, U_{m-1} and their first two derivatives.

It remains to evaluate those two Chebyshev values near $x = -1$, and for that we use their Taylor expansions there. Writing $U_m(x) = \sum_{k=0}^{\infty} a_k\,\delta^k$ in the coordinate $\delta = (p+2q)/(2q) = x+1$, the Chebyshev differential equation $(1-x^2)U_m'' - 3xU_m' + m(m+2)U_m = 0$ yields

$$a_0 = (-1)^m(m+1), \qquad \frac{a_{k+1}}{a_k} = -\frac{(m-k)(m+k+2)}{(2k+3)(k+1)}, \tag{125}$$

with the same recursion at order $m-1$ for U_{m-1}; the coefficients of Q and S are index shifts of these by (124), and the derivative series are read off the same arrays. The implementation carries the coefficients against powers of $\tau = n^2\delta$, $n = m+1$, rather than of δ: none of them then overflows at any m, and it is in τ that the series converges uniformly in the sample size, successive terms falling by a factor of at most $|\tau|/((2k+3)(k+1))$, so that twenty terms reach machine precision on $|\tau| < 5/2$ whatever m is. Like δ, the variable τ is complex along the integration path; at the left endpoint $s = t = 0$, where p and q are real, it is real and negative, $\delta = -(1-\phi)^2/(2\phi)$. The thresholds below are therefore stated in $|\tau|$. Expanding the Chebyshev values rather than the primitives is what buys that bound: the primitives are rational, with poles at the zeros of U_m, so their own series would converge only as far as the nearest one, $\tau^* = 2n^2\sin^2(\pi/(2n)) \to \pi^2/2$, whereas the Chebyshev values are entire and the division in (123) reproduces the poles exactly.

The evaluation is therefore two-regime, the regimes differing only in how the two Chebyshev values are obtained: for $|\tau| \geq 5/2$, the closed forms of Lemma 2 in the roots, computed after dividing the tridiagonal Toeplitz matrix D by $|r_+|$, which divides both roots by $|r_+|$ and

so leaves every power bounded by one in modulus (overflow-free for any T); for $|\tau| < 5/2$, the expansions (125). Both are strictly $O(1)$ per point, and the threshold is not delicate: Table 1 shows the closed forms back to full accuracy by $|\tau| \approx 1$, below it, and the expansions still at machine precision at $|\tau| = 43$, seventeen times above it. That table is one primitive at one sample size; for the toolkit as a whole, every primitive at every order is accurate to 5×10^{-12} or better within the band at every sample size we have tried, out to $m = 9600$, with $p + 2q$ supplied in closed form as discussed below. Along the integration path of Figure 1, at $T = 600$, the evaluation tracks a 100-digit reference at near machine precision. Since $|\tau|$ grows with s along the path, the confluent branch occupies an interval at the origin and nothing more; it is entered at all only when $n(1-\phi) < \sqrt{5\phi}$, and at $T = 120$ and $\phi = 0.99$ it covers $s \leq 0.0011$, about two parts in a thousand of the range on which the integrand is above 10^{-10} of its peak. The expansions are there for the corner that needs them, not for the working range.

What does eventually limit the primitives at large n is the conditioning of the argument rather than anything in the evaluation. On the band $p \approx -2q$, so the numerator of $\delta = (p+2q)/(2q)$ is a difference of nearly equal quantities and δ is delivered with absolute error ε however it is subsequently used; $\tau = n^2\delta$ is then known only to εn^2, and since the primitives vary with $|\tau|$ at an $O(1)$ relative rate, each inherits a relative error of about $\varepsilon n^2/5$—4×10^{-9} at $m = 9600$, and 6×10^{-10} at the largest sample size of Table 2. This is a property of the argument and not of any algorithm: given the same p and q, the closed forms in the roots, the expansions (125), and the pole-corrected expansions of the previous literature all agree to precisely that accuracy. It is also avoidable, and our programs avoid it. Every caller forms $p + 2q$ from a closed form in which nothing cancels—(121) for the distribution of b, and its analog for the joint problem of Section 4—in which $1 - \phi$ is itself exact in floating point, and hands it to the evaluation rather than letting it be recovered by subtraction. Two further cancellations go with it: the primitives are then taken to be those of the matrix with diagonal $(p+2q) - 2q$, so that all five belong to one and the same matrix, and $h^2 = p^2 - 4q^2$ is formed as $[(p + 2q) - 4q](p + 2q)$. At $T = 3840$ and $\phi = 1 - 10^{-7}$ the three together improve the primitives from 5×10^{-10} to 6×10^{-13} on the band, and from 2×10^{-11} to

3×10^{-13} above it. (The evaluation of $|D|^{-\frac{1}{2}}$ raises a separate question—the choice of branch of the complex square root—which we take up, for both regimes at once, in Section 3.3.2.) Figure 1 illustrates the two points of this subsection on the primitive d_1: the closed form of Lemma 2, evaluated in double precision, is accurate for large $|\tau|$ but fails catastrophically as $\tau \to 0$—by $|\tau| = 4 \times 10^{-9}$ it overstates $|d_1|$ by more than eight orders of magnitude—whereas the two-regime evaluation tracks the exact value at near machine precision uniformly in $|\tau|$.

We have carried out the same comparison for d_2, d_3, d_4, and $\log|D|$, and the results reveal a clear hierarchy that is worth recording, because it explains which quantities force the two-regime treatment. The severity of the double-precision failure is governed by the number of divisions by $p+2q$ in the closed forms of Lemma 2. The formulas for d_3 and d_4 involve none: their only degradation comes from the cancellation in $r_+^k - r_-^k$ as the roots coalesce, which loses digits only like $\varepsilon/\sqrt{|\tau|}$, and on the path of Figure 1 their worst double-precision relative errors are 1.3×10^{-12} and 2.6×10^{-10}; the same is true of $\log|D|$ (worst absolute error 2.1×10^{-10}), whose distinct failure mode—overflow of the power $r_\pm^{m+1}$ at larger T—is removed by never forming the power: $\log|D|$ is evaluated from the factorization $|D| = r_+^{m+1}(1-\zeta^{m+1})/h$ with $\zeta = r_-/r_+$, in which $|\zeta| \le 1$ and r_+^{m+1} enters only through $(m+1)\log r_+$ (this factorization is also what carries the phase; see Section 3.3.2). The formula for d_2 involves one such division, and its worst relative error jumps to 2.1×10^{-1}; that for d_1 involves two, and fails by the eight orders of magnitude shown in the figure. Having no such division, d_3 and d_4 barely feel the coalescence, and $|D|$ feels it only logarithmically. By contrast, the two-regime evaluation is accurate to 2.2×10^{-11} or better for all five quantities. The robustness of d_3 and d_4 does not survive differentiation, however. The second- and third-order primitives required for the densities and for the joint problem of Section 4 compound further divisions, roundoff being amplified roughly like $\varepsilon/|\tau|^j$ at the j-th order: at $|\tau| = 7 \times 10^{-7}$, where the closed form gives d_3 and d_4 to 10^{-14} and 4×10^{-13}, its third-order counterparts are wrong by factors of 0.4 and 4. The confluent evaluation is therefore used for all four primitives at all three orders, and we display only the worst case, d_1.

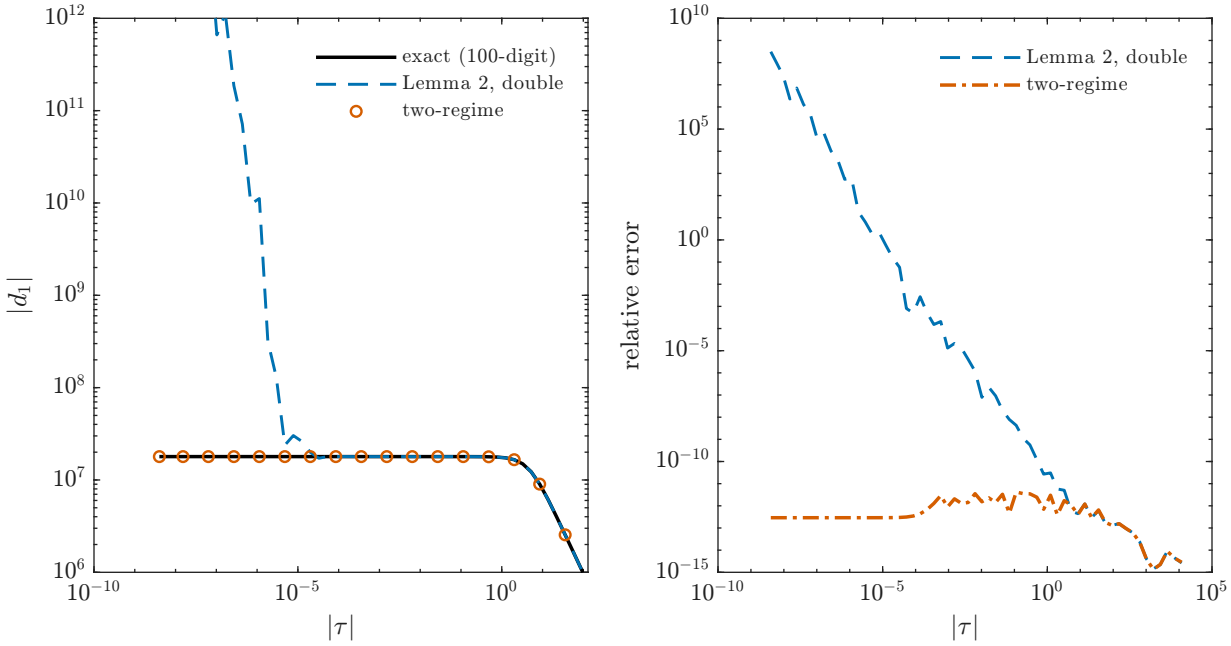


Figure 1: **Two-Regime Evaluation of** d_1

The figure plots $d_1 = 1'_m D^{-1} 1_m$ against $|\tau|$ along the integration path of the cumulative distribution function of b, for $T = 600$, $\phi = 1 - 10^{-7}$, $\rho_{uv} = -0.85$, and $c = 0.1$ (as s ranges over $[10^{-13}, 10^{-0.5}]$, $|\tau|$ ranges over $[4 \times 10^{-9}, 1.2 \times 10^{4}]$). Both d_1 and τ are complex along the path, so the figure is drawn in moduli. The left panel plots $|d_1|$ computed exactly (solid line), by the closed form of Lemma 2 in double precision (dashed line), and by the two-regime algorithm (circles). The right panel plots the relative errors of the latter two against the exact values. As in Table 1, the exact values are (109)–(113) themselves carried in 100-digit arithmetic, so that the dashed line differs from the solid one by the effect of double precision.

3.3.2. Continuity of the Complex Square Root

The inversion formulas (69)–(70) require $|C(s,-cs)|^{-1/2}$ as a *continuous* function of s. The principal square root has a branch cut along the negative real axis, so if $|C|$ is computed first and its square root taken afterward, the result jumps by a sign whenever the path $s \mapsto |C(s,-cs)|$ crosses the negative real axis—and it does so many times, since the phase of $|C|$ grows linearly with T. The eigenvalue-based algorithms of Imhof (1961) and Broda and Paolella (2009) avoid this problem automatically because they apply the square root factor by factor, $\prod_{j=1}^{2T}(1-2it\lambda_j)^{-1/2}$, and each factor stays in the right half plane. Our closed-form evaluation must accomplish the same thing analytically.

The solution is to track the modulus and the phase of $|C|$ separately. The three-factor assembly (104), with $|D|$ written out in the roots of Lemma 2, is

$$|C| = \underbrace{\frac{r_+^{m+1} - r_-^{m+1}}{h}}_{|D|} \cdot \underbrace{(1-\kappa d_1)}_{\text{rank-one}} \cdot \underbrace{\left[\delta_1\delta_3 - \delta_2^2\right]}_{\text{border}}, \tag{126}$$

with δ_1, δ_2, δ_3 as in (105)–(107), and we evaluate

$$|C|^{-\frac{1}{2}} = \chi\, e^{\mathrm{i}\xi}, \tag{127}$$

where the modulus χ is assembled from the moduli of the factors (which involve no branch ambiguity), and the phase is assembled as

$$\xi = -\frac{1}{2}\left[\arg(h^{-1}) + \arg\left(1-\zeta^{m+1}\right) + (m+1)\arg(r_+) + \arg(1-\kappa d_1) + \arg(\text{border})\right], \tag{128}$$

with $\zeta = r_-/r_+$, the labeling of the roots being the one the principal square root delivers; Lemmas 3 and 4 in the Appendix show that this labeling always satisfies $|\zeta| < 1$. The critical term is $(m+1)\arg(r_+)$: the phase of r_+^{m+1} is computed as $m+1$ times the phase of r_+, rather than as the principal argument of the assembled power, which is exactly what keeps that part of the phase continuous in s. Lemmas 3 and 4 in the Appendix show that the terms of (128) associated with $1/|D|$ are continuous, without qualification: the elementary combinations $p \pm 2q$ both lie in the open right half plane along the whole path, from which it follows that the discriminant $p^2 - 4q^2$ never meets the branch cut, that h is continuous

and nonvanishing, that $\mathrm{Re}(r_+) > 0$, and that $|r_-| < |r_+|$ strictly. The remaining two terms are the principal arguments of the rank-one and border factors, and their continuity is a consequence of a structural property of the matrices along the integration path. All the imaginary terms of $C(s, -cs)$ are real symmetric matrices multiplied by i, so

$$\mathrm{Re}\, C(s, -cs) = H^{-1} + s^2 M (V^{11})^{-1} M \succ 0 \qquad \text{for every } s : \tag{129}$$

C is uniformly *accretive* (sectorial) along the path, and, as shown in the Appendix ((A.27)), so is the tridiagonal Toeplitz core D. These two facts pin both factors away from the branch cut:

Proposition 4. *Let C be a complex $T \times T$ matrix with $\mathrm{Re}\, C \succ 0$, partitioned as in (80) with interior block $\Delta = D - \kappa 1_m 1_m'$, and suppose in addition that $\mathrm{Re}\, D \succ 0$. Then (i) the rank-one factor $1 - \kappa d_1 = |\Delta|/|D|$ does not lie on the closed negative real axis, and (ii) the border factor of (126), which is the determinant of the 2×2 Schur complement of Δ in C, does not lie on the closed negative real axis. Consequently their principal arguments are continuous along any continuous path on which the hypotheses hold, and (128) is the continuous phase of $|C|^{-1/2}$.*

The proof, given in the Appendix, is short. Part (i) follows from a convexity argument: $(1-t)D + t\Delta$ is accretive—hence nonsingular—for every $t \in [0, 1]$, while its determinant divided by $|D|$ equals $1 - t\kappa d_1$, which is *linear* in t; if $1 - \kappa d_1$ were a nonpositive real number, this linear function would have to cross zero inside $[0, 1]$. Part (ii) follows because Schur complements inherit accretivity, so the 2×2 Schur complement has both eigenvalues in the open right half plane and its determinant has principal argument in $(-\pi, \pi)$. Both hypotheses are theorems for the problem at hand: (129) gives the first, and (A.27) gives the second.

In the confluent regime of Section 3.3.1, where $1/|D|$ is evaluated through the expansions (125) rather than through the roots, continuity holds by a separate and simpler argument, given in the Appendix. Written as a product over the values of δ at which the eigenvalues of D vanish, $|D|$ is a nonvanishing constant times a product of factors none of which the band

reaches, and the total argument of that product is bounded well inside π. Switching between the two regimes cannot introduce a jump either: on their overlap the two pointwise formulas agree a priori up to a locally constant multiple of 2π, and the multiple is zero because both phases vanish at a real positive reference point. The continuity of the strictly $O(1)$ assembly (126)–(128) is therefore established: what earlier practice had to monitor numerically is here a theorem. The released code also offers an independent $O(T)$ evaluation of the same phase, from the principal arguments of the LU pivots of the accretive matrix (Proposition 9 in the Appendix), and validates the default assembly against it.

3.3.3. Behavior Near the Origin

The integrand of (69) is a $0/0$ form at $s = 0$. By L'Hôpital's rule,

$$\lim_{s\to 0} \frac{\mathrm{Im}(\varphi_{U,W}(s, -cs))}{s} = \mathbb{E}[U - cW] = \mathrm{tr}((A - cB)V_x), \tag{130}$$

and both traces have simple closed forms:

$$\mathrm{tr}(AV_x) = -\frac{\rho_{uv}}{1-\phi} + \frac{\rho_{uv}(1-\phi^T)}{T(1-\phi)^2}, \tag{131}$$

$$\mathrm{tr}(BV_x) = \frac{T-1}{1-\phi^2} - \frac{2\phi}{(1-\phi)(1-\phi^2)}\left[1 - \frac{1-\phi^T}{T(1-\phi)}\right]. \tag{132}$$

In the implementation we replace the integrand on a small interval $[0, s_0]$ by the quadratic through the limit and the computed values at $s_0/2$ and s_0, where s_0 is chosen adaptively by comparing the computed integrand with its local Taylor approximation. For very large s, the modulus of the integrand decays rapidly, and once it falls below machine precision the integrand is set to zero.

3.4. Summary of the Algorithm

For each quadrature point s: (i) form the scalars p, q, κ, α_1, β_1, and the boundary entries of (80); (ii) evaluate the primitives d_1–d_4 (and their second-order analogs for the density) by the two-regime rule of Section 3.3.1: the closed forms of Lemma 2 in the roots when $|\tau| \geq 5/2$ and the Chebyshev expansions (125) when $|\tau| < 5/2$; (iii) assemble $|C|^{-1/2}$ (and $\mathrm{tr}(C^{-1}M)$ for the density) in modulus–phase form using (126)–(128); (iv) accumulate the quadrature sum, patching the removable singularity at the origin as in Section 3.3.3. Every step costs $O(1)$ operations.

4. Fast Computation of the Exact Distributions of $t(\hat{\beta})$ and R^2

The distributions of $t(\hat{\beta})$ and R^2 require the joint characteristic function $\phi_{X_1,X_2}(s,t) = |\mathcal{D}|^{1/2}$ of Section 2.3 and the trace terms in the integrands $h(s,t)$ and $g(s,t)$. The obstacle identified in Section 2.3 is that $\mathcal{D}$ depends on both variables of integration, so no one-time eigenvalue decomposition is available. In this section we show that the same structural reduction as in Section 3 applies: for every (s,t), the matrix $\mathcal{D}^{-1}$ is congruent to the direct sum of a diagonal-plus-rank-one block and a bordered, rank-one-corrected tridiagonal Toeplitz matrix, so $|\mathcal{D}|^{1/2}$, $\mathrm{tr}(\mathcal{D}\tilde{B})$, and $\mathrm{tr}((\mathcal{D}\tilde{B})^2)$ can all be evaluated with $O(1)$ operations using the toolkit of Section 3.2.

4.1. Reduction of the Joint Characteristic Function

Write

$$\tilde{\mathcal{C}} = (V_y^{-1} - 2\mathrm{i}s(A - r_1 B) - 2\mathrm{i}t(C - r_2 B))^{-1} \equiv \begin{bmatrix} \tilde{\mathcal{C}}_{11} & \tilde{\mathcal{C}}_{12} \\ \tilde{\mathcal{C}}_{21} & \tilde{\mathcal{C}}_{22} \end{bmatrix}, \tag{133}$$

so that $\phi_{X_1,X_2}(s,t) = |V_y|^{-1/2}|\tilde{\mathcal{C}}|^{1/2}$, and the integrands of Propositions 1 and 2 become

$$h(s,t) = |V_y|^{-\frac{1}{2}}|\tilde{\mathcal{C}}|^{\frac{1}{2}}\left[(\mathrm{tr}(\tilde{\mathcal{C}}_{22}M))^2 + 2\mathrm{tr}((\tilde{\mathcal{C}}_{22}M)^2)\right], \tag{134}$$

$$g(s,t) = |V_y|^{-\frac{1}{2}}|\tilde{\mathcal{C}}|^{\frac{1}{2}}\,\mathrm{tr}(\tilde{\mathcal{C}}_{22}M), \tag{135}$$

where we use the fact that $\mathrm{tr}(\mathcal{D}\tilde{B}) = \mathrm{tr}(\tilde{\mathcal{C}}_{22}M)$ and $\mathrm{tr}((\mathcal{D}\tilde{B})^2) = \mathrm{tr}((\tilde{\mathcal{C}}_{22}M)^2)$, since B selects the lower-right $T \times T$ block.

As in Section 3.1, the reduction rests on the explicit inverse of V_y. Partitioning V_y^{-1} into $T \times T$ blocks V^{11}, V^{12}, $V^{21} = (V^{12})'$ and V^{22}—the same symbols as in Section 3.1, now redefined as the blocks of V_y^{-1} rather than of V_x^{-1}, which they reduce to at $\tilde{\beta} = 0$—and using (71) and (72),

$$V^{11} = \begin{bmatrix} \frac{1}{1-\rho_{uv}^2} I_{T-1} & 0_{T-1} \\ 0'_{T-1} & 1 \end{bmatrix}, \tag{136}$$

$$V^{12} = -\tilde{\beta}V^{11} - \rho_{uv}V^{11}GH^{-1}$$

$$= \frac{1}{1-\rho_{uv}^2} \begin{bmatrix} \rho_{uv}\phi - \tilde{\beta} & -\rho_{uv} & 0 & \cdots & 0 \\ 0 & \rho_{uv}\phi - \tilde{\beta} & -\rho_{uv} & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \rho_{uv}\phi - \tilde{\beta} & -\rho_{uv} \\ 0 & \cdots & 0 & 0 & -\tilde{\beta}(1-\rho_{uv}^2) \end{bmatrix}, \tag{137}$$

$$V^{22} = \frac{1}{1-\rho_{uv}^2} \begin{bmatrix} 1-\rho_{uv}^2 + (\tilde{\beta} - \rho_{uv}\phi)^2 & \omega & 0 & \cdots & 0 \\ \omega & \eta & \omega & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & \omega & \eta & \omega \\ 0 & \cdots & 0 & \omega & 1 + (1-\rho_{uv}^2)\tilde{\beta}^2 \end{bmatrix}, \tag{138}$$

where $\omega = \tilde{\beta}\rho_{uv} - \phi$, $\eta = 1 + \phi^2 + \tilde{\beta}^2 - 2\tilde{\beta}\rho_{uv}\phi$, and $|V_y| = (1-\rho_{uv}^2)^{T-1}/(1-\phi^2)$. The presence of the $-2\mathrm{i}tM$ term in the upper-left block makes the first stage of the Schur reduction slightly more involved than in Section 3: with $\lambda_1 = (1-\rho_{uv}^2)^{-1} - 2\mathrm{i}t$ and $\lambda_2 = 1 - 2\mathrm{i}t$, the upper-left block is diagonal plus rank one,

$$V^{11} - 2\mathrm{i}tM = \Lambda + \frac{2\mathrm{i}t}{T} 1_T 1_T', \qquad \Lambda = \begin{bmatrix} \lambda_1 I_{T-1} & 0_{T-1} \\ 0_{T-1}' & \lambda_2 \end{bmatrix}, \tag{139}$$

so that

$$|V^{11} - 2\mathrm{i}tM| = \lambda_1^{T-1}\lambda_2 \left[1 + \frac{2\mathrm{i}t}{T}\left(\frac{T-1}{\lambda_1} + \frac{1}{\lambda_2}\right)\right], \tag{140}$$

$$(V^{11} - 2\mathrm{i}tM)^{-1} = \Lambda^{-1} - c\,\Lambda^{-1} 1_T 1_T' \Lambda^{-1}, \qquad c = \frac{2\mathrm{i}t}{T + 2\mathrm{i}t\left(\frac{T-1}{\lambda_1} + \frac{1}{\lambda_2}\right)}. \tag{141}$$

Taking the Schur complement then gives

$$|\tilde{\mathcal{C}}|^{\frac{1}{2}} = |V^{11} - 2\mathrm{i}tM|^{-\frac{1}{2}}\, |F(s,t)|^{-\frac{1}{2}}, \tag{142}$$

where $F(s,t) = \tilde{\mathcal{C}}_{22}^{-1}$ is the $T \times T$ matrix

$$F(s,t) = V^{22} + 2\mathrm{i}(sr_1 + tr_2)M - (V^{21} - \mathrm{i}sM)(V^{11} - 2\mathrm{i}tM)^{-1}(V^{12} - \mathrm{i}sM). \tag{143}$$

Despite its appearance, $F(s,t)$ has exactly the structure of (80): because V^{12} is bidiagonal and $(V^{11} - 2\mathrm{i}tM)^{-1}$ is diagonal plus rank one, the product in (143) is tridiagonal plus rank

one plus boundary corrections. Define the scalars

$$k_1 = \frac{\rho_{uv}\phi - \tilde{\beta}}{1-\rho_{uv}^2} - \mathrm{i}s, \qquad k_2 = \frac{\mathrm{i}s}{T}, \qquad k_4 = -\frac{\rho_{uv}}{1-\rho_{uv}^2},$$
$$k_5 = -(\tilde{\beta} + \mathrm{i}s), \qquad \sigma_\Lambda = \frac{T-1}{\lambda_1} + \frac{1}{\lambda_2}, \qquad k_3 = \frac{k_1}{\lambda_1} + k_2\sigma_\Lambda,$$
$$k_6 = \frac{k_4}{\lambda_1} + \frac{k_5}{\lambda_2} + k_2\sigma_\Lambda. \tag{144}$$

Then

$$F(s,t) = \begin{bmatrix} f_{1,1} & a' & f_{1,T} \\ a & \Delta & b \\ f_{T,1} & b' & f_{T,T} \end{bmatrix}, \qquad \Delta = D - \kappa_F\, 1_{T-2} 1'_{T-2}, \tag{145}$$

where D is the $(T-2)\times(T-2)$ symmetric tridiagonal Toeplitz matrix with

$$p_F = \frac{\eta}{1-\rho_{uv}^2} + 2\mathrm{i}(sr_1 + tr_2) - \frac{k_1^2 + k_4^2}{\lambda_1}, \tag{146}$$
$$q_F = \frac{\omega}{1-\rho_{uv}^2} - \frac{k_1 k_4}{\lambda_1}, \tag{147}$$

the border vectors are again of the form $a = a_1 1_{T-2} + q_F\, e_{1,T-2}$ and $b = b_1 1_{T-2} + q_F\, e_{T-2,T-2}$, and the corner entries $f_{1,1}$, $f_{1,T} = f_{T,1}$, $f_{T,T}$ and the scalars a_1, b_1, κ_F are explicit expressions in $k_1, \ldots, k_6$, c, λ_1, λ_2, and $2\mathrm{i}(sr_1 + tr_2)$:

$$f_{1,1} = 1 + \frac{(\tilde{\beta} - \rho_{uv}\phi)^2}{1-\rho_{uv}^2} + 2\mathrm{i}(sr_1 + tr_2)\left(1 - \frac{1}{T}\right) - \left[\frac{k_1(k_1+2k_2)}{\lambda_1} + k_2^2\sigma_\Lambda\right] + ck_3^2, \tag{148}$$
$$f_{1,T} = -\frac{2\mathrm{i}(sr_1 + tr_2)}{T} - \frac{k_2(k_1+k_4)}{\lambda_1} - \frac{k_2 k_5}{\lambda_2} - k_2^2\sigma_\Lambda + ck_6k_3, \tag{149}$$
$$f_{T,T} = \frac{1}{1-\rho_{uv}^2} + \tilde{\beta}^2 + 2\mathrm{i}(sr_1 + tr_2)\left(1 - \frac{1}{T}\right) - \left[\frac{k_4(2k_2+k_4)}{\lambda_1} + \frac{k_5(2k_2+k_5)}{\lambda_2} + k_2^2\sigma_\Lambda\right] + ck_6^2, \tag{150}$$
$$a_1 = -\frac{2\mathrm{i}(sr_1 + tr_2)}{T} - \frac{k_1 k_2}{\lambda_1} - (k_2 - ck_3)\left[\frac{k_1+k_4}{\lambda_1} + k_2\sigma_\Lambda\right], \tag{151}$$
$$b_1 = -\frac{2\mathrm{i}(sr_1 + tr_2)}{T} - \frac{k_2(k_1+k_4)}{\lambda_1} - k_6\left[k_2 - c\left(\frac{k_1+k_4}{\lambda_1} + k_2\sigma_\Lambda\right)\right], \tag{152}$$
$$\kappa_F = \frac{2\mathrm{i}(sr_1 + tr_2)}{T} + \frac{2k_2(k_1+k_4)}{\lambda_1} + k_2^2\sigma_\Lambda - c\left[\frac{k_1+k_4}{\lambda_1} + k_2\sigma_\Lambda\right]^2. \tag{153}$$

Consequently, $|F(s,t)|$—and with it, via (140) and (142), the joint characteristic function $\phi_{X_1,X_2}(s,t)$—is an explicit scalar expression in $|D|$ and the primitives d_1–d_4 of Lemma 2, now evaluated at (p_F, q_F).

4.2. The Trace Terms

The integrand $g(s,t)$ of Proposition 2 requires $\mathrm{tr}(\tilde{\mathcal{C}}_{22}M) = \mathrm{tr}(F^{-1}M) = \mathrm{tr}(F^{-1}) - 1_T'F^{-1}1_T/T$. By Lemma 1 applied to the partition (145), both $\mathrm{tr}(F^{-1})$ and $1_T'F^{-1}1_T$ are explicit scalar expressions in (i) the first-order primitives d_1–d_4 and $\mathrm{tr}(D^{-1})$ of (116), and (ii) the second-order primitives (117)–(120), all evaluated at (p_F, q_F). The rank-one correction passes through the Sherman–Morrison identity (98) exactly as in Section 3.1, and the border passes through the explicit blocks of Lemma 1. The resulting expressions are lengthy but entirely mechanical, and we do not print them; they are recorded, exactly as assembled here, in the released MATLAB suite.

The integrand $h(s,t)$ of Proposition 1 additionally requires $\mathrm{tr}((F^{-1}M)^2)$. Expanding,

$$\mathrm{tr}((F^{-1}M)^2) = \mathrm{tr}(F^{-2}) - \frac{2\,1_T'F^{-2}1_T}{T} + \frac{(1_T'F^{-1}1_T)^2}{T^2}, \tag{154}$$

so the new objects are $\mathrm{tr}(F^{-2})$ and $1_T'F^{-2}1_T$. Through Lemma 1 and (98) these reduce to third-order primitives—bilinear forms of D^{-3} and $\mathrm{tr}(D^{-2})$—which are again obtained by one further differentiation with respect to p of the closed forms already in hand, since $D^{-3} = \frac{1}{2}\,\partial^2 D^{-1}/\partial p^2$. No object of order higher than three is needed. We therefore have the following analog of Proposition 3.

Proposition 5. *For any (s,t,r_1,r_2), the integrands $h(s,t)$ and $g(s,t)$ in Propositions 1 and 2 can be evaluated with $O(1)$ arithmetic operations, independent of T. Consequently, the density and cumulative distribution functions of $t(\hat{\beta})$ and R^2 can be computed as triple numerical integrals whose total cost is proportional to the number of quadrature points and independent of T.*

Two remarks are in order. First, the numerical issues of Section 3.3 arise here in identical form and are treated identically: the two-regime rule of Section 3.3.1 is applied to the primitives at (p_F, q_F), and the square root in $|F(s,t)|^{-1/2}$ and the factor (140) are evaluated in modulus–phase form, with continuity guaranteed by Proposition 4: part (ii) applies verbatim, and the hypothesis $\mathrm{Re}\,D \succ 0$ of part (i) is established for (p_F, q_F) by Lemma 5 in the Appendix, which proves the exact identity $\mathrm{Re}(p_F \pm 2q_F) = (1 \mp \phi)^2 + (1-\rho_{uv}^2)(s - 2\alpha_\pm t)^2/(1+$

$4(1-\rho_{uv}^2)^2t^2)$ with explicit constants $\alpha_\pm$—the same lower bounds $(1\mp\phi)^2$ as in the b problem, uniformly in $\tilde{\beta}$ and in both transform variables. The branch analysis of Section 3.3.2 therefore transfers to the joint problem in its entirety. For (140) in particular, the phase of λ_1^{T-1} must be computed as $(T-1)\arg(\lambda_1)$; computing the determinant first and taking its square root afterward produces branch-cut discontinuities. Second, the polar transformation (46) remains useful: it converts the inner double integral over (s,t) into an integral over a bounded angular variable and a radial variable along which the integrand decays geometrically. The quadrature variables must, however, be matched to the scales of the problem, which drift with T: since $\mathcal{R}_2$ concentrates around a fixed mean with standard deviation of order $T^{-1/2}$, the characteristic function spreads in t on the conjugate scale $T^{1/2}$, so a quadrature rule with T-independent ranges eventually under-resolves the integrand. We therefore center and scale the outer variable by the closed-form mean and standard deviation of the Cauchy–Schwarz bound $\mathcal{R}_2 \le (|\tilde{\beta}| + \mathcal{Q}^{1/2})^2$ with $\mathcal{Q} = u'Mu/z'Mz$: we set $r_2 = m_2 + s_2 y$, where $m_2 = (|\tilde{\beta}| + \sqrt{\mu_{\mathcal{Q}}})^2$ and s_2 are built from the first-order mean $\mu_{\mathcal{Q}}$ and standard deviation of $\mathcal{Q}$ by the delta method, and we scale the radial variable conjugately ($R = v/s_2$); the Jacobians cancel, and the effective integration ranges are $O(1)$ for all T. This is a change of variables only, and no accuracy claim rests on m_2 and s_2. In our experiments the unscaled quadrature is accurate up to $T \approx 120$ and loses accuracy from there on, failing visibly by $T \approx 240$, while the scaled quadrature is uniformly accurate, at essentially T-independent cost, through the largest sample sizes we consider.

4.3. Truncating the Outer Integral

The outer integrals in Propositions 1 and 2 run over $r_2 \in (0,\infty)$, and the numerical evaluation of $f_{\mathcal{R}_1,\mathcal{R}_2}(r_1,r_2)$ deteriorates for very large r_2. Truncating at some $r_2^{\max}$ discards mass at most $\mathbb{P}[\mathcal{R}_2 > r_2^{\max}]$ from a density that integrates to one, so the truncation error is controlled as soon as that tail probability is. A bound on it is available in closed form, and it is the exact moment generating function (mgf) of the problem that supplies it. For any $\theta > 0$ in the domain of the mgf, Markov's inequality applied to $e^{\theta(y'Cy - r_2^{\max} y'By)}$ gives

$$\mathbb{P}\left[\mathcal{R}_2 > r_2^{\max}\right] = \mathbb{P}\left[y'Cy - r_2^{\max}\, y'By > 0\right] \le \Psi\big(0,\ \theta r_2^{\max},\ -\theta\big), \tag{155}$$

where Ψ is the mgf (163) of Section 5 and B, C are the matrices of (18)–(26), so that $y'By = z'Mz$ and $y'Cy = r'Mr$. The right-hand side of (155) is an instance of the same determinant the rest of the paper evaluates, at a real rather than an imaginary argument, so it costs $O(1)$; its logarithm is convex in θ on the domain of the mgf. That domain is the set of $\theta > 0$ for which the symmetric matrix $V_y^{-1} + 2\theta(r_2^{\max}B - C)$, with V_y of (29), is positive definite, and it is certified exactly, since positive definiteness holds if and only if every pivot of the banded factorization of that matrix and the determinant of its rank-two correction are positive—quantities that the evaluation produces anyway. We therefore choose $r_2^{\max}$ as the smallest value for which the minimum of (155) over the certified domain falls below 10^{-15}. Nothing here is approximated: the bound holds for every T, ϕ, ρ_{uv} and $\tilde{\beta}$, and the bisection that produces $r_2^{\max}$ costs a fraction of a second, once per configuration.

4.4. The Special Case $\rho_{uv} = 0$: An $O(1)$ Algorithm for the Density of $z'Mz$

When $\rho_{uv} = 0$, Section 2.4 shows that the distributions of $t(\hat{\beta})$ and R^2 are mixtures of noncentral t and noncentral beta distributions, mixing over the noncentrality parameter through the density f_Q, $Q = z'Mz$. The characteristic function of Q is

$$\phi_Q(t) = (1-\phi^2)^{\frac{1}{2}}\,|F_Q(t)|^{-\frac{1}{2}}, \qquad F_Q(t) = H^{-1} - 2\mathrm{i}tM, \tag{156}$$

and $F_Q(t)$ is precisely of the form (80): with $c_Q = 2\mathrm{i}t/T$,

$$f_{1,1} = f_{T,T} = 1 - 2\mathrm{i}t\left(1 - \frac{1}{T}\right), \qquad f_{1,T} = f_{T,1} = \frac{2\mathrm{i}t}{T}, \tag{157}$$

$$a = c_Q 1_{T-2} - \phi e_{1,T-2}, \qquad b = c_Q 1_{T-2} - \phi e_{T-2,T-2}, \qquad \Delta = D + c_Q 1_{T-2} 1'_{T-2}, \tag{158}$$

where D is tridiagonal Toeplitz with $p_Q = 1+\phi^2-2\mathrm{i}t$ and $q_Q = -\phi$. The toolkit of Section 3.2 therefore delivers $\phi_Q(t)$, and with it the integrands for the density and distribution functions of Q,

$$f_Q(q) = \frac{(1-\phi^2)^{\frac{1}{2}}}{\pi}\int_0^\infty \mathrm{Re}\left(e^{-\mathrm{i}tq}|F_Q(t)|^{-\frac{1}{2}}\right)\mathrm{d}t, \tag{159}$$

$$\mathbb{P}[Q < q] = \frac{1}{2} - \frac{(1-\phi^2)^{\frac{1}{2}}}{\pi}\int_0^\infty \frac{\mathrm{Im}\left(e^{-\mathrm{i}tq}|F_Q(t)|^{-\frac{1}{2}}\right)}{t}\mathrm{d}t, \tag{160}$$

with $O(1)$ operations per quadrature point. Substituting f_Q into (59)–(62) then gives the exact distributions of $t(\hat{\beta})$ and R^2 as double integrals with $O(1)$ integrands. For the behavior of these integrands near $t = 0$ (a removable singularity for the distribution function), the required limits are

$$\lim_{t\to 0} \frac{\mathrm{Im}(e^{-\mathrm{i}tq}\phi_Q(t))}{t} = \mathrm{tr}(MH) - q, \tag{161}$$

and the local curvature is characterized by

$$\left.\frac{\partial^2 \,\mathrm{Re}(e^{-\mathrm{i}tq}\phi_Q(t))}{\partial t^2}\right|_{t=0} = -q^2 + 2q\,\mathrm{tr}(MH) - [\mathrm{tr}(MH)]^2 - 2\,\mathrm{tr}((MH)^2), \tag{162}$$

which we use to select the width of the origin patch as in Section 3.3.3.

4.5. Computational Performance

Table 2 reports the time required to evaluate the exact density and cumulative distribution function of b, $t(\hat{\beta})$, and R^2 at a single point, for sample sizes ranging from $T = 120$ to $T = 3840$, under the illustrative parameters $\phi = 0.9805$, $\rho_{uv} = -0.856$, and $\beta = 0$. Two implementations are compared: the $O(1)$ algorithms of Sections 3–4, and an eigenvalue-based implementation in the spirit of Imhof (1961) and Broda and Paolella (2009), which diagonalizes the underlying matrices once per integrand configuration and evaluates the characteristic function as a product over eigenvalues. For the distribution of b a single decomposition serves the entire univariate integral, so the eigenvalue approach remains usable at all sample sizes shown, although its cost grows roughly cubically while the cost of the $O(1)$ algorithm is flat—by $T = 3840$ the two differ by roughly four orders of magnitude. For $t(\hat{\beta})$ and R^2 the decomposition must be recomputed at every value of the outer integration variables, and the eigenvalue approach becomes infeasible almost immediately: cells whose projected cost exceeds half an hour are extrapolated by the T^3 operation count of the decomposition (flagged with an asterisk), and already at $T = 240$ a single density evaluation is projected to take between one and a half and two and a half hours, against under five minutes—and barely growing in T—for the $O(1)$ algorithm. At $T = 960$ the projected eigenvalue cost is between four and a half days and just over six days per point, against under four minutes, and at $T = 3840$ it is between nine and thirteen months, against under four minutes. With

the $O(1)$ algorithms of this paper, the sample size ceases to be the binding constraint on exact finite sample inference in predictive regressions.

Two qualifications on how to read the table. First, every eigenvalue entry for $t(\hat{\beta})$ and R^2 beyond the smallest sample size is a projection and not a measurement, as are the comparisons of the previous paragraph that rest on them; the smallest sample size in each row is always measured, and every unflagged entry is a measurement. Second, the times are per evaluation point, not per curve and not per inferential task. A plotted density of $t(\hat{\beta})$ on a grid of sixty points costs sixty times the entry—two to three hours—and the panels of Section 6 were produced overnight. What the $O(1)$ reduction removes is the dependence of that cost on T, not the cost itself. Where the statistic is b the distinction does not arise: the whole of Table 3, forty parameter configurations, each requiring two exact tail probabilities and two quantiles located by bisection on the exact distribution function, was computed in six seconds. That, rather than the per-point timings of the t-ratio, is the scale at which an $O(1)$ integrand changes what is possible.

5. Exact Moments

The distributions of the previous sections answer questions about tail probabilities and quantiles. Many questions of interest are instead about moments: the bias and variance of $\hat{\beta}$ (Stambaugh, 1999), the mean of the t-ratio under the null, which measures the size distortion at its source, or the expected R^2, which calibrates how much explanatory power a predictive regression produces mechanically. This section derives integral representations for the exact moments of all three statistics and shows that the machinery of Sections 3 and 4 computes them with $O(1)$ operations per integrand evaluation. The representations for the moments of b (and hence of $\hat{\beta}$) and of R^2 follow from the classical device of linearizing each quadratic form in a denominator by a gamma integral (Sawa, 1978; Magnus, 1986; Meng, 2005); the representation for the moments of $t(\hat{\beta})$ is, to the best of our knowledge, new.

Table 2: **Computation Time per Evaluation Point**

The table reports the time in seconds to evaluate the exact density (pdf) and cumulative distribution function (cdf) at a single point, for $\phi = 0.9805$, $\rho_{uv} = -0.856$, and $\tilde{\beta} = 0$, using the $O(1)$ algorithms of this paper (new) and an eigenvalue-based implementation (eig). Each entry is one evaluation, not an average over a grid: the argument is 0.15 for the distribution of b, 2 for $t(\hat{\beta})$, and 0.02 for R^2, each of them a point in the body of the relevant distribution. Because the quadrature for $t(\hat{\beta})$ and R^2 is adaptive, its cost varies with the argument; the times below are representative rather than worst case. Entries flagged with * are extrapolated, not measured: from the second sample size onwards a cell is run only if its cost, projected from the last measured one by the T^3 count of the decomposition, falls below half an hour. The smallest sample size is always measured, there being nothing to project from; every other unflagged entry is a measurement. In the b panel, where one evaluation takes a few milliseconds, the entry is the median of nine repetitions after a warm-up call, a single timing there being dominated by first-call overhead rather than by the computation. Times are wall-clock seconds in MATLAB R2025b on a desktop PC with an AMD Ryzen 9 5950X 16-core processor and 64 GB of RAM.

	T					
b	120	240	480	960	1920	3840
pdf, new	0.002	0.001	0.001	0.001	0.002	0.003
pdf, eig	0.089	0.063	0.306	0.822	4.772	37.311
cdf, new	0.001	0.001	0.001	0.001	0.002	0.002
cdf, eig	0.028	0.037	0.223	0.636	3.469	13.950
	T					
$t(\hat{\beta})$ and R^2	120	240	480	960	1920	3840
$t(\hat{\beta})$ pdf, new	101.1	148.9	159.2	162.3	167.9	208.3
$t(\hat{\beta})$ pdf, eig	766.5	$6.1\times10^{3*}$	$4.9\times10^{4*}$	$3.9\times10^{5*}$	$3.1\times10^{6*}$	$2.5\times10^{7*}$
$t(\hat{\beta})$ cdf, new	28.3	26.6	39.6	43.8	48.5	46.6
$t(\hat{\beta})$ cdf, eig	382.5	$3.1\times10^{3*}$	$2.4\times10^{4*}$	$2.0\times10^{5*}$	$1.6\times10^{6*}$	$1.3\times10^{7*}$
R^2 pdf, new	146.4	258.5	244.6	219.7	128.3	137.2
R^2 pdf, eig	1040.0	$8.3\times10^{3*}$	$6.7\times10^{4*}$	$5.3\times10^{5*}$	$4.3\times10^{6*}$	$3.4\times10^{7*}$
R^2 cdf, new	54.0	59.4	60.5	67.7	81.7	115.2
R^2 cdf, eig	656.5	$5.3\times10^{3*}$	$4.2\times10^{4*}$	$3.4\times10^{5*}$	$2.7\times10^{6*}$	$2.2\times10^{7*}$

5.1. The Moment Generating Function

Define, for $s, t \geq 0$ and γ in a complex neighborhood of the real interval on which the expectation is finite,

$$\Psi(\gamma, s, t) = \mathbb{E}\left[e^{\gamma\, y'Ay - s\, y'By - t\, y'Cy}\right] = |I_{2T} - 2V_y(\gamma A - sB - tC)|^{-\frac{1}{2}}, \tag{163}$$

with A, B, C as in (18) and (26), so that $\Psi(\gamma, -s, -t)$ is the joint moment generating function of the three quadratic forms $y'Ay$, $y'By$ and $y'Cy$ out of which the three statistics of (173) are built. It is the only object any of the moment representations below requires. Setting $t = 0$ leaves the corresponding transform of $(y'Ay, y'By)$, which is all that the moments of b require. No separate object is needed for them: the distribution of b is determined by (T, ϕ, ρ_{uv}) alone and does not depend on β, so its moments may be computed at $\tilde{\beta} = 0$, where $y = x$ and V_y of (29) reduces to V_x of (19). We therefore write $\Psi(\gamma, s, 0)$ throughout. The determinant in (163) is the one whose evaluation Sections 3 and 4 reduce to $O(1)$ operations; the only difference is that the argument of the determinant is now real rather than imaginary. This difference works entirely in our favor. For $s, t \geq 0$ and real γ in the domain of (163), the matrix $V_y^{-1} - 2\gamma A + 2sB + 2tC$ is symmetric positive definite, so every pivot of its LDL' factorization is a positive real number, and the branch questions that required care in Section 3.3.2 cannot arise at all: a moment generating function is positive. The closed-form primitives of Section 3 apply verbatim at real arguments, where they are simpler—the oscillation that produces their two-regime structure is absent—and the details are collected in the Online Appendix.

5.2. Evaluating the Derivatives

Every representation below involves a partial derivative of Ψ at a point on the real axis, and none of them is of any use until that derivative can be produced at the same $O(1)$ cost as Ψ itself. This subsection says how.

The organizing fact is that the derivatives of Ψ are one easy step away from the cumulants of $y'Ay$. Write

$$\widetilde{W}(\gamma, s, t) = V_y^{-1} - 2\gamma A + 2sB + 2tC, \qquad \widetilde{W}_0 \equiv \widetilde{W}(\gamma_0, s, t), \tag{164}$$

so that $\Psi = (|V_y|\,|\widetilde{W}|)^{-1/2}$, where γ_0 denotes the base point at which a given representation evaluates the derivatives: $\gamma_0 = 0$ for the moments of b and R^2, and nonzero base points for those of the t-ratio (Section 5.5). The γ-derivatives of $\log\Psi$ at the base point,

$$\kappa_j \;=\; \frac{\partial^j \log\Psi}{\partial\gamma^j}\bigg|_{\gamma=\gamma_0}, \qquad j \geq 1, \tag{165}$$

are the cumulants of $y'Ay$ under the Gaussian measure tilted by $\exp(\gamma_0\, y'Ay - s\, y'By - t\, y'Cy)$, and the derivatives of Ψ itself follow from the classical moment–cumulant recursion: writing $D_p = \Psi^{-1}\partial^p\Psi/\partial\gamma^p$,

$$D_0 = 1, \qquad D_p \;=\; \sum_{j=1}^{p} \binom{p-1}{j-1} \kappa_j\, D_{p-j}, \qquad \frac{\partial^p\Psi}{\partial\gamma^p}\bigg|_{\gamma=\gamma_0} = \Psi(\gamma_0, s, t)\, D_p. \tag{166}$$

The recursion is used exactly as written—it costs a triangle of scalar multiplications, and it is what the quadratures of this section consume—so the entire question is how to obtain the cumulants.

They are available in closed form. Differentiating $\log|\widetilde{W}|$ repeatedly gives, with $X = \widetilde{W}_0^{-1}A$,

$$\kappa_j \;=\; 2^{j-1}(j-1)!\,\mathrm{tr}\big(X^j\big), \qquad j \geq 1. \tag{167}$$

The formula identifies the object. The traces are real—$\widetilde{W}_0$ is symmetric and positive definite for $s, t \geq 0$ and real γ_0 in the domain of (163), so X is similar to the symmetric matrix $\widetilde{W}_0^{-\frac{1}{2}} A \widetilde{W}_0^{-\frac{1}{2}}$ and its spectrum lies on the real line—and through (166) the quantity in (169), (171) and (175) is an explicit polynomial in them: a tilted moment, which is exactly what those representations ask for. But (167) is not what we compute, because evaluating it costs $O(T)$ per order: what Sections 3 and 4 reduce to $O(1)$ operations is the *determinant* of $\widetilde{W}$, not its inverse, and $\mathrm{tr}(X^j)$ is a sum over $2T$ terms. This is the route Kan and Pan (2022) take, and at their sample sizes it is the sensible one. For any fixed j the sum can be collapsed further—the entries of the resolvent of a banded Toeplitz matrix are finite combinations of geometric sequences, which is how the traces collected in the Appendix are obtained—but that reduction has to be redone at each order. We obtain the cumulants instead from values of the determinant alone.

The first cumulant is a complex step (Squire and Trapp, 1998). Since $\log \Psi$ is analytic in γ and real on the real axis,

$$\kappa_1 \;=\; \frac{\operatorname{Im}\left(\log \Psi(\gamma_0 + \mathrm{i}h, s, t)\right)}{h} \;+\; O(h^2), \tag{168}$$

and—this is the point of the device—no subtraction of nearly equal numbers occurs anywhere in it, so h may be taken as small as 10^{-60} times the scale of γ and (168) is exact to machine precision. A real central difference of the same width would return zero.

The higher cumulants are obtained by Taylor-mode arithmetic. The $O(1)$ evaluation of $\log|\widetilde{W}|$ is a finite composition of analytic operations on scalars: form the coefficients of the symbol, take the roots of a quadratic, take square roots, logarithms and quotients, sum geometric series, and add the corner and rank-two corrections. Replace every scalar in it by its Taylor polynomial of degree p about γ_0, carried in powers of $(\gamma - \gamma_0)$—a vector of $p+1$ coefficients, which is all that the derivatives at γ_0 up to order p can depend on—and run the same assembly in the arithmetic of such polynomials, multiplying and composing them with everything beyond degree p discarded. The result is the degree-p Taylor polynomial of $\log \Psi$ itself, so its coefficients are the cumulants (165) exactly, through order p, and $D_1, \ldots, D_p$ follow from (166). Although this method can also deliver κ_1, we prefer to use (168) when only the first derivative is wanted, since it needs nothing beyond the scalar evaluator.

Three facts make this cheap. First, γ enters the assembly only through the argument of the characteristic function: the moment generating function is the assembly of Section 4 continued to the imaginary argument $-\mathrm{i}\gamma$ in place of s, so the only quantities that carry γ are the scalars k_1, k_2 and k_5 of (144), each affine in it, and everything built from them is a quadratic polynomial in γ whose series is exact and finite. Second, the only transcendental step is the set of tridiagonal Toeplitz primitives, and each of the operations they are built from has a standard series recursion. Third, only the coefficients of order one and above are wanted, so the constant coefficient of every logarithm may be discarded, and with it every branch question; $\log \Psi$ itself comes from the assembly of Section 4, which settles its own branch.

Two precautions are needed, and both concern the tridiagonal Toeplitz primitives. The first is a matter of scale: the primitives are rewritten in $w = q/r_+$, the ratio of the roots'

geometric mean to the larger root, which satisfies $|w| \le 1$. In that variable the T-th powers appear only as w^{2T} and are exponentially small, so no intermediate quantity in the recursion is larger than the answer it contributes to.

The second concerns the confluent point of the primitives, where the two roots of the symbol coincide. Section 3.3.1 evaluates everything there from the Taylor expansions (125) of the two Chebyshev values, and that evaluation is itself a finite composition of the same four operations, so it runs in series arithmetic verbatim: the same coefficients, the same index shifts, the same single division by U_m. Both regimes of the primitives are therefore available at every order of the series, and they remain interchangeable: where both are accurate they agree to 10^{-12} of the largest derivative at the node through order six, and the agreement survives to thirty times the switching threshold.

Against a reference computed from the dense determinant in forty-digit arithmetic, the derivatives so assembled are at machine precision at every order, with relative error below 5×10^{-14} through order six, in the confluent neighborhood as well as away from it.

5.3. Moments of $\hat{\beta}$

The moments of b exist if and only if $p < T - 1$, and the moments of $\hat{\beta}$ follow from those of b through the exact identity $\hat{\beta} = \beta + (\sigma_u/\sigma_v)b$. The representation below, and that of Section 5.4 for R^2, are the $k = 1$ specializations of the corresponding results of Kan and Pan (2022) for long-horizon regressions, and we refer the reader there for their proofs; what is new here is that the integrands can be evaluated in $O(1)$ operations.

Proposition 6. *For an integer $1 \le p < T - 1$, and with Ψ of (163) taken at $\tilde{\beta} = 0$,*

$$\mathbb{E}[b^p] = \frac{1}{\Gamma(p)} \int_0^\infty s^{p-1} \left. \frac{\partial^p \Psi(\gamma, s, 0)}{\partial \gamma^p} \right|_{\gamma=0} \mathrm{d}s, \tag{169}$$

and

$$\mathbb{E}[\hat{\beta}^p] = \sum_{j=0}^{p} \binom{p}{j} \beta^{p-j} \left(\frac{\sigma_u}{\sigma_v} \right)^j \mathbb{E}[b^j]. \tag{170}$$

A single sweep over the s-grid delivers all the moments up to order p at once, because the series carried at each node produces every derivative up to order p simultaneously. The

cost of the sweep is therefore set by the highest order wanted rather than by how many are wanted: the series has length $p+1$, that length enters only through the products in the assembly, and so the work at a node grows quadratically in p and not at all in the sample size.

5.4. Moments of R^2

Since $0 \le R^2 \le 1$, all of its moments exist. Linearizing the two quadratic forms in the denominator by two gamma integrals gives the following representation.

Proposition 7. *For an integer $p \ge 1$,*

$$\mathbb{E}[(R^2)^p] = \frac{1}{\Gamma(p)^2} \int_0^\infty \int_0^\infty s^{p-1} t^{p-1} \left. \frac{\partial^{2p} \Psi(\gamma, s, t)}{\partial \gamma^{2p}} \right|_{\gamma=0} \mathrm{d}s \, \mathrm{d}t. \tag{171}$$

5.5. Moments of $t(\hat{\beta})$

The t-ratio is where the classical device stops working. Collect the three quadratic forms in the 2×2 Gram matrix of the demeaned data,

$$\mathcal{G} \;=\; [z \;\; r]' M [z \;\; r] \;=\; \begin{bmatrix} y'By & y'Ay \\ y'Ay & y'Cy \end{bmatrix}, \qquad |\mathcal{G}| = (y'By)(y'Cy) - (y'Ay)^2, \tag{172}$$

so that, with $\nu = T - 2$,

$$\tilde{\beta} + b = \frac{y'Ay}{y'By}, \qquad R^2 = \frac{(y'Ay)^2}{(y'By)(y'Cy)}, \qquad t(\hat{\beta}) = \frac{\sqrt{\nu}\; y'Ay}{|\mathcal{G}|^{\frac{1}{2}}} \tag{173}$$

(the first identity reduces to $b = y'Ay/y'By$ at $\tilde{\beta} = 0$, where $y = x$ and the moments of b are computed). The three statistics differ only in their denominators: a single quadratic form for b, a product of two for R^2, and the determinant $|\mathcal{G}|$ for $t(\hat{\beta})$. The scalar identity $V^{-q} = \Gamma(q)^{-1} \int_0^\infty s^{q-1} e^{-sV} \mathrm{d}s$ absorbs one quadratic form per integration variable while keeping the integrand an exponential of a quadratic form—this is what produces (169) and (171)—but it cannot absorb $|\mathcal{G}|$, which is a quartic in y. Expanding $(1 - R^2)^{-p/2}$ instead expresses $\mathbb{E}[t^p]$ as an infinite series in the moments of R^2, each term a double integral; the series converges slowly precisely when R^2 has appreciable mass near one, which is the persistent-predictor, small-T region where exact methods are needed.

The resolution is the matrix generalization of the scalar identity: the matrix-variate gamma integral over the cone of 2×2 positive definite matrices (Muirhead, 2005, Theorem 2.1.11),

$$|\mathcal{G}|^{-a} = \frac{1}{\Gamma_2(a)} \int_{S \succ 0} \operatorname{etr}(-S\mathcal{G})\, |S|^{a-\frac{3}{2}}\, \mathrm{d}S, \qquad a > \tfrac{1}{2}, \tag{174}$$

where $\Gamma_2(a) = \sqrt{\pi}\,\Gamma(a)\Gamma(a-\frac{1}{2})$, $\operatorname{etr}(\cdot) = e^{\operatorname{tr}(\cdot)}$, and the integral runs over the three free entries of S. The identity restores exactly what the scalar device requires: $\operatorname{tr}(S\mathcal{G}) = s_{11}\, y'By + 2s_{12}\, y'Ay + s_{22}\, y'Cy$ is again a quadratic form in y, so the tilted expectation is again a derivative of (163). Parametrizing the cone by its Cholesky factorization $S = LL'$, where L is lower triangular with diagonal (ℓ_1, ℓ_3) and off-diagonal entry ℓ_2, removes the singular weight: $|S|^{(p-3)/2}\mathrm{d}S = 4\ell_1^{p-1}\ell_3^{p-2}\,\mathrm{d}\ell_1\mathrm{d}\ell_2\mathrm{d}\ell_3$, a smooth monomial for $p \geq 2$.

Proposition 8. *Let $\nu = T - 2$. The moments of $t(\hat{\beta})$ exist if and only if $p < \nu$. For an integer $2 \leq p < \nu$,*

$$\mathbb{E}[t(\hat{\beta})^p] = \frac{4\,\nu^{p/2}}{\Gamma_2(p/2)} \int_0^\infty \int_{-\infty}^\infty \int_0^\infty \ell_1^{p-1}\ell_3^{p-2}\, \frac{\partial^p \Psi(\gamma, s, t)}{\partial \gamma^p}\bigg|_{\substack{\gamma=-2\ell_1\ell_2 \\ s=\ell_1^2,\ t=\ell_2^2+\ell_3^2}} \mathrm{d}\ell_1\, \mathrm{d}\ell_2\, \mathrm{d}\ell_3. \tag{175}$$

For odd p the cone integral reduces to a two-dimensional one. Writing $m_i = \frac{p+1}{2} - i$,

$$\mathbb{E}[t(\hat{\beta})^p] = \frac{(-1)^{\frac{p+1}{2}}\,\nu^{p/2}}{\sqrt{\pi}\,\Gamma(p/2)} \sum_{i=0}^{\frac{p-1}{2}} \frac{p!}{i!\,(p-2i)!\,4^i} \int_0^\infty \int_{-\infty}^\infty \lambda^{\frac{p-1}{2}-i}\theta^{p-2i}\, \frac{\partial^{m_i}\Psi(\gamma, s, t)}{\partial s^{m_i}}\bigg|_{\substack{\gamma=2\lambda\theta \\ s=\lambda\theta^2,\ t=\lambda}} \mathrm{d}\theta\, \mathrm{d}\lambda, \tag{176}$$

the inner integral being taken first. For $p = 1$, after a polar transformation, this simplifies to

$$\mathbb{E}[t(\hat{\beta})] = \frac{\sqrt{\nu}}{2\pi} \int_0^{2\pi} \int_0^\infty \frac{\partial \Psi(\gamma, s, t)}{\partial \gamma}\bigg|_{\substack{\gamma=-v\sin 2\psi \\ s=v\cos^2\psi,\ t=v\sin^2\psi}} \mathrm{d}v\, \mathrm{d}\psi. \tag{177}$$

The two two-dimensional forms arise differently from (175). Equation (177) is a face of the cone, and the cleanest way to see it is to parametrize S by its spectral decomposition rather than by its Cholesky factor. Writing $S = R(\psi)\,\mathrm{diag}(v_1, v_2)\,R(\psi)'$ with $R(\psi)$ the rotation through ψ, the measure becomes

$$|S|^{\frac{p-3}{2}}\mathrm{d}S = \tfrac{1}{2}(v_1v_2)^{\frac{p-3}{2}}|v_1 - v_2|\,\mathrm{d}v_1\mathrm{d}v_2\mathrm{d}\psi, \qquad \psi \in [0, \pi),\ v_1, v_2 > 0, \tag{178}$$

and $\mathrm{tr}(S\mathcal{G})$ is the quadratic form with

$$\gamma = -(v_1 - v_2)\sin 2\psi, \qquad s = v_1\cos^2\psi + v_2\sin^2\psi, \qquad t = v_1\sin^2\psi + v_2\cos^2\psi. \tag{179}$$

At $p = 1$ the factor $(v_1v_2)^{-1}$ makes this weight non-integrable at $v_2 = 0$: the mass concentrates on the rank-one matrices $S = v_1w_1w_1'$, and setting $v_2 = 0$ in (179) reproduces the arguments of (177) exactly, with $v = v_1$. In that sense (177) is the $p = 1$ face of (175) written in the same coordinates. We nevertheless state and compute (175) in Cholesky coordinates, because the spectral weight carries both the eigenvalue-repulsion factor $|v_1 - v_2|$ and the singularity $(v_1v_2)^{(p-3)/2}$, on which a tensor-product rule converges appreciably more slowly, whereas the Cholesky weight is a smooth monomial.

For $p \geq 3$ there is no such concentration, and (176) comes instead from an identity for the Hermite polynomials: once a scalar gamma integral has linearized the denominator of $t(\hat\beta)$, $\mathbb{E}[He_p(Z)e^{aZ}] = a^pe^{a^2/2}$ converts the p-th power of the numerator into a single Gaussian tilt. The exponent m_i that this produces is a positive integer exactly when p is odd, which is why the reduction is available there and not for even p. Note that (176) differentiates Ψ in its second argument rather than its first, and that the quantity being tilted is $\lambda\|M(\theta z - r)\|^2$: λ and θ are the squared scale and the direction of the same rank-one S.

At the default quadratures of the released programs, (175) uses roughly 2×10^4 integrand evaluations and delivers about eight correct digits, (176) roughly 6×10^3, and (177) roughly 3×10^3. The main moment routine uses (177) for the mean and (175) for every $p \geq 2$, odd as well as even, and does not use (176) at all. The reason is that one sweep of (175) delivers every order up to p simultaneously—the series at each node carries all the derivatives at once—so whenever any even moment is wanted, the odd ones arrive with it at no additional cost, and (176), which computes one odd moment per quadrature, has nothing left to save. It is the more economical route only for a user who wants a single odd moment in isolation, where it is three to four times cheaper; the released suite implements it, for exactly that use, as a separate routine. Because the integrand is $O(1)$, the sample size drops out of the cost entirely. At $\phi = 0.95$ and $\rho_{uv} = -0.856$, one call takes about 0.002 seconds for $\mathbb{E}[b],\dots,\mathbb{E}[b^4]$, 0.03 seconds for the first two moments of R^2, 0.005 seconds for $\mathbb{E}[t(\hat\beta)]$ and 0.3 seconds for $\mathbb{E}[t(\hat\beta)],\dots,\mathbb{E}[t(\hat\beta)^4]$, and these figures are the same at $T = 120$ and at

$T = 491{,}520$. The spread across statistics is the number of quadrature nodes, which ranges from 64 for the moments of b to $26 \times 34 \times 26$ for the higher moments of the t-ratio. The released suite implements (169)–(177), and its output has been validated against the closed forms available in the central case $\rho_{uv} = 0$—where $t(\hat{\beta}) \sim t_{T-2}$ and $R^2 \sim \text{Beta}(\frac{1}{2}, \frac{T-2}{2})$ exactly—and against direct simulation elsewhere.

5.6. The Exact Moments as Functions of the Sample Size

Figures 2–4 put the moment machinery to work on the first two moments of the three statistics for $\phi = 0.95$ and $\rho_{uv} \in \{0, -0.5, -0.856\}$, as functions of the sample size, against the fixed-ϕ limits of Section 6.1.

Three observations. First, the mean of b converges to its first-order bias approximation slowly when the predictor is persistent: at $\phi = 0.95$ and $\rho_{uv} = -0.856$, the Stambaugh approximation understates the exact bias by about eleven percent at $T = 96$ and still by about four percent at $T = 480$. Second, and more strikingly, the mean of the t-ratio under the null does not vanish unless $\rho_{uv} = 0$: it decays only like $T^{-1/2}$, because the bias of $\hat{\beta}$ is of order T^{-1} while its standard deviation is of order $T^{-1/2}$. At $\phi = 0.95$ and $\rho_{uv} = -0.856$, the exact mean of the t-ratio under the null is 1.06 at $T = 24$ and still 0.73 at $T = 96$: the entire distribution of the test statistic is shifted by the better part of a standard deviation before any predictability is present, which is the size distortion of Section 6 seen at its source. Third, the expected R^2 under the null is far from negligible at the sample sizes at issue: at $\phi = 0.95$ and $\rho_{uv} = -0.856$ it is 0.076 at $T = 24$ and still 0.015 at $T = 96$, against a population R^2 of two percent under the alternative. The comparison with the alternative is stark: the null mean accounts for about three quarters of the expected R^2 under the alternative at $T = 24$, one half at $T = 60$, and one third at $T = 96$, and falls below a tenth only by $T = 480$. For a persistent predictor and a sample of the length typically available, a substantial part of the R^2 that a predictive regression reports is generated by the estimation itself rather than by predictability. The dotted line in the bottom left panel shows that even the size of this mechanical component is understated by the asymptotic value $1/T$, by a factor of 1.8 at $T = 24$.

The second moments, in the right panel of Figure 2 and in Figure 4, tell a different

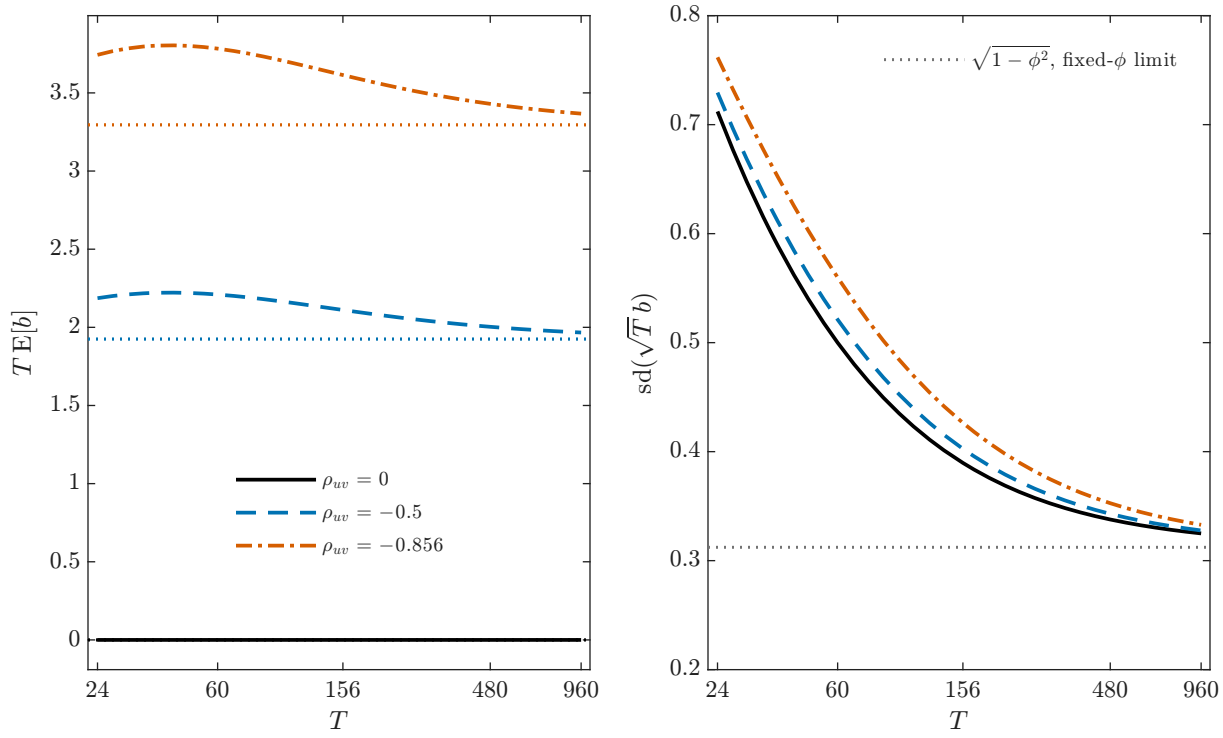


Figure 2: **Exact Mean and Standard Deviation of** b

The figure plots the exact mean and standard deviation of $b = (\sigma_v/\sigma_u)(\hat{\beta} - \beta)$ for $\phi = 0.95$ and three values of ρ_{uv}. Left panel: the mean, scaled by the sample size, against the first-order bias approximation $T\,\mathbb{E}[b] \approx -\rho_{uv}(1 + 3\phi)$ of Stambaugh (1999) (dotted lines of the matching color). Right panel: the standard deviation of $\sqrt{T}\,b$, against its fixed-ϕ limit $(1-\phi^2)^{1/2}$ from (180) (dotted). The distribution of b does not depend on β. The sample size is on a logarithmic scale.

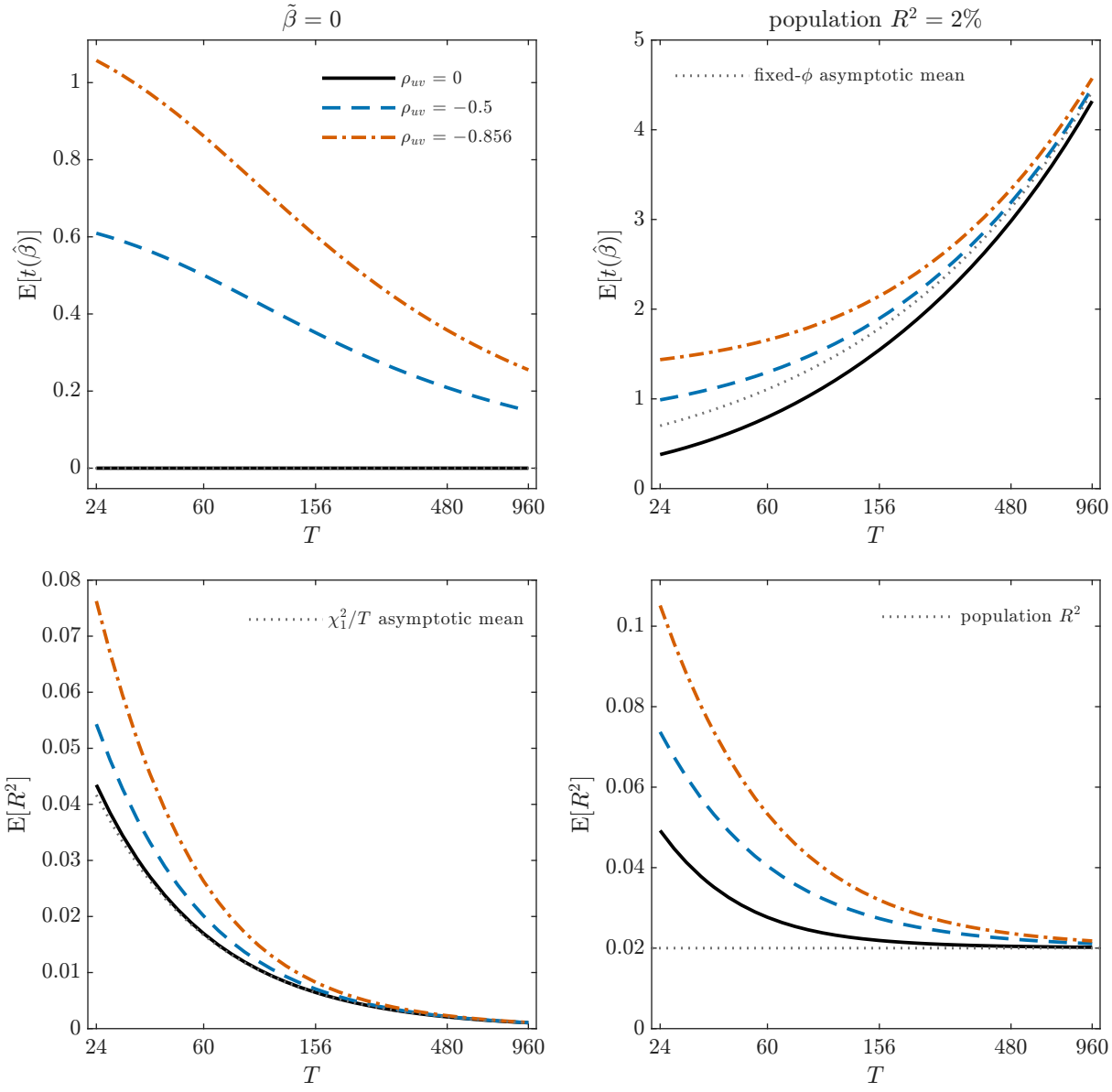


Figure 3: **Exact Means of $t(\hat{\beta})$ and R^2**

The figure plots the exact means of $t(\hat{\beta})$ (top row) and R^2 (bottom row) as functions of the sample size, for $\phi = 0.95$ and three values of ρ_{uv}, under the null in the left column and under the alternative in the right. In the top right panel the dotted line is the fixed-ϕ asymptotic mean $\sqrt{T}\tilde{\beta}/(1-\phi^2)^{1/2}$; in the bottom left panel the dotted line is the fixed-ϕ asymptotic mean $1/T$; in the bottom right panel the dotted line is the population R^2. Each panel carries its own vertical scale. The sample size is on a logarithmic scale.

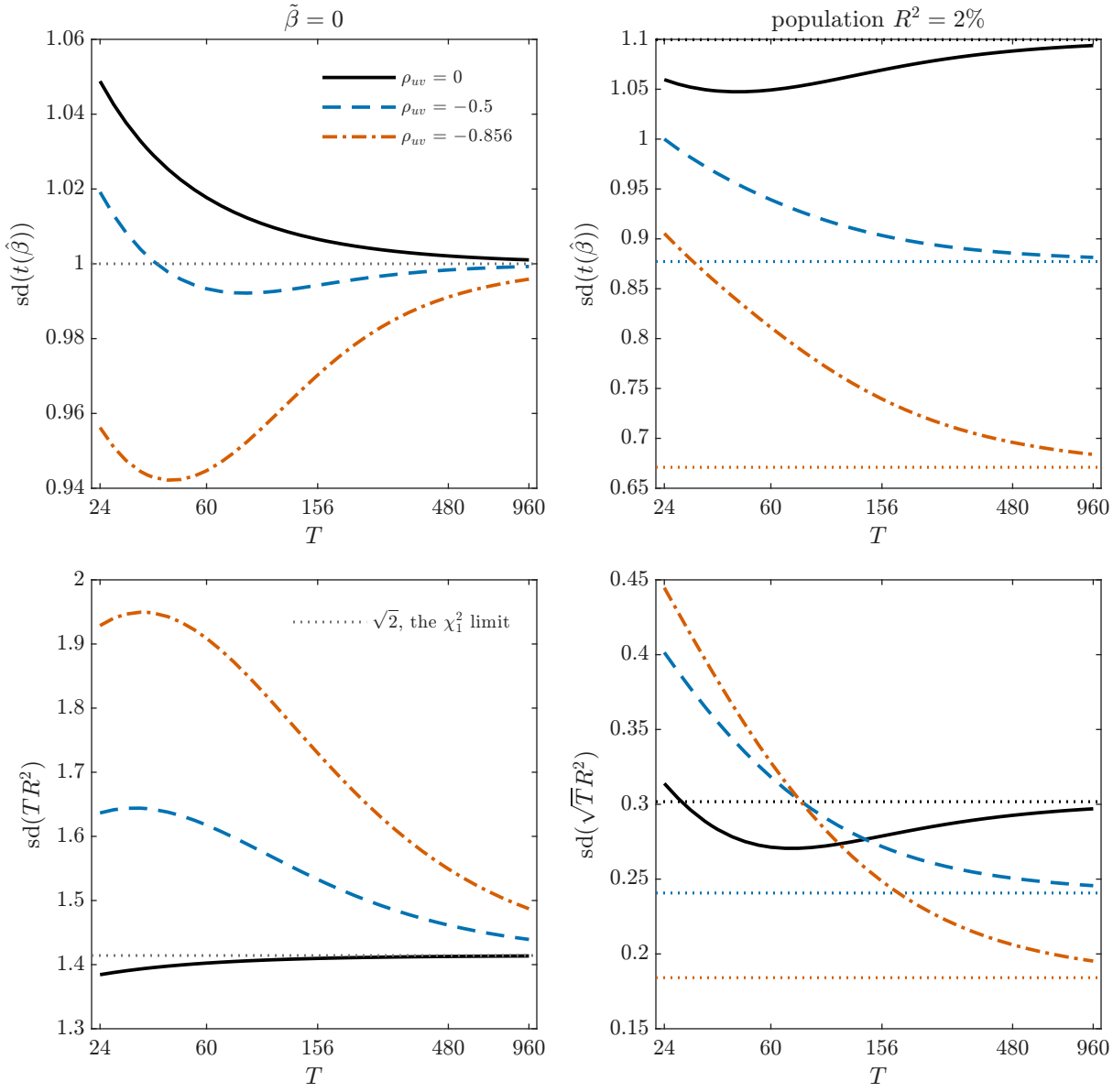


Figure 4: **Exact Standard Deviations of $t(\hat{\beta})$ and R^2**

The figure plots the exact standard deviations of $t(\hat{\beta})$ (top row) and R^2 (bottom row) as functions of the sample size, for $\phi = 0.95$ and three values of ρ_{uv}, in the layout of Figure 3. Dotted lines are the fixed-ϕ limits of Section 6.1, in the matching color where the limit depends on ρ_{uv} and in gray where it does not: 1 and $V(t)^{1/2}$ of (185) for the t-ratio, $\sqrt{2}$ and $V(R^2)^{1/2}$ of (184) for the R^2. The two R^2 panels are scaled differently—TR^2 under the null and $\sqrt{T}R^2$ under the alternative—because the limit is a χ^2_1 in the first case and a normal in the second, the discontinuity at $\beta = 0$ noted in Section 6.1. The sample size is on a logarithmic scale.

story from the first, and in one respect a more consequential one. The dispersion of $\hat{\beta}$ is understated by its asymptotic approximation by far more, and for far longer, than its center is. At $\phi = 0.95$ the exact standard deviation of $\sqrt{T}\,b$ exceeds the fixed-ϕ limit $(1-\phi^2)^{1/2}$ by a factor of 2.4 at $T = 24$ and by 55 percent at $T = 96$, and it is still six percent high at $T = 960$, where the Stambaugh approximation to the mean is accurate to about two percent. A conventional standard error therefore understates the sampling variability of the slope estimate substantially at every sample size a predictive regression is likely to be run on, and does so almost irrespective of ρ_{uv}: the three curves in the right panel of Figure 2 are close together, whereas those in the left panel are not. Bias and dispersion are distorted by different mechanisms, and only the first is removed by a bias correction.

For the t-ratio the two moments separate cleanly, and the separation is useful. Under the null its standard deviation stays within six percent of one at every sample size and every ρ_{uv} shown—the top left panel of Figure 4 is drawn on a vertical range of 0.94 to 1.06, far narrower than the other panels—while its mean is displaced by the better part of a standard deviation. The size distortion of the t-test is thus almost purely a location effect: the statistic is shifted, not stretched, and a correction that recenters it will do most of the work. Under the alternative the picture reverses. The standard deviation is then well away from one, 0.91 at $T = 24$ and 0.68 at $T = 960$ for $\rho_{uv} = -0.856$, and it converges to 0.67, the square root of the fixed-ϕ limiting variance $V(t)$ derived in Section 6.1, rather than to unity. The unit dispersion that a normal approximation to power carries over from the null therefore overstates the spread of the statistic under this alternative by about a third.

The R^2 behaves like neither. Under the null the χ^2_1 approximation understates its mean at every ρ_{uv} shown, and at $\rho_{uv} = -0.856$ it understates its dispersion as well: the exact standard deviation of TR^2 exceeds $\sqrt{2}$ by 36 percent at $T = 24$ and is still five percent above it at $T = 960$, so both the location and the width of the null distribution of the reported R^2 are larger than the asymptotic approximation admits. Two of these panels also serve as a visible check on the machinery, since at $\rho_{uv} = 0$ the exact standard deviations are available in closed form from the t_{T-2} and $\mathrm{Beta}(\frac{1}{2}, \frac{T-2}{2})$ distributions of Section 2.4—$(\nu/(\nu-2))^{1/2}$ for $t(\hat{\beta})$ and $T\sqrt{2(T-2)}/((T-1)\sqrt{T+1}\,)$ for TR^2—and the computed curves reproduce

them to the last digit plotted.

5.7. End-to-End Certification

The checks reported so far, for the distributions and for the moments alike, are of the pieces: the primitives against their spectral representations, the $O(1)$ integrands against dense evaluation, the mgf against a dense determinant. A distribution function must also pass checks that do not refer to how it was computed, and it should pass them where the computation is hard rather than where it is comfortable. We therefore ran the following at six configurations chosen for difficulty—small samples, persistence up to $\phi = 0.995$, innovation correlations of both signs with $|\rho_{uv}|$ up to 0.99 and equal to zero, and a negative ϕ—and report what they returned.

The density is a density and the distribution function is its integral. Over a window running from the exact 10^{-10} quantile to the exact $1 - 10^{-10}$ quantile, so that essentially the whole support is covered, the computed density of b is nonnegative at every point of every configuration, and integrates to between 0.999999911 and 1.000000000 by Simpson's rule on 801 points. The Richardson estimate of that rule's own error is 10^{-9} or smaller in every case, so the shortfall from one is a property of the density values and not of the check: it places their accuracy at about 10^{-7}, which is what the requested tolerances imply. The distribution function is monotone to 7×10^{-8}, two orders of magnitude inside the 10^{-6} relative tolerance of its quadrature.

The three statistics are mutually consistent. For $t(\hat{\beta})$ and R^2 the same checks were run at the two hardest configurations. Both distribution functions are monotone on their grids, the t-ratio running from 2.8×10^{-7} at -4 to $1 - 2.2 \times 10^{-5}$ at 5, and R^2 from 0.336 at 0.005 to $1 - 10^{-10}$ at 0.4. Comparing each density against a central difference of its own distribution function gives agreement to 3×10^{-4} for the t-ratio. For R^2 the agreement is one to two percent, which is the accuracy of the comparison and not of the routines: the difference carries a truncation error $h^2 f''/6$, and the R^2 density near the origin is sharply curved, falling by a factor of 400 between $w = 0.05$ and $w = 0.15$ on top of its $w^{-1/2}$ singularity at zero. Halving h reduces the gap by a factor of 4.007, against the 4 that an $O(h^2)$ truncation predicts, and Richardson extrapolation of the two differences reproduces

the density to 7×10^{-6}.

The answers do not depend on the numerical settings. Tightening the quadrature tolerance from its default 10^{-4} to 10^{-6} moves a distribution function value by at most 4×10^{-7}, and doubling the truncation point $r_2^{\max}$—already certified to discard less than 10^{-15}—moves it by at most 3×10^{-9}. The second figure is the one worth having: it confirms empirically what (155) bounds a priori.

One practical limit is worth recording, since it is a property of the quadrature rather than of the algorithms. The triple integrals default to an absolute tolerance of 10^{-4}; the distribution functions will usually deliver 10^{-6}, but the density of R^2 near the origin, where it is unbounded, generally will not, and below its floor MATLAB's adaptive routine exhausts its interval budget and returns a value together with a warning. The released validation script negotiates the tolerance rather than assuming one, and reports the tolerance that produced each number.

6. Comparison with Asymptotic Distributions

6.1. The Two Limiting Distributions

Two asymptotic frameworks are used in the literature to approximate the finite sample distributions of $\hat{\beta}$, $t(\hat{\beta})$, and R^2, and we lay them out explicitly before comparing them with the exact distributions.

The first is the standard *fixed-ϕ* asymptotic: ϕ is held fixed with $|\phi| < 1$ and $T \to \infty$. In the notation $b = (\sigma_v/\sigma_u)(\hat{\beta} - \beta)$ and $\tilde{\beta} = (\sigma_v/\sigma_u)\beta$, and with $\Rightarrow$ denoting convergence in distribution as $T \to \infty$ (here and throughout this section), Lemma 1 of Kan and Pan (2022) with $k = 1$ gives the limits

$$\sqrt{T}\, b \;\Rightarrow\; \mathcal{N}(0,\, 1 - \phi^2), \tag{180}$$

$$t(\hat{\beta}) - \frac{\sqrt{T}\,\tilde{\beta}}{\sqrt{1-\phi^2}} \;\Rightarrow\; \mathcal{N}(0,\, V(t)), \tag{181}$$

$$TR^2 \;\Rightarrow\; \chi^2_1 \qquad (\beta = 0), \tag{182}$$

$$\sqrt{T}\,(R^2 - \rho^2) \;\Rightarrow\; \mathcal{N}(0,\, V(R^2)) \qquad (\beta \neq 0), \tag{183}$$

where $\rho^2 = \tilde{\beta}^2/(\tilde{\beta}^2 + 1 - \phi^2)$ is the population R^2 and

$$V(R^2) = \frac{4\rho^2(1-\rho^2)\left[(1-\rho^2) - (\phi - \rho_r(1))^2\right]}{1-\phi^2}, \qquad \rho_r(1) = \frac{\tilde{\beta}^2\phi + \tilde{\beta}\rho_{uv}(1-\phi^2)}{\tilde{\beta}^2 + 1 - \phi^2}, \tag{184}$$

with $\rho_r(1)$ the first-order autocorrelation of r_t. The limit for the t-ratio follows by the delta method, accounting for the estimation of both σ_z and σ_u and for the covariance between the numerator of $\hat{\beta}$ and $\hat{\sigma}_z^2$, and has

$$V(t) = 1 + \frac{2\rho_{uv}\phi\tilde{\beta}}{1-\phi^2} + \frac{\tilde{\beta}^2}{1-\phi^2}\left[\frac{1+\phi^2}{2(1-\phi^2)} + \frac{1}{2} - \rho_{uv}^2\right]. \tag{185}$$

The result (181) holds for any β: when $\beta = 0$ the centering term vanishes and $V(t) = 1$, so it reduces to the familiar standard normal limit $t(\hat{\beta}) \Rightarrow \mathcal{N}(0,1)$ under the null. Note the discontinuity of the R^2 limit at $\beta = 0$: the approximating family switches from a scaled χ_1^2 under the null to a normal under the alternative, whereas the exact finite sample distribution moves continuously in β. Throughout, $t(\hat{\beta})$ is the usual statistic testing $H_0 : \beta = 0$; under the alternative it diverges at rate $\sqrt{T}$, and (181) gives the corresponding centered and scaled limit.

The second is the *local-to-unit-root* (LUR) asymptotic: $\phi = 1 + c/T$ for fixed $c < 0$ and $T \to \infty$. Since the regression contains an intercept, the three statistics are invariant to the location of z_t, and $\mu_z = 0$ can be assumed without loss of generality. Let W_1 and W_2 be standard Brownian motions on $[0,1]$ with correlation ρ_{uv}, let J_c be the *stationary* Ornstein–Uhlenbeck process $\mathrm{d}J_c(s) = cJ_c(s)\mathrm{d}s + \mathrm{d}W_2(s)$ with $J_c(0) \sim \mathcal{N}(0, 1/(-2c))$ drawn independently of the increments of (W_1, W_2), and let $J_c^{\mu}(s) = J_c(s) - \int_0^1 J_c(\tau)\mathrm{d}\tau$ denote its demeaned version.

The initial condition deserves a comment, because the limits below are usually stated with $J_c(0) = 0$. That version corresponds to a fixed—or, more generally, $o(\sqrt{T})$—initial observation: the limiting functionals are invariant to any fixed z_0 (Perron, 1991). Our predictor, however, is initialized at its stationary distribution (Section 2.1), whose standard deviation grows with the sample size along the LUR sequence: $\mathrm{Var}(z_0) = \sigma_v^2/(1-\phi^2) \approx \sigma_v^2 T/(-2c)$, so $z_0/\sqrt{T} \Rightarrow \mathcal{N}(0, 1/(-2c))$ and the initial condition survives into the limit. The weak limit of $z_{\lfloor Ts \rfloor}/\sqrt{T}$ is therefore the stationary Ornstein–Uhlenbeck process, not the one started at zero. Perron (1991) documents that the $J_c(0) = 0$ approximation is

adequate only when the initial observation is at its mean, and Elliott (1999) and Müller and Elliott (2003) develop unit-root asymptotics under exactly this stationary initialization. The distinction is quantitatively real in the region this section studies: at $c = -1.2$, the most persistent configuration below, the distribution functions of the two versions of (186) differ by up to 3.5 percentage points, several times the approximation error that remains once the initialization is matched (Section 6.2). With J_c stationary, then, by Lemma 3 of Kan and Pan (2022) with $k = 1$, adjusted to the stationary initialization—only the weak limit of $z_0/\sqrt{T}$ changes in the argument (the result for $\hat{\beta}$ with $J_c(0) = 0$ is due to Hjalmarsson, 2011),

$$T\,b \;\Rightarrow\; \frac{\int_0^1 J_c^{\mu}(s)\,\mathrm{d}W_1(s)}{\int_0^1 J_c^{\mu}(s)^2\,\mathrm{d}s}, \tag{186}$$

$$t(\hat{\beta}) \;\Rightarrow\; \frac{\int_0^1 J_c^{\mu}(s)\,\mathrm{d}W_1(s)}{\left[\int_0^1 J_c^{\mu}(s)^2\,\mathrm{d}s\right]^{1/2}} \qquad (\beta = 0), \tag{187}$$

$$\frac{t(\hat{\beta})}{T} \;\Rightarrow\; \tilde{\beta}\left[\int_0^1 J_c^{\mu}(s)^2\,\mathrm{d}s\right]^{1/2} \qquad (\beta \neq 0), \tag{188}$$

$$TR^2 \;\Rightarrow\; \frac{\left[\int_0^1 J_c^{\mu}(s)\,\mathrm{d}W_1(s)\right]^2}{\int_0^1 J_c^{\mu}(s)^2\,\mathrm{d}s} \qquad (\beta = 0), \tag{189}$$

$$T(R^2 - 1) \;\Rightarrow\; -\frac{1}{\tilde{\beta}^2 \int_0^1 J_c^{\mu}(s)^2\,\mathrm{d}s} \qquad (\beta \neq 0), \tag{190}$$

where (187) follows from the convergence results behind (186) by the continuous mapping theorem. Note that (186) holds under both the null and the alternative, so the LUR limit of $\hat{\beta}$ is continuous at $\beta = 0$. Unlike (180), the limit (186) is not normal: for $\rho_{uv} \neq 0$ it is shifted and skewed, reflecting the finite-sample bias that the fixed-ϕ theory misses.

The alternative-hypothesis statements (188) and (190) require care, because they are statements about rates rather than about shapes: under a fixed alternative $t(\hat{\beta})$ diverges at rate T and R^2 converges to one, and the $O(1)$ sampling noise that gives either statistic its dispersion has been scaled away. A non-degenerate limit is obtained instead by letting the alternative shrink at the rate at which the statistic diverges. Along the same sequence

$\phi = 1 + c/T$, take the Pitman drift $\tilde{\beta} = \mathfrak{b}/T$ for a fixed $\mathfrak{b}$. Then

$$t(\hat{\beta}) \;\Rightarrow\; \mathfrak{b}\left[\int_0^1 J_c^{\mu}(s)^2\,\mathrm{d}s\right]^{1/2} + \frac{\int_0^1 J_c^{\mu}(s)\,\mathrm{d}W_1(s)}{\left[\int_0^1 J_c^{\mu}(s)^2\,\mathrm{d}s\right]^{1/2}} \;\equiv\; \mathcal{T}_{\mathfrak{b},c}, \tag{191}$$

$$TR^2 \;\Rightarrow\; \mathcal{T}_{\mathfrak{b},c}^2, \tag{192}$$

both of which are genuine distributions, and both of which reduce at $\mathfrak{b} = 0$ to the null limits (187) and (189). The second term of $\mathcal{T}_{\mathfrak{b},c}$ is exactly that null limit. It is what (188) and (190) discard, and while it is invisible against a fixed alternative it carries essentially the whole dispersion of the statistic: at the four configurations of Section 6.2 ($\phi \in \{0.5, 0.99\}$, $T \in \{120, 960\}$, $\rho_{uv} = -0.856$, with $\tilde{\beta}$ set in each case so that the population R^2 is $\rho^2 = 0.02$) the drift term alone has standard deviation between 0.12 and 0.82, against between 0.58 and 0.94 for the exact distribution.

The two families of statements are also tied together by an exact algebraic identity, $R^2 = t(\hat{\beta})^2/(\nu + t(\hat{\beta})^2)$ with $\nu = T - 2$, so an approximation to one determines the approximation to the other. Equation (190) is the branch of that identity valid when $t(\hat{\beta})^2 \gg \nu$, and (192) is the branch valid when $t(\hat{\beta})^2 = O(1)$. At the configurations above $t(\hat{\beta})$ is of order one while ν is of order T, so it is the second branch that applies; the first is wrong by two orders of magnitude, returning mean values between -65 and -257 across the four configurations for a quantity confined to $[0, 1]$.

We therefore evaluate the local-to-unit-root approximation under the alternative from (191) with $\mathfrak{b} = T\tilde{\beta}$, and obtain the corresponding approximation to R^2 from $R^2 = t(\hat{\beta})^2/(\nu + t(\hat{\beta})^2)$ rather than from its leading term (192), since the identity is exact and keeps R^2 inside $[0, 1]$. So evaluated, the approximation is accurate: the mean of R^2 is 0.026, 0.016, 0.034 and 0.021 across the four configurations, against exact values of 0.031, 0.021, 0.034 and 0.021. To evaluate the LUR limits at a stationary parameter value ϕ, we set $c = T(\phi - 1)$ and simulate the functionals (200,000 paths of J_c on a grid of 1,000 steps, initialized at the stationary distribution $J_c(0) \sim \mathcal{N}(0, 1/(-2c))$ and stepping with the exact Ornstein–Uhlenbeck transition so that the only discretization is in the quadrature of the integrals, and using the representation $\int_0^1 J_c^{\mu}\,\mathrm{d}W_1 = \rho_{uv}\int_0^1 J_c^{\mu}\,\mathrm{d}W_2 + (1 - \rho_{uv}^2)^{1/2}[\int_0^1 (J_c^{\mu})^2]^{1/2} Z$ with $Z \sim \mathcal{N}(0, 1)$ independent of W_2).

A feature of the fixed-ϕ framework deserves emphasis. Its approximating family is *discontinuous* at $\beta = 0$: the R^2 limit switches from the scaled χ^2_1 of (182) to the normal of (183), and the t-ratio limit from a standard normal to a drifting one. The local-to-unit-root family is discontinuous only in the form in which its limits are usually stated: written as (191) and (192) it passes continuously through $\mathfrak{b} = 0$, where it returns the null limits. The exact finite sample distributions likewise move continuously in $\tilde{\beta}$. The practical consequence is that the alternative-hypothesis limits are necessarily poor approximations when β is close to zero: by continuity, the exact distributions are then still close to their null shapes, and the quality of the alternative limits is governed not by T alone but by the strength of the signal, essentially $T\rho^2$. For the weak predictability typical of return regressions (ρ^2 of a few percent), $T\rho^2$ is of order one even for T in the hundreds—$T\rho^2 = 2.4$ for $\rho^2 = 0.02$ and $T = 120$—so the data sit in precisely the transition region where neither branch of either asymptotic family applies. The exact distribution involves no branch choice at all.

6.2. Exact versus Asymptotic Densities

Figures 5–9 contrast the exact finite sample densities of b, $t(\hat{\beta})$, and R^2 with the two asymptotic approximations, for $\phi \in \{0.5, 0.99\}$ and $T \in \{120, 960\}$, with $\rho_{uv} = -0.856$. Figure 5 presents b, whose limits in both frameworks are the same under the null and under the alternative; Figures 6 and 7 present $t(\hat{\beta})$ under the null and under the alternative, and Figures 8 and 9 present R^2 under the null and under the alternative. Under the alternative, $\tilde{\beta}$ is chosen so that the population R^2 is $\rho^2 = 0.02$.

Under the null (Figures 5–8) the fixed-ϕ approximation behaves as its derivation suggests. When the predictor is only moderately persistent ($\phi = 0.5$) it is accurate already at $T = 120$ and essentially exact at $T = 960$; measured as the largest deviation relative to the peak of the exact density, its error is 0.10 and 0.03 for b, 0.11 and 0.04 for $t(\hat{\beta})$, and below 0.02 for R^2. When the predictor is highly persistent ($\phi = 0.99$)—the empirically relevant case for valuation ratios—it fails badly, and the failure does not go away with the sample size: the corresponding errors are 1.44 and 0.36 for b, 0.62 and 0.32 for $t(\hat{\beta})$, and 0.94 and 0.15 for R^2. The exact density of b is shifted and right-skewed relative to the asymptotic normal, the exact density of $t(\hat{\beta})$ is centered near $+1$ rather than 0, and the exact density of R^2 carries

far more mass in the right tail than χ^2_1/T.

The LUR approximation is, under the null, the more accurate of the two almost everywhere. It is excellent at $\phi = 0.99$ (errors 0.02 and 0.005 for b, 0.013 and 0.004 for $t(\hat{\beta})$, and below 0.001 for R^2), which is what it is designed for. The stationary initialization of Section 6.1 matters here: with the conventional $J_c(0) = 0$ the errors for b at $\phi = 0.99$ are 0.09 and 0.03, four to six times larger, so at this persistence most of what would be recorded as approximation error is in fact an initial-condition mismatch. The approximation also remains good at $\phi = 0.5$ for $t(\hat{\beta})$ and R^2, where it is in fact more accurate than the fixed-ϕ approximation; its one visible weakness there is the density of b, with an error of about 0.15. This accuracy comes at a price that the numbers do not show: the LUR limits must be evaluated by simulation.

Figures 7 and 9 turn to the alternative, with $\tilde{\beta}$ set so that the population R^2 is two percent. Here the two frameworks must be compared with care, for the reason set out in Section 6.1: the alternative-hypothesis limits (188) and (190) describe a *rate* of divergence rather than a distribution, having scaled away the $O(1)$ sampling noise that gives the statistic its dispersion, and used literally they are not merely imprecise but meaningless at these sample sizes. Restoring that term, as in (191), and carrying it to R^2 through the exact identity $R^2 = t(\hat{\beta})^2/(\nu + t(\hat{\beta})^2)$, gives an approximation that is accurate in three of the four panels of Figures 7 and 9, and visibly shifted only at $\phi = 0.5$, $T = 960$, where $c = T(\phi - 1) = -480$ is far outside the regime the approximation is built for. The lesson is not that the local-to-unit-root framework fails under the alternative but that its published limits must not be truncated before they are used.

The fixed-ϕ approximation, by contrast, is genuinely strained here, and its performance is governed by the signal $T\rho^2$ rather than by T alone. At $T = 960$, where $T\rho^2 = 19$, the normal limit (183) describes the exact density of R^2 well at both persistences: the exact standard deviation exceeds the asymptotic one by only two percent at $\phi = 0.5$ and eight percent at $\phi = 0.99$, with a mild residual right skew. At $T = 120$, where $T\rho^2 = 2.4$, it fails, and it fails in two different ways. At $\phi = 0.5$ the exact density still has its mode at the origin—the sample is not informative enough for the distribution to have separated from

its null shape—while the normal limit places a symmetric bump around ρ^2. At $\phi = 0.99$ the exact density is broad and strongly right-skewed, with a standard deviation 75 percent larger than the asymptotic one and appreciable mass out to five times ρ^2. For $t(\hat{\beta})$ the fixed-ϕ approximation tracks the exact density reasonably at $\phi = 0.5$ and is visibly shifted at $\phi = 0.99$. This is the discontinuity of Section 6.1 in visible form: the alternative-hypothesis limits are good descriptions only once $T\rho^2$ is large, and at a population R^2 of two percent that takes a sample of roughly a thousand observations.

Neither asymptotic family is therefore reliable across the range of persistence encountered in practice, and the empirically most relevant region—ϕ near one with T in the hundreds to thousands—is precisely where the fixed-ϕ approximation is at its worst. The exact finite sample distribution resolves the difficulty uniformly: it requires no judgment about which asymptotic regime applies, and with the algorithms of this paper each panel of Figures 5–9 is computed in seconds (for b) to minutes (for $t(\hat{\beta})$ and R^2) at any sample size.

6.3. Size and Quantiles over a Grid

Density overlays are not sufficient for inference, and four configurations are not a grid. What inference uses is a quantile and a rejection probability, and both are available exactly. Table 3 reports, for the null $\tilde{\beta} = 0$, the exact probability that a nominal five percent two-sided test rejects when its critical value is taken from the fixed-ϕ normal approximation evaluated at the *true* ϕ, together with the ratio of the exact ninety-fifth percentile of b to that approximation's. Nothing is estimated, so the table isolates the error of the limiting approximation itself. It covers five values of ϕ including a negative one, four innovation correlations running from -0.95 to zero, and two sample sizes. Computing the whole of it takes six seconds.

Three things are visible that the four-panel comparison could not show. First, persistence rather than the innovation correlation is what creates the distortion: at $\rho_{uv} = 0$, where the estimator is exactly median-unbiased and the two tails contribute equally, the exact size at $\phi = 0.99$ and $T = 120$ is already 0.338. The correlation then compounds it, monotonically in $|\rho_{uv}|$ and to 0.582 at $\rho_{uv} = -0.95$, and it governs how the excess rejection is *split* between the tails: the total is symmetric in the sign of ρ_{uv}, since changing that sign reflects the

distribution of b about the origin, but at $\rho_{uv} = -0.95$ almost all of the 0.582 sits in the right tail. Second, negative persistence is harmless: at $\phi = -0.5$ the exact size never leaves $[0.049, 0.054]$ at either sample size, so the failure is a one-sided phenomenon in ϕ and not a symmetric small-sample effect. Third, the sample size buys much less than one would hope where it is needed: quadrupling T from 120 to 480 takes the worst cell from 0.582 only to 0.299, and the ninety-fifth percentile is still off by a factor of 2.77. A quantile that is wrong by a factor of five, as it is at $T = 120$, is not a mis-calibration to be absorbed into a robust standard error.

7. Conclusion

Exact finite sample inference of the OLS slope coefficient in predictive regressions has been available in principle since Stambaugh (1999), and that of the t-ratio and R^2 since Kan and Pan (2022), but not in practice: the integrands of the classical inversion formulas involve determinants and traces of large complex matrices, and for the t-ratio and R^2 these must be re-evaluated at every node of a triple integral. This paper removes the computational obstacle. We show that after an analytical reduction, every determinant and trace that arises is built from a tridiagonal Toeplitz matrix modified by a rank-one update and a low-dimensional border, so that each integrand can be evaluated with a fixed number of arithmetic operations, independent of the sample size. The result is that the exact density and distribution function of $\hat{\beta}$ cost a univariate integral whose integrand is essentially free, and those of $t(\hat{\beta})$ and R^2 cost triple integrals that remain a matter of minutes at any sample size—where the eigenvalue-based alternative is already projected in hours at $T = 240$.

Making the closed forms reliable in floating point required three further ingredients, each of which we expect to be useful beyond this paper. First, the closed-form primitives degenerate when the two roots of the characteristic equation coalesce—precisely the empirically relevant nearly-integrated region—and we restore uniform accuracy by recognizing every primitive as a rational function of two Chebyshev polynomials, which are entire, and evaluating those near the confluent point from their explicit Taylor expansions there, at $O(1)$ cost and with worst observed error below 5×10^{-10}. Second, the square root of a complex

Table 3: **Exact Size and Quantile Error over a Grid**

The table reports, for the null $\tilde{\beta} = 0$, the exact probability that a nominal five percent two-sided test of $H_0 : \beta = 0$ rejects when the critical value is taken from the fixed-ϕ normal approximation with the true ϕ, and the ratio $q_{.95}(b)_{\text{exact}}/q_{.95}(b)_{\text{asymptotic}}$ of the exact 95th percentile of b to the percentile of that approximation. Nothing is estimated. Changing the sign of ρ_{uv} reflects the distribution of b about the origin, so the size is unchanged and only non-positive values are shown; the entry the right panel would carry at $|\rho_{uv}|$ is the corresponding ratio for the fifth percentile.

	exact size				$q_{.95}(b)_{\text{exact}}/q_{.95}(b)_{\text{asymptotic}}$			
	ρ_{uv}				ρ_{uv}			
ϕ	-0.95	-0.80	-0.50	0.00	-0.95	-0.80	-0.50	0.00
$T = 120$								
-0.50	0.049	0.051	0.053	0.054	0.92	0.94	0.97	1.01
0.00	0.049	0.050	0.052	0.053	1.05	1.04	1.03	1.01
0.50	0.062	0.061	0.058	0.057	1.22	1.19	1.13	1.03
0.90	0.161	0.141	0.110	0.090	1.87	1.75	1.52	1.15
0.99	0.582	0.516	0.408	0.338	5.16	4.67	3.72	2.26
$T = 480$								
-0.50	0.050	0.050	0.051	0.051	0.96	0.97	0.98	1.00
0.00	0.050	0.050	0.051	0.051	1.02	1.02	1.02	1.00
0.50	0.053	0.053	0.052	0.052	1.11	1.09	1.06	1.01
0.90	0.082	0.075	0.066	0.060	1.38	1.32	1.21	1.04
0.99	0.299	0.259	0.193	0.150	2.77	2.53	2.08	1.39

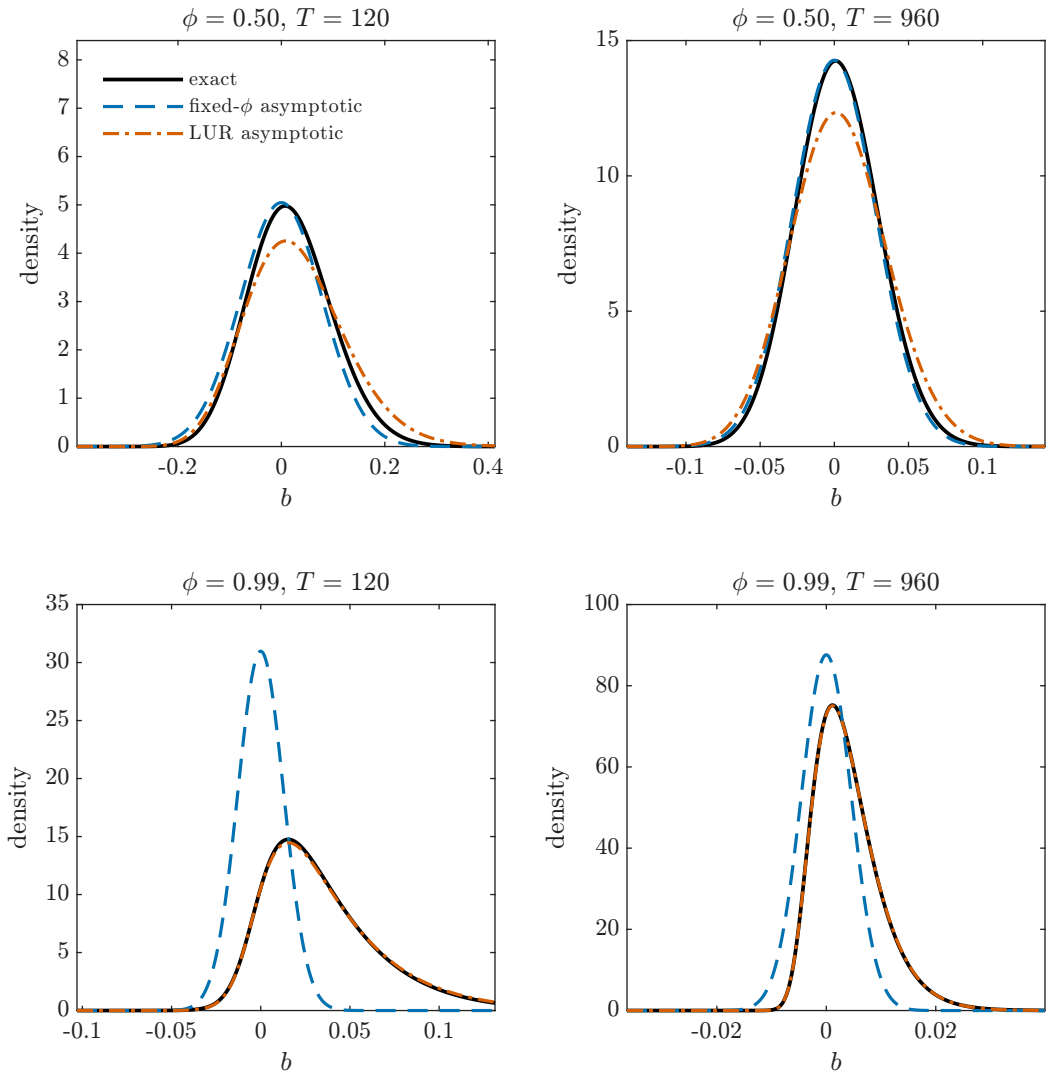


Figure 5: **Exact and Asymptotic Densities of** b

The figure plots the exact finite sample density of $b = (\sigma_v/\sigma_u)(\hat{\beta} - \beta)$ (solid line), the fixed-ϕ asymptotic approximation $\mathcal{N}(0, (1-\phi^2)/T)$ (dashed line), and the local-to-unit-root asymptotic approximation with $c = T(\phi - 1)$ (dash-dot line), for $\phi \in \{0.5, 0.99\}$ and $T \in \{120, 960\}$, with $\rho_{uv} = -0.856$ and $\beta = 0$. The LUR limiting distribution is simulated with 200,000 paths.

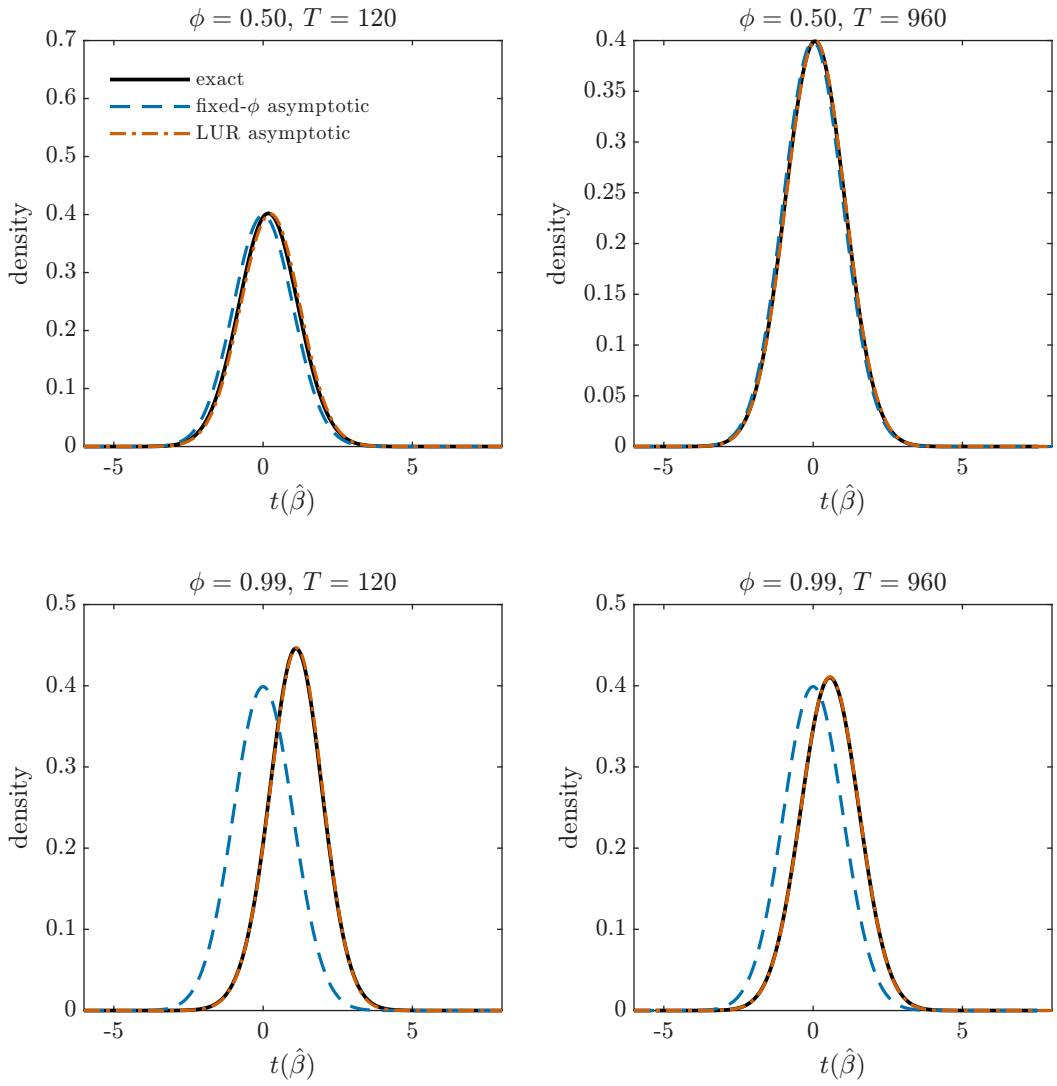


Figure 6: **Exact and Asymptotic Densities of $t(\hat{\beta})$ under the Null**

The figure plots the exact finite sample density of $t(\hat{\beta})$ (solid line), the fixed-ϕ asymptotic approximation $\mathcal{N}(0,1)$ (dashed line), and the local-to-unit-root asymptotic approximation with $c = T(\phi-1)$ (dash-dot line), for $\phi \in \{0.5, 0.99\}$ and $T \in \{120, 960\}$, with $\rho_{uv} = -0.856$ and $\beta = 0$. The LUR limiting distribution is simulated with 200,000 paths.

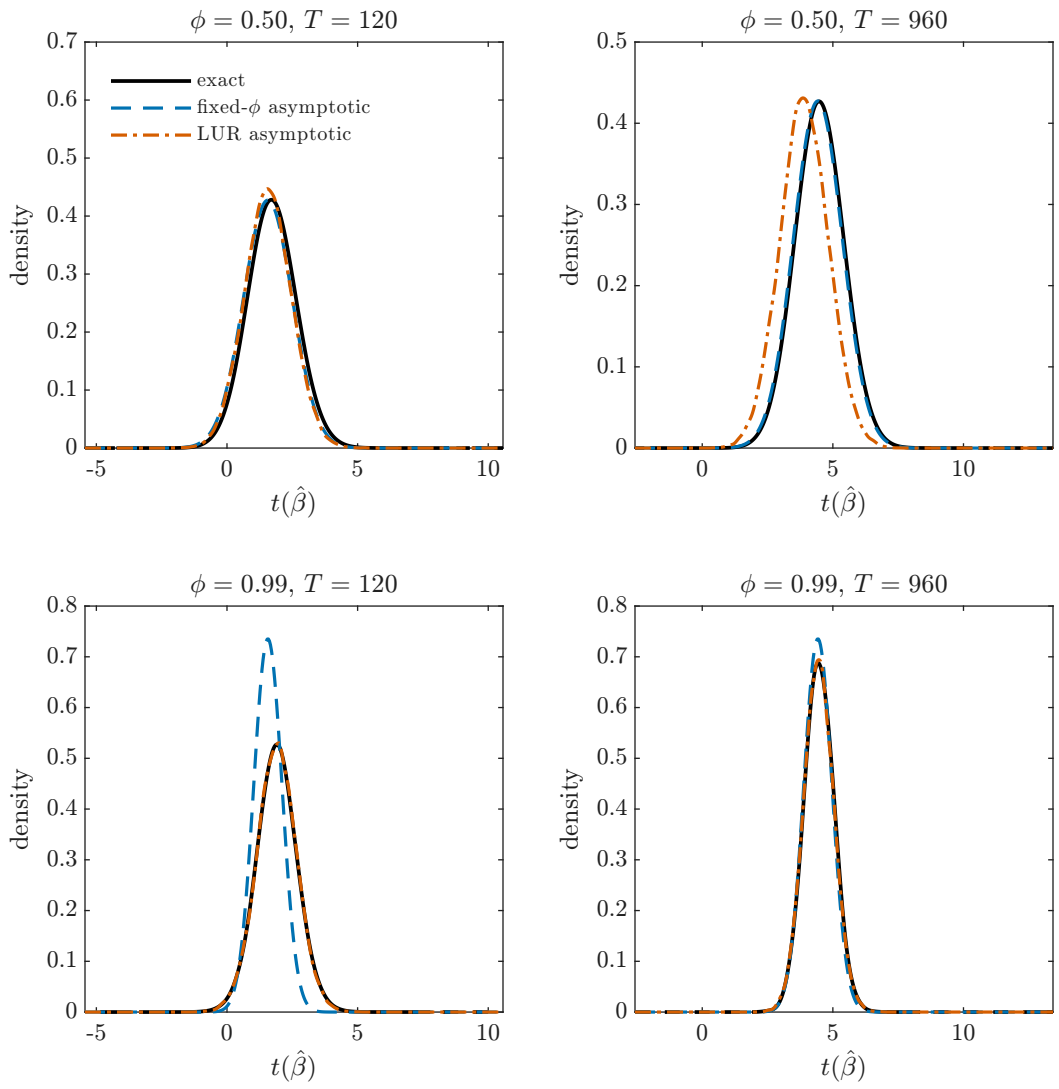


Figure 7: **Exact and Asymptotic Densities of $t(\hat{\beta})$ under the Alternative**

The figure plots the exact finite sample density of $t(\hat{\beta})$ (solid line), the fixed-ϕ asymptotic approximation $\mathcal{N}(\sqrt{T}\tilde{\beta}/\sqrt{1-\phi^2},\, V(t))$ of (181)–(185) (dashed line), and the local-to-unit-root asymptotic approximation (191) with $c = T(\phi - 1)$ and $\mathfrak{b} = T\tilde{\beta}$ (dash-dot line; the density of $\mathcal{T}_{\mathfrak{b},c}$, simulated with 200,000 paths). The degenerate fixed-alternative form (188) is not used here, for the reason given in Section 6.1. The configurations are $\phi \in \{0.5, 0.99\}$ and $T \in \{120, 960\}$, with $\rho_{uv} = -0.856$ and $\tilde{\beta}$ set so that the population R^2 is $\rho^2 = 0.02$ ($\tilde{\beta} = 0.1237$ for $\phi = 0.5$ and $\tilde{\beta} = 0.0202$ for $\phi = 0.99$).

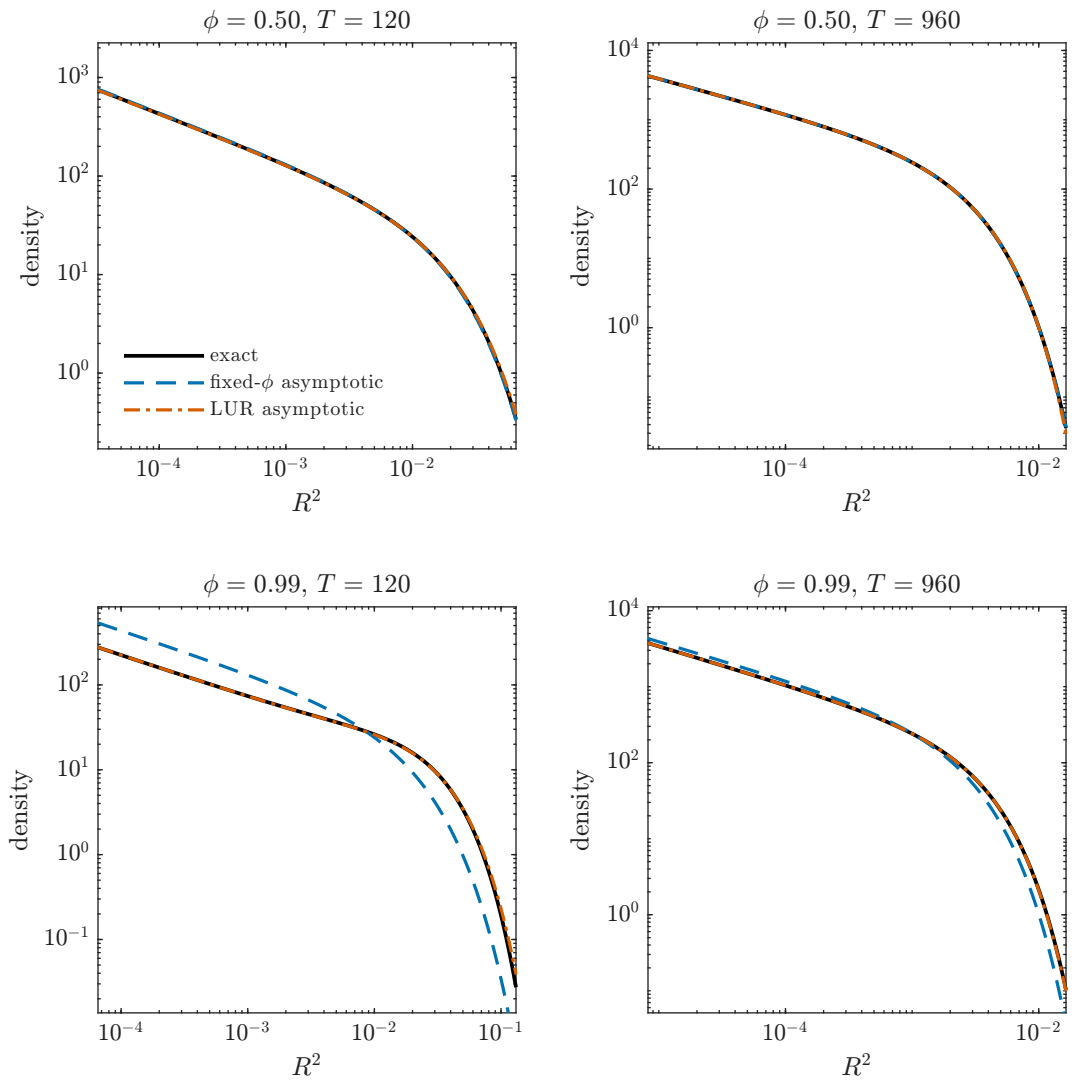


Figure 8: **Exact and Asymptotic Densities of R^2 under the Null**

The figure plots the exact finite sample density of R^2 (solid line), the fixed-ϕ asymptotic approximation (the density of χ^2_1/T, dashed line), and the local-to-unit-root asymptotic approximation with $c = T(\phi - 1)$ (dash-dot line), for $\phi \in \{0.5, 0.99\}$ and $T \in \{120, 960\}$, with $\rho_{uv} = -0.856$ and $\beta = 0$. The plotted range starts at the χ^2_1/T 0.5th percentile scale; the exact and asymptotic densities are both integrable but unbounded at $R^2 = 0$; both axes are therefore logarithmic, since on a linear scale the singularity at the origin would be the only visible feature. The LUR limiting distribution is simulated with 200,000 paths.

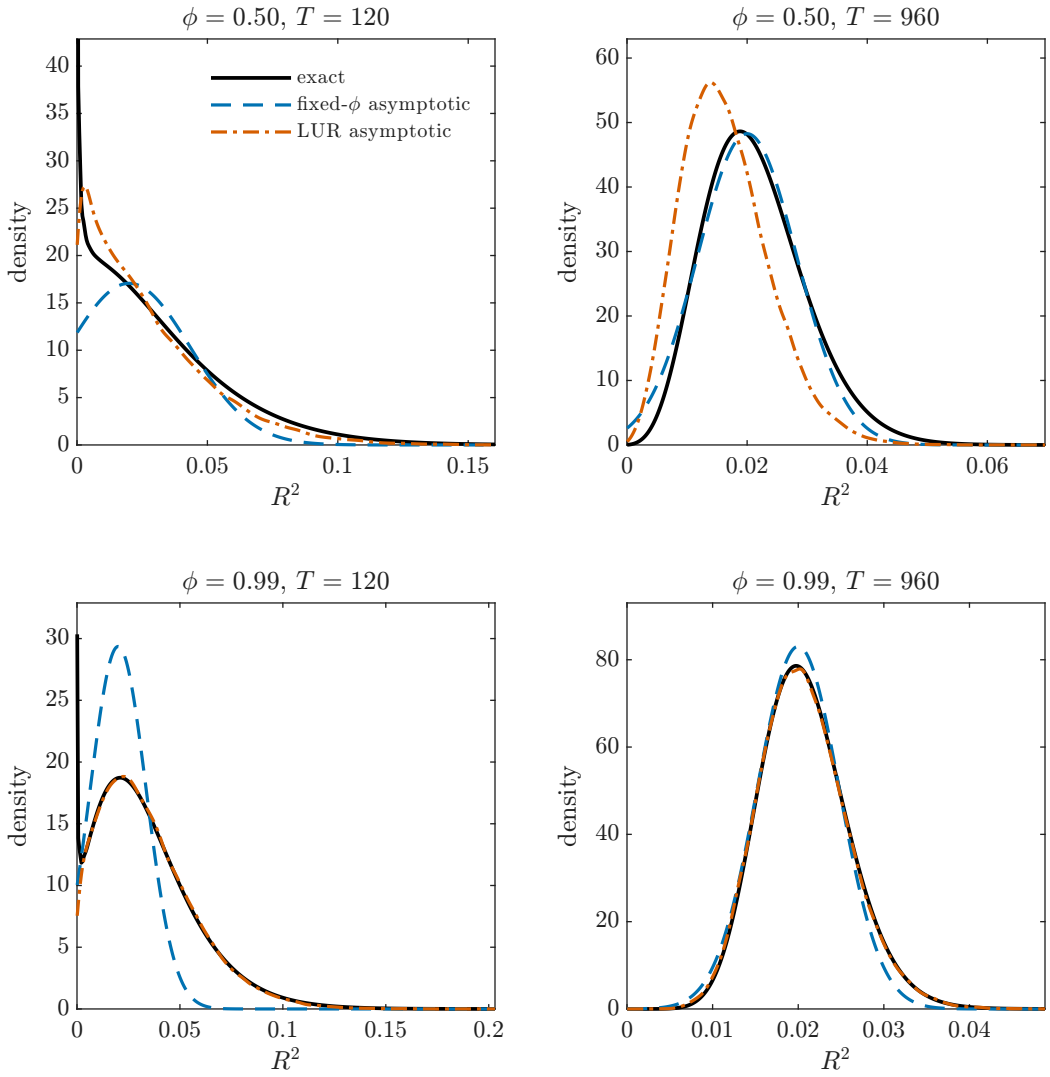


Figure 9: **Exact and Asymptotic Densities of R^2 under the Alternative**

The figure plots the exact finite sample density of R^2 (solid line), the fixed-ϕ asymptotic approximation $\mathcal{N}(\rho^2, V(R^2)/T)$ of (183)–(184) (dashed line), for $\phi \in \{0.5, 0.99\}$ and $T \in \{120, 960\}$, with $\rho_{uv} = -0.856$ and $\tilde{\beta}$ set so that the population R^2 is $\rho^2 = 0.02$ ($\tilde{\beta} = 0.1237$ for $\phi = 0.5$ and $\tilde{\beta} = 0.0202$ for $\phi = 0.99$). The dash-dot line is the local-to-unit-root approximation with $c = T(\phi - 1)$ and $\mathfrak{b} = T\tilde{\beta}$, obtained from (191) through $R^2 = t(\hat{\beta})^2/(\nu + t(\hat{\beta})^2)$ and simulated with 200,000 paths; the fixed-alternative form (190) is not usable here, for the reason given in Section 6.1. The axes are linear, unlike Figure 8, and the vertical range of each panel is set from the body of the distribution, so at $T = 120$ the exact density leaves the top of the box near the origin, where it is unbounded.

determinant must be taken continuously in the integration variable, and we *prove* that our assembled phase is continuous by exploiting the accretivity of the underlying matrices: Schur complements of accretive matrices are accretive, determinant ratios of rank-one updates stay off the branch cut by a convexity argument, and the two roots of the tridiagonal core are ordered with a discriminant that never meets the cut, because $p \pm 2q$ both lie in the open right half plane (Lemma 4). Since essentially every application of the inversion theorem to quadratic forms in normal variables faces the same branch-cut problem, and the prevailing practice is to detect crossings numerically, we regard this as a contribution of independent interest. Third, the outer integral must be truncated and its variable matched to a scale that drifts with T; the scale comes from the moments of a Cauchy–Schwarz bound, and the truncation point is certified by a Chernoff bound computed from the exact moment generating function, so that the discarded mass is bounded a priori rather than checked after the fact.

The exact distributions settle a question that asymptotic theory cannot: which approximation to trust. We find that the standard fixed-ϕ asymptotic is excellent for moderately persistent predictors and fails badly for the persistent predictors typical of return predictability studies, even at $T = 960$. The local-to-unit-root asymptotic repairs that failure and is the more accurate of the two whenever the predictor is genuinely persistent, but it must be evaluated by simulation, and under the alternative its published limits are rates of divergence that have to be restored to their $O(1)$ terms before they describe a distribution at all. Since the practitioner does not know which regime her data inhabit, and since choosing a framework means choosing a value of c or committing to a fixed ϕ, the exact finite sample distribution—now computable in seconds to minutes—is the natural default. That distribution does condition on ϕ, which near the unit root is not estimable at the usual rate, so a procedure that wants to be exact in ϕ as well must take a conservative envelope over a confidence set for it, in the manner of Cavanagh et al. (1995) and Campbell and Yogo (2006). What this paper supplies for such a procedure is the ingredient that made it expensive: an exact critical value at each ϕ in a grid, at negligible cost per point.

Two extensions seem within reach. For predictive regressions with overlapping long-

horizon returns, the exact distribution theory of Kan and Pan (2022) leads to matrices with the same Toeplitz-plus-low-rank flavor, but with bandwidth governed by the return horizon k rather than tridiagonal. The prospects there seem good: Trench (1985) gives the characteristic polynomial and the eigenvectors of a Toeplitz band matrix in terms of the zeros of a polynomial of degree equal to the bandwidth, with a complexity that depends on the bandwidth and *not* on the order of the matrix, so the primitives should again admit an evaluation whose cost grows with k but not with T. For regressions with multiple persistent predictors, the characteristic function involves block-Toeplitz structures for which analogous closed forms are less complete; developing them would bring exact inference to the multivariate predictive regressions that applied work increasingly employs.

Appendix A. Proofs

Proof of Proposition 1: By writing the cumulative distribution functions of $t(\hat{\beta})$ and R^2 as

$$\mathbb{P}[t(\hat{\beta}) < c] = \mathbb{P}\left[\frac{(T-2)^{\frac{1}{2}}\mathcal{R}_1}{(\mathcal{R}_2 - \mathcal{R}_1^2)^{\frac{1}{2}}} < c\right] = \int_0^\infty \int_{-\sqrt{r_2}}^{\frac{c\sqrt{r_2}}{\sqrt{T-2+c^2}}} f_{\mathcal{R}_1,\mathcal{R}_2}(r_1, r_2)\mathrm{d}r_1\mathrm{d}r_2, \tag{A.1}$$

$$\mathbb{P}[R^2 < c] = \mathbb{P}[\mathcal{R}_1^2 < \mathcal{R}_2 c] = \int_0^\infty \int_{-\sqrt{r_2 c}}^{\sqrt{r_2 c}} f_{\mathcal{R}_1,\mathcal{R}_2}(r_1, r_2)\mathrm{d}r_1\mathrm{d}r_2 \quad \text{for } 0 < c \leq 1, \tag{A.2}$$

we can differentiate them with respect to c to obtain the density functions of $t(\hat{\beta})$ and R^2 as

$$f_{t(\hat{\beta})}(c) = \frac{(T-2)}{(T-2+c^2)^{\frac{3}{2}}} \int_0^\infty \sqrt{r_2} f_{\mathcal{R}_1,\mathcal{R}_2}\left(\frac{c\sqrt{r_2}}{\sqrt{T-2+c^2}}, r_2\right) \mathrm{d}r_2, \tag{A.3}$$

$$f_{R^2}(c) = \frac{1}{2\sqrt{c}} \int_0^\infty \sqrt{r_2}[f_{\mathcal{R}_1,\mathcal{R}_2}(-\sqrt{cr_2}, r_2) + f_{\mathcal{R}_1,\mathcal{R}_2}(\sqrt{cr_2}, r_2)]\mathrm{d}r_2 \quad \text{for } 0 < c \leq 1. \tag{A.4}$$

□

Proof of Proposition 2: Instead of using (A.1) and (A.2) to compute the cumulative distribution functions of $t(\hat{\beta})$ and R^2 (which require evaluation of quadruple integrals), we introduce a function which is based on the derivative of $F_{\mathcal{R}_1,\mathcal{R}_2}(r_1, r_2)$ defined in (31):

$$\tilde{G}(r_1, r_2) = \frac{\partial F_{\mathcal{R}_1,\mathcal{R}_2}(r_1, r_2)}{\partial r_2} = \frac{f_{\mathcal{R}_2}(r_2)}{2} - \frac{1}{2\pi^2} \int_0^\infty \int_0^\infty \frac{\mathrm{Im}(g(s,t) + g(s,-t))}{s} \mathrm{d}s\mathrm{d}t, \tag{A.5}$$

where $g(s,t) = |\mathcal{D}|^{\frac{1}{2}}\mathrm{tr}(\mathcal{D}\tilde{B})$. Note that by differentiating (31) under the integral sign (valid because the differentiated integrand remains dominated by an integrable function of (s,t)

locally uniformly in r_2) and using the Leibniz integral rule, we can show that

$$\tilde{G}(r_1, r_2) = \int_{-\sqrt{r_2}}^{r_1} f_{\mathcal{R}_1,\mathcal{R}_2}(u, r_2)\mathrm{d}u = \mathbb{P}[\mathcal{R}_1 < r_1 | \mathcal{R}_2 = r_2] f_{\mathcal{R}_2}(r_2). \tag{A.6}$$

Using $\tilde{G}(r_1, r_2)$, we can obtain the cumulative distribution function of $t(\hat{\beta})$ using

$$\begin{aligned}
\mathbb{P}[t(\hat{\beta}) < c] &= \int_0^\infty \mathbb{P}\left[\mathcal{R}_1 < \frac{c\sqrt{r_2}}{\sqrt{T-2+c^2}} \middle| \mathcal{R}_2 = r_2\right] f_{\mathcal{R}_2}(r_2)\mathrm{d}r_2 \\
&= \int_0^\infty \tilde{G}\left(\frac{c\sqrt{r_2}}{\sqrt{T-2+c^2}}, r_2\right)\mathrm{d}r_2 \\
&= \int_0^\infty \left[\frac{f_{\mathcal{R}_2}(r_2)}{2} - G\left(\frac{c\sqrt{r_2}}{\sqrt{T-2+c^2}}, r_2\right)\right]\mathrm{d}r_2 \\
&= \frac{1}{2} - \int_0^\infty G\left(\frac{c\sqrt{r_2}}{\sqrt{T-2+c^2}}, r_2\right)\mathrm{d}r_2,
\end{aligned} \tag{A.7}$$

where

$$G(r_1, r_2) = \frac{1}{2\pi^2}\int_0^\infty \int_0^\infty \frac{\mathrm{Im}(g(s,t) + g(s,-t))}{s}\mathrm{d}s\mathrm{d}t. \tag{A.8}$$

Similarly, the cumulative distribution of R^2 can be obtained by using

$$\begin{aligned}
\mathbb{P}[R^2 < c] &= \int_0^\infty \mathbb{P}[\mathcal{R}_1^2 < r_2 c | \mathcal{R}_2 = r_2] f_{\mathcal{R}_2}(r_2)\mathrm{d}r_2 \\
&= \int_0^\infty [\tilde{G}(\sqrt{r_2 c}, r_2) - \tilde{G}(-\sqrt{r_2 c}, r_2)]\mathrm{d}r_2 \\
&= \int_0^\infty [G(-\sqrt{r_2 c}, r_2) - G(\sqrt{r_2 c}, r_2)]\mathrm{d}r_2.
\end{aligned} \tag{A.9}$$

□

Proof of Lemma 1: The lemma follows from the standard partitioned determinant and inverse formulas. Writing $\tilde{\mathcal{A}}$ for the leading $(n-1)\times(n-1)$ block of $\mathcal{A}$ and $\tilde{c} = [a_{1n},\ c']'$, we have

$$|\mathcal{A}| = |\tilde{\mathcal{A}}|\,(a_{nn} - \tilde{c}'\tilde{\mathcal{A}}^{-1}\tilde{c}), \tag{A.10}$$

$$|\tilde{\mathcal{A}}| = |\mathcal{A}_{22}|(a_{11} - b'\mathcal{A}_{22}^{-1}b), \tag{A.11}$$

$$\tilde{\mathcal{A}}^{-1} = \begin{bmatrix} 0 & 0'_{n-2} \\ 0_{n-2} & \mathcal{A}_{22}^{-1} \end{bmatrix} + \frac{1}{a_{11} - b'\mathcal{A}_{22}^{-1}b}\begin{bmatrix} 1 \\ -\mathcal{A}_{22}^{-1}b \end{bmatrix}\begin{bmatrix} 1 \\ -\mathcal{A}_{22}^{-1}b \end{bmatrix}', \tag{A.12}$$

from which $\tilde{c}'\tilde{\mathcal{A}}^{-1}\tilde{c} = c'\mathcal{A}_{22}^{-1}c + \delta_2^2/\delta_1$ and hence $|\mathcal{A}| = |\mathcal{A}_{22}|\delta_1(\delta_3 - \delta_2^2/\delta_1) = |\mathcal{A}_{22}|\delta_4$. The blocks of $\mathcal{A}^{-1}$ follow by applying the partitioned inverse formula once more. □

Proof of Lemma 2: Closed-form expressions of this type are classical (see, e.g., da Fonseca and Petronilho, 2001); we include the short derivation for completeness. The determinants $|D_j|$ of the leading principal submatrices satisfy the recursion $|D_{j+1}| = p|D_j| - q^2|D_{j-1}|$ with $|D_0| = 1$ and $|D_1| = p$; when $p \neq \pm 2q$ the characteristic roots $r_\pm$ of the recursion are distinct and (109) follows. For (110) and (111), the cofactor expansion of D^{-1} gives

$$e'_{1,m} D^{-1} e_{1,m} = \frac{|D_{m-1}|}{|D_m|}, \qquad e'_{1,m} D^{-1} e_{m,m} = \frac{(-q)^{m-1}}{|D_m|}, \tag{A.13}$$

which simplify to the stated forms. For (112) and (113), note the row-sum identity

$$D 1_m = (p + 2q) 1_m - q(e_{1,m} + e_{m,m}). \tag{A.14}$$

Multiplying (A.14) by $e'_{1,m} D^{-1}$ gives $1 = (p+2q)d_2 - q(d_3 + d_4)$, which is (112); multiplying by $1'_m D^{-1}$ and using persymmetry gives $m = (p + 2q)d_1 - 2qd_2$, which is (113). □

The Chebyshev reduction (Section 3.3.1): The passage from the determinant recursion (114) to (122) is carried out in the text. Comparing (122) with (109), $(r_+^{m+1} - r_-^{m+1})/h = q^m U_m(x)$ and likewise $(r_+^m - r_-^m)/h = q^{m-1} U_{m-1}(x)$, and substituting into (110)–(111) gives the first two forms of (123). For (124), the values $U_m(-1) = (-1)^m(m+1)$ follow from $U_m(\cos\theta) = \sin((m+1)\theta)/\sin\theta$ as $\theta \to \pi$; hence $U_m + U_{m-1} + (-1)^{m+1}$ vanishes at $x = -1$ and the factor theorem yields the first identity. Differentiating it at $x = -1$ gives $Q(-1) = U'_m(-1) + U'_{m-1}(-1) = (-1)^{m+1} m(m+1)$ (the derivative values follow from (125) at $k = 0$), so $m\,U_m + Q$ vanishes at $x = -1$ as well, and the factor theorem yields the second identity; the last two forms of (123) are then (112)–(113) with $p + 2q = 2q(x + 1)$ canceled. For the coefficients, substitute $U_m = \sum_{k=0}^{\infty} a_k \delta^k$, $\delta = x + 1$, into the Chebyshev equation $(1 - x^2)U''_m - 3xU'_m + m(m+2)U_m = 0$, whose leading coefficient is $(2-\delta)\delta$ at $x = -1+\delta$; collecting the coefficient of δ^k gives

$$(k+1)(2k+3)\, a_{k+1} = -\left[m(m+2) - k(k+2)\right] a_k = -(m-k)(m+k+2)\, a_k, \tag{A.15}$$

which is (125). The identities (124) shift the coefficient index by one, and term-by-term differentiation of the same series supplies U'_m and U''_m, hence, through $D^{-2} = -\partial D^{-1}/\partial p$ and $2D^{-3} = \partial^2 D^{-1}/\partial p^2$, every second- and third-order primitive on the band. □

Spectral representations of the primitives (Section 3.3.1): Several statements of Sections 3.3.1 and 3.3.2—the location of the primitives' poles, the weight each primitive places on the vanishing eigenvalue, the product form of W, and the arbitrary-precision reference of Figure 1—rest on the exact representation of each primitive as a weighted sum over the eigenvalues of D. The eigenvalues and eigenvectors of an $m \times m$ symmetric tridiagonal Toeplitz matrix with diagonal p and off-diagonal q are classical (Trench, 1985, Section 4): with $n = m + 1$ and $\delta_j = 2\sin^2(\frac{j\pi}{2n})$,

$$\lambda_j = p + 2q\cos\left(\frac{j\pi}{n}\right) = 2q(\delta - \delta_j), \qquad 1 \le j \le m, \tag{A.16}$$

$$u_j = \sqrt{\frac{2}{n}}\left[\sin\left(\frac{j\pi}{n}\right),\ \sin\left(\frac{2j\pi}{n}\right),\ \ldots,\ \sin\left(\frac{mj\pi}{n}\right)\right]'. \tag{A.17}$$

Substituting the spectral decomposition $D^{-\ell} = \sum_{j=1}^{m} \lambda_j^{-\ell} u_j u_j'$ into each primitive, the inner products $1_m' u_j$, $e_{1,m}' u_j$, and $e_{m,m}' u_j$ evaluate in closed form; in particular $\sum_{k=1}^{m} \sin(\frac{kj\pi}{n})$ equals $\cot(\frac{j\pi}{2n})$ for odd j and 0 for even j, which is why only odd-indexed eigenvalues enter the sums involving 1_m. Since $D = pI_m + q\Lambda_0$ with Λ_0 the fixed 0–1 tridiagonal matrix, the eigenvectors u_j do not depend on p or q; the weights below are therefore the same for every power, and the representations hold for every integer $\ell \ge 1$ (indeed, with $\lambda_j^{-\ell}$ replaced by $f(\lambda_j)$, for any function f defined on the spectrum):

$$1_m' D^{-\ell} 1_m = \sum_{j=1}^{\lfloor (m+1)/2 \rfloor} \frac{2}{n}\cot^2\left(\frac{(2j-1)\pi}{2n}\right)\lambda_{2j-1}^{-\ell}, \tag{A.18}$$

$$e_{1,m}' D^{-\ell} 1_m = \sum_{j=1}^{\lfloor (m+1)/2 \rfloor} \frac{4}{n}\cos^2\left(\frac{(2j-1)\pi}{2n}\right)\lambda_{2j-1}^{-\ell}, \tag{A.19}$$

$$e_{1,m}' D^{-\ell} e_{1,m} = \sum_{j=1}^{m} \frac{2}{n}\sin^2\left(\frac{j\pi}{n}\right)\lambda_j^{-\ell}, \tag{A.20}$$

$$e_{1,m}' D^{-\ell} e_{m,m} = \sum_{j=1}^{m} (-1)^{j-1}\frac{2}{n}\sin^2\left(\frac{j\pi}{n}\right)\lambda_j^{-\ell}, \tag{A.21}$$

$$\operatorname{tr}(D^{-\ell}) = \sum_{j=1}^{m} \lambda_j^{-\ell}, \qquad |D| = \prod_{j=1}^{m} \lambda_j. \tag{A.22}$$

Setting $\ell = 1$ recovers d_1–d_4; $\ell = 2$ and $\ell = 3$ give the higher-order primitives used for the density and for the joint problem of Section 4, and no higher order is needed here. The last

representation, in the form $\log|D| = \sum_{j=1}^{m} \log \lambda_j$, furnishes the arbitrary-precision reference values used in Figure 1; factoring out its value at $\delta = 0$ gives $|D| = (-q)^m\, n \prod_{j=1}^{m}(1 - \delta/\delta_j)$, the product whose argument (A.29) bounds. □

Correctness of the phase decomposition (Section 3.3.2): We first construct the rigorous reference branch. By (78),

$$|C(s,-cs)| = (1-\phi^2)\left|I_{2T} - 2\mathrm{i}s(\tilde{A} - c\tilde{B})\right| = (1-\phi^2)\prod_{j=1}^{2T}(1 - 2\mathrm{i}s\lambda_j^*), \tag{A.23}$$

where $\lambda_1^*, \ldots, \lambda_{2T}^*$ are the eigenvalues of the *real symmetric* matrix $\tilde{A} - c\tilde{B}$. Every factor $1 - 2\mathrm{i}s\lambda_j^*$ has real part equal to one, so each principal square root $(1-2\mathrm{i}s\lambda_j^*)^{-1/2}$ is continuous in s, and

$$\Phi_0(s) = (1-\phi^2)^{-\frac{1}{2}} \prod_{j=1}^{2T}(1 - 2\mathrm{i}s\lambda_j^*)^{-\frac{1}{2}} \tag{A.24}$$

is a continuous square root of $1/|C(s,-cs)|$ that equals $(1-\phi^2)^{-1/2}$ at $s = 0$: this is the classical construction of Imhof (1961), and it is the branch that defines the correct integrand. Since two continuous pointwise square roots of the same nonvanishing continuous function that agree at one point agree everywhere on a connected domain, the assembly (128) is correct if and only if it is continuous in s; the question is therefore reduced to the continuity of each term of (128).

For the terms coming from $1/|D|$ everything follows from a single elementary fact about the two combinations $p \pm 2q$: that they lie in the open right half plane. We record the consequences once, for arbitrary p and q, and afterward verify that one hypothesis in each of the two problems the paper treats.

Lemma 3. *Let p and q be complex numbers for which $a_+ = p+2q$ and $a_- = p-2q$ both have positive real part, let $h = (p^2 - 4q^2)^{1/2}$ be the principal square root, and let $r_\pm = (p \pm h)/2$. Then*

(i) $p^2 - 4q^2 = a_+ a_-$ does not lie on $(-\infty, 0]$, so h is nonvanishing and $h = a_+^{1/2} a_-^{1/2}$ with both factors principal;

(ii) $\mathrm{Re}(h) > 0$ and $\mathrm{Re}(r_+) > 0$;

(iii) $\mathrm{Re}(p\,\bar{h}) > 0$, *equivalently* $|r_-| < |r_+|$ *strictly.*

Consequently, along any path on which a_+ *and* a_- *vary continuously and remain in the open right half plane,* h *and* r_+ *are continuous and nonvanishing and* $\arg(r_+)$ *is continuous with values in* $(-\pi/2, \pi/2)$.

Proof. Both $\arg a_+$ and $\arg a_-$ lie in $(-\pi/2, \pi/2)$ and $|a_+ a_-| > 0$. Then $p^2 - 4q^2 = a_+ a_-$ has $\arg(a_+ a_-) = \arg a_+ + \arg a_- \in (-\pi, \pi)$, so $a_+ a_-$ avoids the cut $(-\infty, 0]$ of the principal root; the principal root is continuous and nonvanishing off that cut, and $\arg h = \frac{1}{2}(\arg a_+ + \arg a_-)$, which is (i). Since $|\arg h| < \pi/2$ we have $\mathrm{Re}(h) > 0$, and with $\mathrm{Re}(p) = \frac{1}{2}[\mathrm{Re}(a_+) + \mathrm{Re}(a_-)] > 0$ this gives $\mathrm{Re}(r_+) = \frac{1}{2}[\mathrm{Re}(p) + \mathrm{Re}(h)] > 0$, which is (ii). For (iii), $p = (a_+ + a_-)/2$ is a positive combination of two vectors whose arguments lie in an interval of length $|\arg a_+ - \arg a_-| < \pi$, so $\arg p$ lies between them, while $\arg h$ is the midpoint of that interval; hence

$$|\arg p - \arg h| \;\le\; \tfrac{1}{2}\,|\arg a_+ - \arg a_-| \;<\; \frac{\pi}{2}, \tag{A.25}$$

so $\mathrm{Re}(p\,\bar{h}) = |p|\,|h| \cos(\arg p - \arg h) > 0$. Finally $|r_+|^2 - |r_-|^2 = \mathrm{Re}(p\,\bar{h})$, so $|r_-| < |r_+|$. The last statement holds because h and r_+ are continuous functions of (a_+, a_-) on that region, by (i), and $\mathrm{Re}(r_+) > 0$ there. □

For the distribution of b the hypothesis is the following identity.

Lemma 4. *Let* p *and* q *be as in (90), with* $|\phi| < 1$, $|\rho_{uv}| < 1$ *and* (s, t) *real. Then*

$$p \pm 2q = (1 \pm q)^2 + s^2 - 2\mathrm{i}t = (1 \mp \phi)^2 + (1 - \rho_{uv}^2)s^2 \mp 2\mathrm{i}\,[(1 \mp \phi)\rho_{uv} s \pm t]\,, \tag{A.26}$$

so that $\mathrm{Re}(p \pm 2q) = (1 \mp \phi)^2 + (1 - \rho_{uv}^2)s^2 > 0$ *and the conclusions of Lemma 3 hold for* (p, q) *at every point of the integration path.*

Proof. A direct computation from (90) gives $q^2 = \phi^2 - \rho_{uv}^2 s^2 + 2\mathrm{i}\phi\rho_{uv} s$ and hence the identity $p = 1 + q^2 + s^2 - 2\mathrm{i}t$, from which (A.26) follows. The real parts are strictly positive for $|\phi| < 1$ and $|\rho_{uv}| < 1$. □

The eigenvalues of D are $\lambda_j = p + 2q\cos(j\pi/n)$, $j = 1, \ldots, m$, and along the integration path

$$\mathrm{Re}(\lambda_j) = 1 + \phi^2 + (1 - \rho_{uv}^2)s^2 - 2\phi\cos(j\pi/n) \ge (1 - |\phi|)^2 + (1 - \rho_{uv}^2)s^2 > 0, \tag{A.27}$$

which is (A.26) at $\cos(j\pi/n) = \pm 1$ together with the linearity of $\mathrm{Re}(\lambda_j)$ in $\cos(j\pi/n)$. By Lemma 3(i)–(ii), $h(s)$ is continuous and $\mathrm{Re}(r_+(s)) > 0$, so $\arg(r_+(s))$ is a continuous function taking values in $(-\pi/2, \pi/2)$ and the accumulated term $(m+1)\arg(r_+)$ is continuous. By Lemma 3(iii) the labeling with $|\zeta| = |r_-/r_+| < 1$ is the one the principal root already delivers—no exchange of the roots is ever required—and the factor $1-\zeta^{m+1}$ satisfies $\mathrm{Re}(1-\zeta^{m+1}) \geq 1 - |\zeta|^{m+1} > 0$, so its principal argument is continuous as well. Note that h never vanishes, so the roots never actually coalesce on the integration path; the Chebyshev evaluation of Section 3.3.1 is invoked near $\delta = 0$ to control the numerical cancellation described there, not to repair a branch, and its phase is in any case continuous by the product bound (A.29) below.

For the confluent regime of Section 3.3.1, where $1/|D|$ is evaluated through the expansions (125), the argument is the following. The eigenvalues of D are $\lambda_j = p + 2q\cos(j\pi/n) = 2q(\delta - \delta_j)$, where $\delta_j = 1 - \cos(j\pi/n) = 2\sin^2(j\pi/(2n))$ are the values of δ at which they vanish, so $|D| = \prod_{j=1}^m \lambda_j = (2q)^m \prod_{j=1}^m (\delta - \delta_j)$. At the confluent point this product equals $n(-q)^m$, and factoring that value out of it gives

$$|D| = n\,(-q)^m\, W, \qquad W = \prod_{j=1}^{m} \left(1 - \frac{\delta}{\delta_j}\right), \qquad W|_{\delta=0} = 1, \tag{A.28}$$

and on the band

$$|\arg W| \;\leq\; \sum_{j=1}^{m} \left|\log\left(1 - \frac{\delta}{\delta_j}\right)\right| \;\leq\; -\sum_{j=1}^{m} \log\left(1 - \frac{5/2}{n^2\delta_j}\right) \;<\; 1.1 \;<\; \pi, \tag{A.29}$$

since $n^2\delta_1 \geq 4$ and $n^2\delta_j$ grows like j^2. The principal argument of W is therefore the continuous determination, and $-[\,m\arg(-q) + \arg W\,]$ is the continuous phase of $1/|D|$, $\arg(-q)$ itself being continuous on the band because $-q = \phi + \mathrm{i}\rho_{uv}s$ has real part bounded away from zero there (the band requires $p + 2q \approx 0$, hence ϕ near one). Each regime therefore carries its own continuity proof, and switching between them cannot introduce a jump: the two pointwise formulas agree a priori up to a locally constant multiple of 2π on the overlap $5/2 \leq |\tau| < \tau^*$, and the multiple is zero because both phases vanish at a real positive reference point.

The remaining two terms of (128)—the arguments of the rank-one factor $1 - \kappa d_1$ and of the border factor of (126)—are covered by Proposition 4, which we now prove.

Proof of Proposition 4: (i) For $t \in [0,1]$ write $\Delta(t) = D - t\kappa 1_m 1_m' = (1-t)D + t\Delta$. The interior principal block of $\operatorname{Re} C$ is $\operatorname{Re}\Delta$, so $\operatorname{Re}\Delta \succ 0$, and by hypothesis $\operatorname{Re} D \succ 0$; hence $\operatorname{Re}\Delta(t) = (1-t)\operatorname{Re} D + t\operatorname{Re}\Delta \succ 0$ for every $t \in [0,1]$. A matrix with positive definite real part is nonsingular, since $\Delta(t)x = 0$ with $x \neq 0$ would give $0 = \operatorname{Re}(x^*\Delta(t)x) = x^*[\operatorname{Re}\Delta(t)]x > 0$. By the matrix determinant lemma,

$$\frac{|\Delta(t)|}{|D|} = 1 - t\kappa\, 1_m' D^{-1} 1_m = 1 - t\kappa d_1, \tag{A.30}$$

a linear function of t that equals 1 at $t = 0$ and does not vanish on $[0,1]$. If $1 - \kappa d_1$ were real and nonpositive, then κd_1 would be real with $\kappa d_1 \geq 1$, and $t^* = 1/(\kappa d_1) \in (0,1]$ would give $|\Delta(t^*)| = 0$, a contradiction. Hence $1 - \kappa d_1 \notin (-\infty, 0]$.

(ii) Order the rows and columns of C so that the two border indices come last,

$$C = \begin{pmatrix} \Delta & E \\ E' & C_b \end{pmatrix}, \qquad C_b = \begin{pmatrix} c_{1,1} & c_{1,T} \\ c_{1,T} & c_{T,T} \end{pmatrix}, \qquad E = (a,\ b), \tag{A.31}$$

so that the border factor of (126) is $|S|$ with $S = C_b - E'\Delta^{-1}E$, the Schur complement of Δ in C. For any matrix A with $\operatorname{Re} A \succ 0$ one has $\operatorname{Re}(A^{-1}) = (A^{-1})^*(\operatorname{Re} A)A^{-1} \succ 0$, and principal submatrices inherit positive definite real part; applying these two facts to $S^{-1} = (C^{-1})_{bb}$ (the border principal block of C^{-1}) gives $\operatorname{Re}(S^{-1}) \succ 0$ and hence $\operatorname{Re} S \succ 0$. The eigenvalues μ_1, μ_2 of S lie in its numerical range $\{x^*Sx/x^*x : x \neq 0\}$, whose real part is positive, so $\arg(\mu_i) \in (-\pi/2, \pi/2)$ for $i = 1, 2$ and $\arg(\mu_1) + \arg(\mu_2) \in (-\pi, \pi)$. Since $|S| = \mu_1\mu_2 \neq 0$ and its principal argument equals $\arg(\mu_1) + \arg(\mu_2)$ whenever that sum lies in $(-\pi, \pi]$, we conclude $|S| \notin (-\infty, 0]$. □

First, a remark on part (ii): it uses only the accretivity of C, so it applies verbatim to the joint problem of Section 4, whose matrix $F(s,t)$ is accretive along the integration path by the same Schur-complement inheritance from the underlying $2T$-dimensional characteristic-function matrix. Part (i) additionally requires $\operatorname{Re} D \succ 0$, that is, $\operatorname{Re}(\lambda_j) > 0$ for the m eigenvalues of the tridiagonal Toeplitz core. For the distribution of b this is exactly (A.27). For the joint problem it holds as well, by the following analog of Lemma 4.

Lemma 5. *Let p_F and q_F be as in (146)–(147), with $|\rho_{uv}| < 1$ and s, t, r_1, r_2, $\tilde{\beta}$ real. Then, writing $w = 1 - \rho_{uv}^2$,*

$$\mathrm{Re}(p_F \pm 2q_F) \;=\; (1 \mp \phi)^2 + \frac{w\,(s - 2\alpha_\pm t)^2}{1 + 4w^2t^2}, \tag{A.32}$$

where

$$\alpha_+ = -[\rho_{uv}(1-\phi) + \tilde{\beta}], \qquad \alpha_- = \rho_{uv}(1+\phi) - \tilde{\beta}, \tag{A.33}$$

so that $\mathrm{Re}(p_F + 2q_F) \geq (1-\phi)^2$ and $\mathrm{Re}(p_F - 2q_F) \geq (1+\phi)^2$ uniformly in s, t, r_1, r_2 and $\tilde{\beta}$. Consequently $\min_j \mathrm{Re}(\lambda_j) > \mathrm{Re}(p_F) - 2|\mathrm{Re}(q_F)| = \min_\pm \mathrm{Re}(p_F \pm 2q_F) \geq (1 - |\phi|)^2 > 0$ for $|\phi| < 1$, and $\mathrm{Re}\, D \succ 0$ for every T.

Proof. The term $2\mathrm{i}(sr_1 + tr_2)$ in (146) is purely imaginary and drops from every real part. From the definitions of η, ω, k_1, k_4 and λ_1,

$$p_F \pm 2q_F = \frac{\eta \pm 2\omega}{w} - \frac{(k_1 \pm k_4)^2}{\lambda_1}, \tag{A.34}$$

$$\eta \pm 2\omega = [(1 \mp \phi) \pm \tilde{\beta}\rho_{uv}]^2 + \tilde{\beta}^2 w \;=:\; A_\pm, \tag{A.35}$$

$$k_1 \pm k_4 = \frac{\alpha_\pm}{w} - \mathrm{i}s, \tag{A.36}$$

and expanding both squares gives the identity $A_\pm - \alpha_\pm^2 = (1 \mp \phi)^2\, w$. For real a, s, t and $L > 0$,

$$\mathrm{Re}\left[\frac{(a - \mathrm{i}s)^2}{L - 2\mathrm{i}t}\right] = \frac{L(a^2 - s^2) + 4ast}{L^2 + 4t^2}, \tag{A.37}$$

so with $a = \alpha_\pm / w$, $L = 1/w$, and $Lw = 1$,

$$\mathrm{Re}(p_F \pm 2q_F) = \frac{L\left[(A_\pm - \alpha_\pm^2)(L^2 + 4t^2) + (s - 2\alpha_\pm t)^2\right]}{L^2 + 4t^2} = (1 \mp \phi)^2 + \frac{L\,(s - 2\alpha_\pm t)^2}{L^2 + 4t^2}, \tag{A.38}$$

where the numerator was completed to a square using $s^2 - 4\alpha_\pm st + 4A_\pm t^2 = (s - 2\alpha_\pm t)^2 + 4t^2(A_\pm - \alpha_\pm^2)$. Since $L/(L^2 + 4t^2) = w/(1 + 4w^2t^2)$, this is (A.32). □

The relation of Lemma 5 to Lemma 4 is containment, not analogy. At $t = 0$ and $\tilde{\beta} = 0$ the joint problem specializes to the marginal problem of b—the second quadratic form drops out and $X_1 = z'Mr - r_1\, z'Mz$ is $U - r_1 W$—and a direct computation from (144)–(147) gives exactly $p_F = p$ and $q_F = q$ of (90), with the t of (90) equal to $-sr_1$. Lemma 4 is therefore the $t = 0$ case of (A.32), the two computations being one identity read in different coordinates;

and for $t \neq 0$ the identity says that, in its real part, the core of the joint problem is the core of the b problem with s replaced by $(s - 2\alpha_{\pm}t)/\sqrt{1 + 4w^2t^2}$. Since $\mathrm{Re}(p_F \pm 2q_F) > 0$, Lemma 3 applies to (p_F, q_F) exactly as it applies to (p, q), and together with Proposition 4, whose hypotheses are now verified for the joint problem, every factor of the assembled phase of $|F(s,t)|$ is continuous: the joint problem stands on the same footing as the distribution of b.

Finally, for the special case $\rho_{uv} = 0$ the positivity of the factors at $s = 0$ can also be seen directly, e.g. $1 - a'\Delta^{-1}a|_{s=0} = (1 - \phi^2)/(1 - \phi^{2n}) > 0$ with $n = T - 1$.

The following result underlies the optional pivot evaluation of the continuous phase,

$$\xi(s) = -\frac{1}{2}\sum_{k=1}^{T} \arg\big(p_k(s)\big), \qquad p_k = \frac{|C_k|}{|C_{k-1}|}, \tag{A.39}$$

where C_k is the k-th leading principal submatrix of C.

Proposition 9. *Let $\{C(s)\}_{s\geq 0}$ be a continuous family of complex symmetric $T \times T$ matrices with $\mathrm{Re}\, C(s) \succ 0$ for every s and $C(0)$ real, and let $\mu_1(s), \ldots, \mu_T(s)$ denote the eigenvalues of $C(s)$ and $p_k(s) = |C_k(s)|/|C_{k-1}(s)|$ its LU pivots, where C_k is the k-th leading principal submatrix ($|C_0| \equiv 1$). Then: (i) every $\mu_j(s)$ and every $p_k(s)$ lies in the open right half plane, so all their principal arguments lie in $(-\pi/2, \pi/2)$; (ii) $\sum_{j=1}^{T} \arg(\mu_j(s)) = \sum_{k=1}^{T} \arg(p_k(s))$ for every s; and (iii) this common value is the unique continuous determination of the phase of $|C(s)|$ that vanishes at $s = 0$. Consequently $|C(s)|^{-1/2} = \exp\{-\frac{1}{2}\sum_{k=1}^{T}[\log|p_k(s)| + \mathrm{i}\arg(p_k(s))]\}$ is the continuous square root that is positive at $s = 0$.*

Proof of Proposition 9: (i) For the eigenvalues, if $Cx = \mu x$ with $x^*x = 1$ then $\mathrm{Re}(\mu) = \mathrm{Re}(x^*Cx) = x^*(\mathrm{Re}\, C)x > 0$. For the pivots, note first that each C_k is accretive (its Hermitian part is a principal submatrix of $\mathrm{Re}\, C$) and hence invertible, so the LU factorization exists; moreover, for any invertible M with $\mathrm{Re}\, M \succ 0$, $\mathrm{Re}(M^{-1}) = (M^{-1})^*(\mathrm{Re}\, M)M^{-1} \succ 0$, and by the standard identity $p_k = 1/[(C_k^{-1})_{kk}]$ we get $\mathrm{Re}[(C_k^{-1})_{kk}] > 0$ and hence $\mathrm{Re}(p_k) > 0$. (ii) Deform $C(s)$ to its Hermitian part along $C^{(t)} = t\, C(s) + (1-t)\,\mathrm{Re}\, C(s)$, $t \in [0,1]$; every $C^{(t)}$ is accretive because $\mathrm{Re}\, C^{(t)} = \mathrm{Re}\, C(s)$ for all t. Both $\sum_{j=1}^{T} \arg(\mu_j)$ and $\sum_{k=1}^{T} \arg(p_k)$ are continuous in t by part (i) (each summand is a principal argument of a quantity confined

to the open right half plane, and eigenvalue sums are continuous under symmetric-function labeling), both are determinations of the phase of $|C^{(t)}|$ and hence differ by an element of $2\pi\mathbb{Z}$, and both vanish at $t = 0$ (where $C^{(0)}$ has positive real eigenvalues and positive real pivots). A continuous $2\pi\mathbb{Z}$-valued function vanishing at an endpoint is identically zero, so the two sums agree at $t = 1$. (iii) The same argument applied along the path parameter s shows that $s \mapsto \sum_{k=1}^{T} \arg(p_k(s))$ is continuous and vanishes at $s = 0$; any two continuous determinations of the phase of $|C(s)|$ differ by a constant element of $2\pi\mathbb{Z}$, which is zero here. □

The hypothesis of Proposition 9 holds along all the integration paths of this paper: for the distribution of b, $\operatorname{Re} C(s, -cs) = H^{-1} + s^2 M (V^{11})^{-1} M \succ 0$ by (129); for the joint problem of Section 4, the $2T \times 2T$ matrix $V_y^{-1} - 2is(A - r_1 B) - 2it(C - r_2 B)$ has Hermitian part $V_y^{-1} \succ 0$, and accretivity passes to the blocks $V^{11} - 2itM$ (Hermitian part V^{11}) and to the Schur complement $F(s,t)$, since $\operatorname{Re}(M^{-1}) \succ 0$ implies $\operatorname{Re}[(M^{-1})_{22}] \succ 0$ and F^{-1} is precisely the corresponding block of the inverse. Because $C(s, -cs)$ (and likewise $F(s,t)$) is tridiagonal up to corrections of rank at most three, its pivots can be computed by scalar forward recursions in $O(T)$ operations per point, giving a certified continuous branch at $O(T)$ cost. Our implementation offers this branch as an optional phase engine, and validates the $O(1)$ assembly (128)—which is the default—against it exactly.

Closed-form traces used in Section 4.3: For completeness, we record the closed forms

$$\operatorname{tr}(H) = \frac{T}{1-\phi^2}, \tag{A.40}$$

$$1_T' G 1_T = \frac{T}{1-\phi} + \frac{\phi^T - 1}{(1-\phi)^2}, \tag{A.41}$$

$$\frac{1_T' H 1_T}{T} = \frac{1}{1-\phi^2} + \frac{2\phi 1_T' G 1_T}{T(1-\phi^2)}, \tag{A.42}$$

$$\operatorname{tr}(MH) = \operatorname{tr}(H) - \frac{1_T' H 1_T}{T}, \tag{A.43}$$

$$\operatorname{tr}(GH) = \frac{\phi[T(1-\phi^2) + \phi^{2T} - 1]}{(1-\phi^2)^3}, \tag{A.44}$$

$$\operatorname{tr}(H^2) = \frac{T}{(1-\phi^2)^2} + \frac{2\phi \operatorname{tr}(GH)}{1-\phi^2}, \tag{A.45}$$

$$1_T' G H 1_T = \frac{T(1-\phi^2)(1+\phi+\phi^T) - (1-\phi^T)(1+3\phi+3\phi^2-\phi^{T+1})}{(1-\phi)^2(1-\phi^2)^2}, \tag{A.46}$$

$$\frac{1_T' H^2 1_T}{T} = \frac{1_T' H 1_T}{T(1-\phi^2)} + \frac{2\phi 1_T' G H 1_T}{T(1-\phi^2)}, \tag{A.47}$$

$$\operatorname{tr}((MH)^2) = \operatorname{tr}(H^2) - \frac{2\, 1_T' H^2 1_T}{T} + \frac{(1_T' H 1_T)^2}{T^2}, \tag{A.48}$$

$$\operatorname{tr}(GG') = \frac{T(1-\phi^2) - 1 + \phi^{2T}}{(1-\phi^2)^2}, \tag{A.49}$$

$$1_T' G G' 1_T = \frac{T}{(1-\phi)^2} - \frac{(1-\phi^T)(1+2\phi-\phi^T)}{(1-\phi)^2(1-\phi^2)}. \tag{A.50}$$

Proof of Proposition 8: Write $X = y'Ay/(y'By)^{1/2}$ and $Y = |\mathcal{G}|/y'By$, so that $t(\hat{\beta}) = \sqrt{\nu}\, XY^{-1/2}$. For existence, condition on z. Since z reveals $v_1, \ldots, v_{T-1}$ but not v_T, under (12) the return vector r is conditionally normal with covariance matrix $\Sigma_z = (1-\rho_{uv}^2) I_T + \rho_{uv}^2 e_{T,T} e_{T,T}'$. Let $q_z = Mz/(z'Mz)^{1/2}$ and $P_z = M - q_z q_z'$, an idempotent matrix of rank ν with $P_z q_z = 0_T$. Then $X = q_z' r$ and $Y = r' P_z r$. Because $P_z \Sigma_z P_z \succeq (1-\rho_{uv}^2) P_z$, Anderson's inequality bounds the small-ball probabilities of $P_z r$ by those of $(1-\rho_{uv}^2)^{1/2}$ times a standard normal vector on the range of P_z, so that $\mathbb{E}[Y^{-q/2} \,|\, z] \leq (1-\rho_{uv}^2)^{-q/2} \mathbb{E}[(\chi_\nu^2)^{-q/2}] < \infty$ for $q < \nu$, uniformly in z. Conditionally on z, X is normal with variance at most one and a mean that is linear in z and $v_1, \ldots, v_{T-1}$, so Hölder's inequality gives $\mathbb{E}|t(\hat{\beta})|^p < \infty$ for $p < \nu$. Conversely, the conditional covariance matrix of Mr is nonsingular on the range of M, so $X = c_z' P_z r + \eta$ with η normal, independent of $P_z r$ and of positive variance. By Anderson's inequality once more, $\mathbb{E}[|X|^p \,|\, z, P_z r] \geq \mathbb{E}[|\eta - \mathbb{E}(\eta)|^p \,|\, z] > 0$, while the conditional density of Y is of exact order $y^{\nu/2-1}$ near zero, so that $\mathbb{E}[Y^{-p/2} \,|\, z] = \infty$ for $p \geq \nu$; hence $\mathbb{E}|t(\hat{\beta})|^p = \infty$ for $p \geq \nu$. When $\rho_{uv} = 0$, $\Sigma_z = I_T$, X and Y are conditionally independent, and $t(\hat{\beta})$ is conditionally noncentral t, as in Section 2.4. When $\rho_{uv} \neq 0$ they are not, because $\operatorname{Cov}(X, P_z r \,|\, z) = P_z \Sigma_z q_z = \rho_{uv}^2 (e_{T,T}' q_z) P_z e_{T,T}$ is nonzero in general, but the argument above does not require independence, and neither does anything that follows.

For (175), apply (174) with $a = p/2$, which requires $p \geq 2$, to $|\mathcal{G}|^{-p/2}$; this is valid pathwise because $\mathcal{G} \succ 0$ almost surely. Then

$$\mathbb{E}[t^p] = \nu^{p/2}\, \mathbb{E}\left[(y'Ay)^p |\mathcal{G}|^{-p/2}\right] = \frac{\nu^{p/2}}{\Gamma_2(p/2)} \int_{S \succ 0} |S|^{\frac{p-3}{2}}\, \mathbb{E}\left[(y'Ay)^p \operatorname{etr}(-S\mathcal{G})\right] \mathrm{d}S, \tag{A.51}$$

the interchange being justified by Tonelli's theorem applied separately to the positive and negative parts of $(y'Ay)^p$, both of which are integrable because $\mathbb{E}[|y'Ay|^p |\mathcal{G}|^{-p/2}] < \infty$

for $p < \nu$. Since $\operatorname{tr}(S\mathcal{G}) = s_{11}\, y'By + 2s_{12}\, y'Ay + s_{22}\, y'Cy$, the inner expectation equals $\partial^p \Psi(\gamma, s, t)/\partial\gamma^p$ evaluated at $\gamma = -2s_{12}$, $s = s_{11}$, $t = s_{22}$. The change of variables to the Cholesky factorization, $s_{11} = \ell_1^2$, $s_{12} = \ell_1\ell_2$, $s_{22} = \ell_2^2 + \ell_3^2$ with $\ell_1, \ell_3 > 0$, has Jacobian $4\ell_1^2\ell_3$ (Muirhead, 2005, Theorem 2.1.9), and $|S| = \ell_1^2\ell_3^2$, which gives the weight $4\ell_1^{p-1}\ell_3^{p-2}$ in (175).

For (176), start again from $t(\hat{\beta}) = \sqrt{\nu}\, XY^{-1/2}$. Every step below is an identity that holds pathwise, so no independence between X and Y is needed; the interchanges of expectation and integration are justified by Tonelli's theorem as before, together with the bound $\mathbb{E}_Z[|He_p(Z)|e^{aZ}] \le c_p(1 + |a|)^p e^{a^2/2}$ for the Hermite step. The scalar gamma identity applied to $Y^{-p/2}$ gives $\mathbb{E}[t^p] = \nu^{p/2}\Gamma(p/2)^{-1}\int_0^\infty \lambda^{p/2-1}\mathbb{E}[X^p e^{-\lambda Y}]\mathrm{d}\lambda$. Since $Y = y'Cy - X^2$, the tilt is $e^{-\lambda\, y'Cy}e^{\lambda X^2}$, and the Hermite identity $\mathbb{E}[He_p(Z)e^{aZ}] = a^p e^{a^2/2}$ with $a = \sqrt{2\lambda}X$ gives $X^p e^{\lambda X^2} = (2\lambda)^{-p/2}\mathbb{E}_Z[He_p(Z)e^{\sqrt{2\lambda}XZ}]$ for an independent standard normal Z. Substituting $Z = \theta\sqrt{2\lambda\, y'By}$ turns the Z-expectation into an integral over $\theta \in \mathbb{R}$ whose exponent is $-\lambda(\theta^2\, y'By - 2\theta\, y'Ay + y'Cy) = -\lambda\|M(\theta z - r)\|^2$, and whose Jacobian contributes a factor $(y'By)^{1/2}$. Expanding $He_p(x) = \sum_i (-1)^i p!\, x^{p-2i}/(i!(p-2i)!2^i)$ leaves powers $(y'By)^{(p+1)/2-i}$, which are nonnegative integers precisely when p is odd; collecting terms and writing each $\mathbb{E}[(y'By)^m e^{\cdot}]$ as $(-1)^m\partial^m\Psi/\partial s^m$ gives (176), the sign $(-1)^i(-1)^{m_i} = (-1)^{(p+1)/2}$ being free of i.

For (177), use instead the two-dimensional Gaussian integral $|\mathcal{G}|^{-1/2} = \pi^{-1}\int_{\mathbb{R}^2} e^{-\xi'\mathcal{G}\xi}\mathrm{d}\xi$, valid pathwise for $\mathcal{G} \succ 0$, interchange by the same argument (valid for $p = 1 < \nu$), and substitute $\xi = \sqrt{v}\,(\cos\psi, \sin\psi)'$ with $\mathrm{d}\xi = \frac{1}{2}\mathrm{d}v\,\mathrm{d}\psi$, noting $\xi'\mathcal{G}\xi = v\cos^2\psi\; y'By + v\sin 2\psi\; y'Ay + v\sin^2\psi\; y'Cy$. At $p = 1$ this agrees with (176) after one integration by parts in θ, which replaces $\theta\, y'By$ by $y'Ay$. □

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic's Claude to assist with drafting and editing the exposition, with the implementation and testing of the accompanying MATLAB programs, and with checking derivations and numerical results. After using this tool the authors reviewed and edited the content as needed and take full responsibility for

the content of the publication.

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